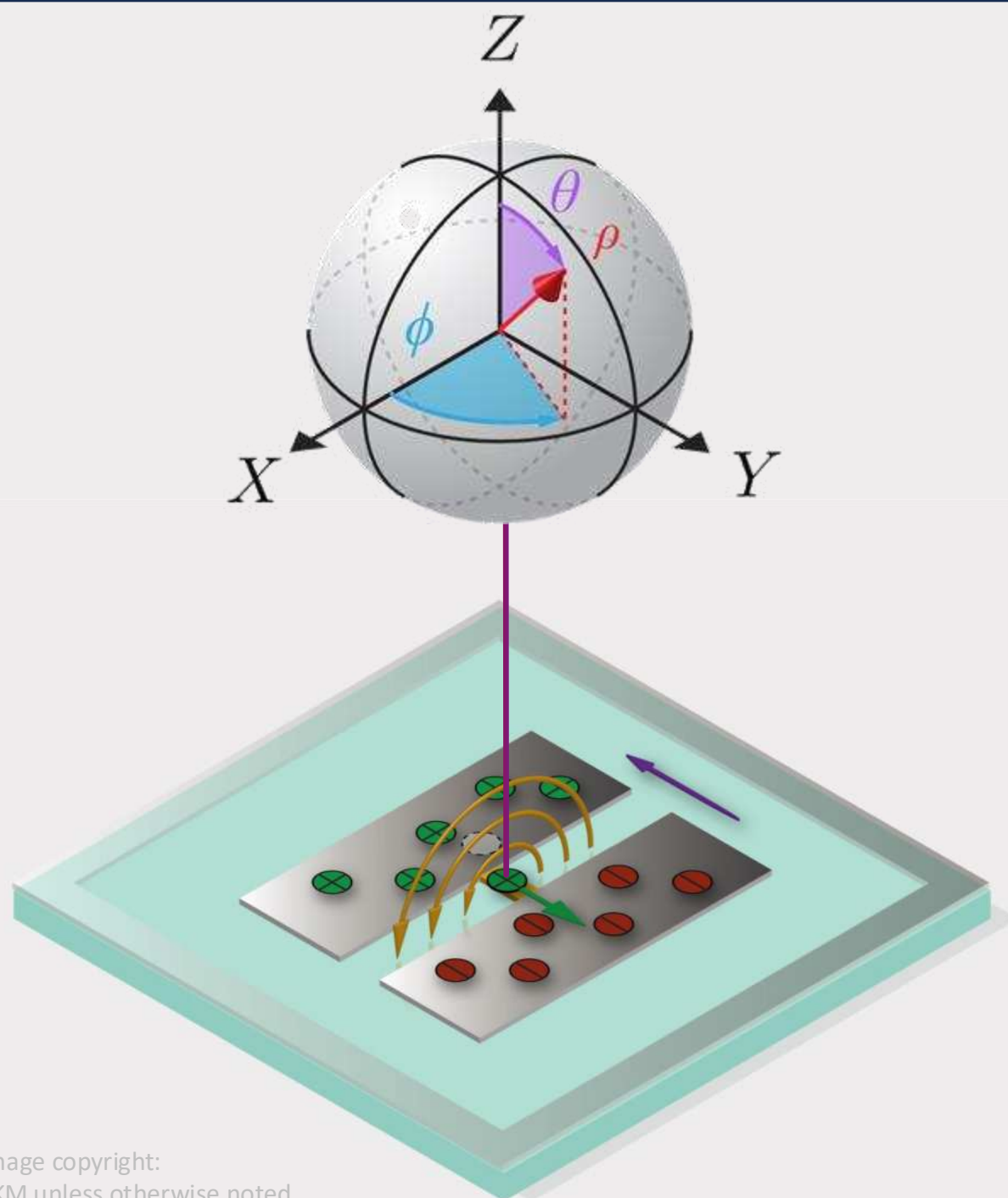
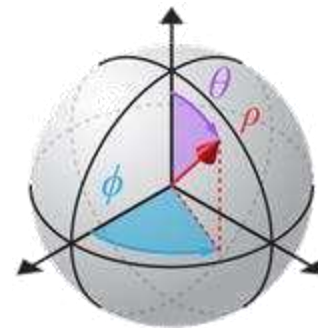
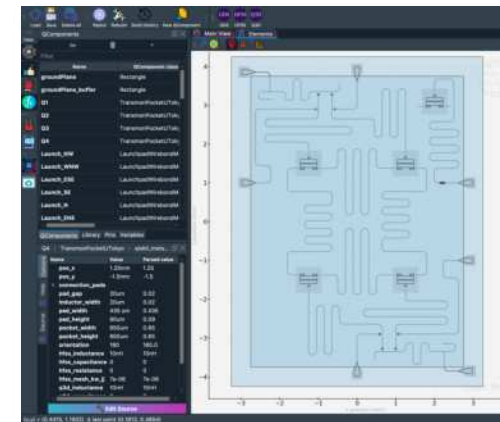




Overview of Quantum Hardware Design

Energy, Circuits, and Metal



Slides!

Zlatko K. Minev

Current Affiliation:

Google Quantum AI; Assoc. Fellow @ CIFAR

Former Affiliation: IBM Quantum; Yale University



@zlatko_minev



zlatko-minev.com/blog

Huge THANK YOU to the organizers and broader team!

Meet the Team

This is the organizational team responsible for making this workshop happen. The combined efforts between representatives from UCLA, USC and Google Quantum AI is responsible for the stellar lineup and broad range of topics. If you have any questions or concerns please reach out to:

quantum.ucla@gmail.com



Eli Levenson-Falk



Zlatko Minev



Mark Gyure



Alex Jürgens



Nicolas Dirneger



Cody Fan



Murat C. Sarihan



Samuel Oh



Drew Downing



Meet Our Speakers



Dr. Ofer Naaman
Ofer Naaman (he/him) is a research scientist with Google Quantum AI, where he leads the readout hardware team. Ofer holds a BSc degree from Tel Aviv University and a PhD in physics from UC San Diego. Prior to joining Google, Ofer was at MIT, Boulder, UC Berkeley, and Northrop Grumman, where he has worked on topics ranging from single-electron transistors, quantum computing, and parametric amplifiers, to superconducting logic and cryogenic memory.



Professor Jens Koch
Professor Jens Koch is a theoretical physicist specializing in quantum information and condensed matter physics. He is a faculty member at Northwestern University, where his research focuses on superconducting circuits, quantum electronics (QED), and the design of qubits for quantum computing. Prof. Koch is widely recognized for co-developing the transmon qubit, a more robust type of superconducting qubit that has become a foundational element in many quantum computing architectures.



Benjamin Jarvis-Frain
Ben is a quantum integrated circuit designer at Rigetti Computing where he works on modeling and design of superconducting quantum processors. Ben's work at Rigetti has focused on developing scalable simulation techniques for quantum processors in 3D integrated environments, and understanding parasitic, multi-qubit interactions that arise in these systems.



David Pahl
David is a PhD student at MIT under Prof. Will Oliver. Before coming to MIT, David studied Electrical Engineering and Information Technology at ETH Zurich, where he received a BSc in 2022 with distinction. At MIT, his research aims to develop scalable hardware architectures for fast and high-fidelity qubit readout. He is also designing quantum devices, capable of near-term quantum error correction.



Meet Our Speakers



Dr. Murat Can Sarihan
Dr. Murat Can Sarihan is a research scientist at Google Quantum AI, where he leads the readout hardware team. He holds a BSc degree from Tel Aviv University and a PhD in physics from UC San Diego. Prior to joining Google, Murat was at MIT, Boulder, UC Berkeley, and Northrop Grumman, where he has worked on topics ranging from single-electron transistors, quantum computing, and parametric amplifiers, to superconducting logic and cryogenic memory.



Lukas Pahl
Lukas is a graduate student researcher in the Engineering Quantum Systems group at MIT, advised by Prof. Will Oliver. He received his BSc in Electrical Engineering and Information Technology at ETH Zurich with distinction. He is currently working to explore techniques that will deliver superconducting quantum computers at scale. His research involves improving qubit gate fidelities through realistic control as well as the development and demonstration of low-overhead quantum error correction architectures.



Dung Pham
Dung (Strong) is a Ph.D. student in the Quantum Engineering group at Princeton University, advised by Prof. Hakan Türeci. He earned his B.Sc. degree in Physics and Electrical & Computer Engineering from Worcester Polytechnic Institute (WPI) at Princeton. His research focuses on developing theoretical techniques for modeling charged quantum fluids and superconducting quantum circuits, along with building computational tools for large-scale, efficient simulation of these devices.



Dr. Daniel Sank
Dr. Dan Sank is a physicist and researcher at Google Quantum AI, where he works on the development of superconducting qubits and scalable quantum computing systems. He has been a key contributor to the engineering of Google's quantum processors, including the Sycamore chip used in their demonstration of quantum supremacy. Dr. Sank focuses on improving qubit coherence, microwave control, and system integration, bridging the gap between quantum hardware and practical computation.



Meet Our Speakers



Professor Eli Levenson-Falk
Professor Eli Levenson-Falk is a physicist and researcher at Google Quantum AI, where he works on advancing the field of quantum computing through innovations in superconducting qubit technologies and quantum control. He is renowned for his pioneering experiments that captured and realized a quantum state in real time, offering profound insight into the nature of quantum entanglement and quantum physics research.



Dr. Zlatko Minev
Dr. Zlatko Minev is a physicist and researcher at Google Quantum AI, where he works on advancing the field of quantum computing through innovations in superconducting qubit technologies and quantum control. He is renowned for his pioneering experiments that captured and realized a quantum state in real time, offering profound insight into the nature of quantum entanglement and quantum physics research.



Dr. Jin-Sung Kim
An experimental quantum physicist by training, Dr. Kim is currently responsible for developing the ecosystem of the NISQ Quantum Platform. He manages technical collaborations and external partnerships, and ensures that users have seamless access to quantum devices, providing, precision, enterprise users are enabled with the best tools for developing useful quantum applications. He received his PhD from Princeton University studying electron spin control in metal-oxide-semiconductor quantum dots.



Professor Hakan Türeci
Professor Hakan Türeci is a theoretical physicist at Princeton University, specializing in quantum optics, quantum information, and the physics of open quantum systems. His research explores the fundamental behavior of light-matter interactions, non-equilibrium systems, and quantum technologies such as superconducting circuits. Prof. Türeci has made significant contributions in understanding how coherence and dissipation interplay in complex quantum systems, with applications in quantum computing and photonics.



Professor Michel Devoret
Professor Michel Devoret is a renowned experimental physicist who recently joined UC Santa Barbara and Google Quantum AI, following a distinguished career at Yale University. He is a pioneer in the field of superconducting quantum circuits, best known for co-developing the transmon qubit and advancing quantum circuit architectures. His research has been instrumental in shaping the architecture of today's quantum processors, blending deep theoretical insight with experimental innovation.



Meet Our Speakers



Dr. Loren D. Aiegria
Dr. Loren D. Aiegria is a staff scientist at Lawrence Livermore National Laboratory, specializing in the materials science understanding of low-dissipation information processing technologies. As part of the Quantum Coherent Device Physics Group, he focuses on the design, manufacturing, and testing of innovative superconducting architectures to explore the fundamental limits of scalability in quantum information processing.



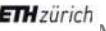
Sadman Ahmed Shanto
Sadman Ahmed Shanto is a Ph.D. candidate in Physics at the University of Southern California, working in the Levenson-Falk Lab on superconducting quantum hardware. His research focuses on quasiparticle dynamics in superconducting circuits, and the optimization of quantum device design workflows.



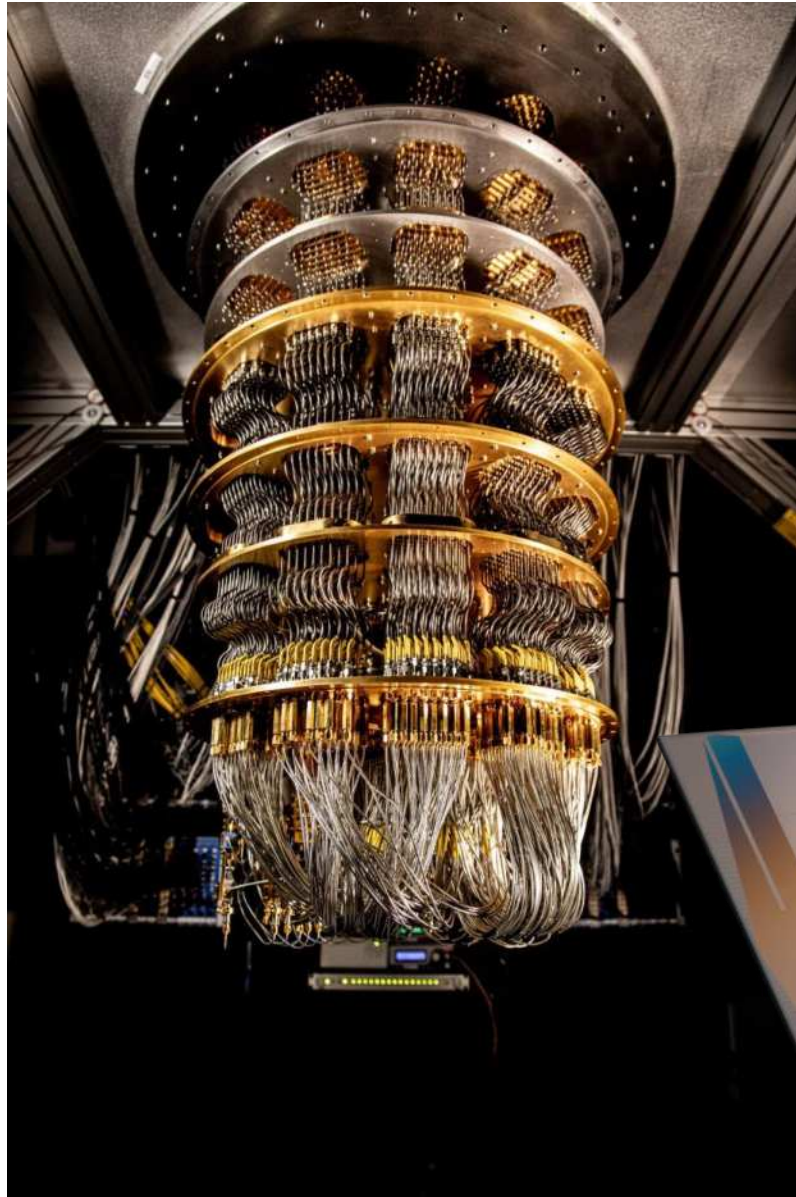
Dr. Arpit Arora
Arpit is a quantum scientist at Nanoscale, UCLA, working at the interface of quantum materials, devices, and computing. His research focuses on understanding emergent phenomena in novel materials to inform the design of next-generation components for utility-scale quantum computers. Currently, he investigates photon-matter hybrid systems to advance coherent quantum technologies and enable new capabilities in quantum control and information processing.



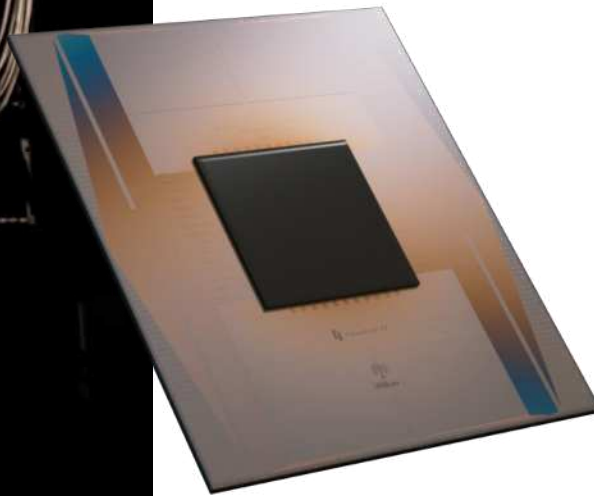
Professor Andreas Waltraff
Prof. Andreas Waltraff conducts research in quantum information processing and quantum optics. He has taught as a professor at ETH Zurich in Zurich since October 2008. He worked at a research scientist with Robert J. Schoelkopf at Yale University from 2002 to 2008. His current work at ETH Zurich focuses on hybrid quantum systems combining superconducting electronic circuits with semiconductor quantum dots and individual Rydberg atoms as well as quantum error

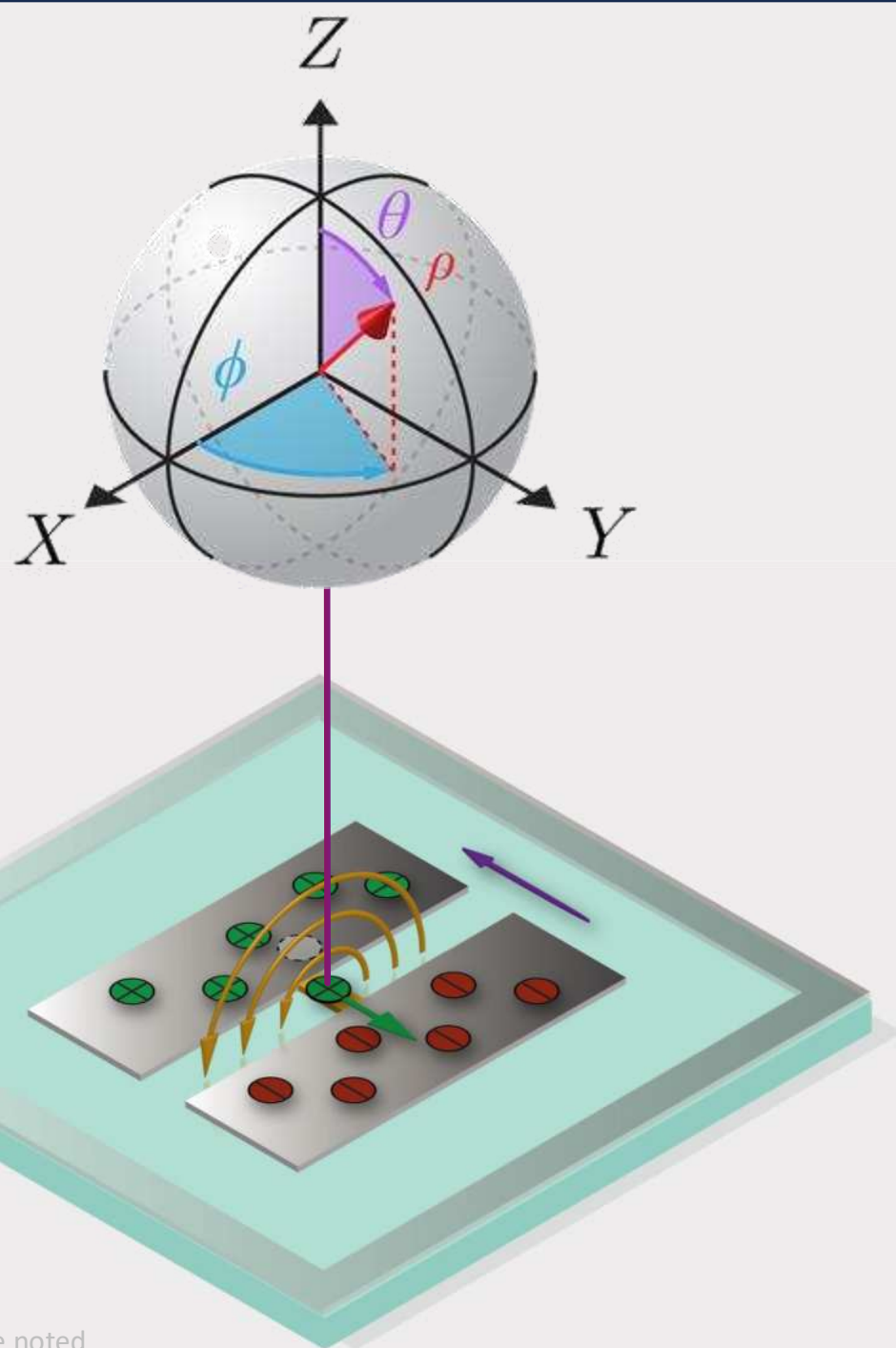


Let's get to know each other 😊



1. Have you ever used a quantum device?
2. Analyzed/designed/made one?

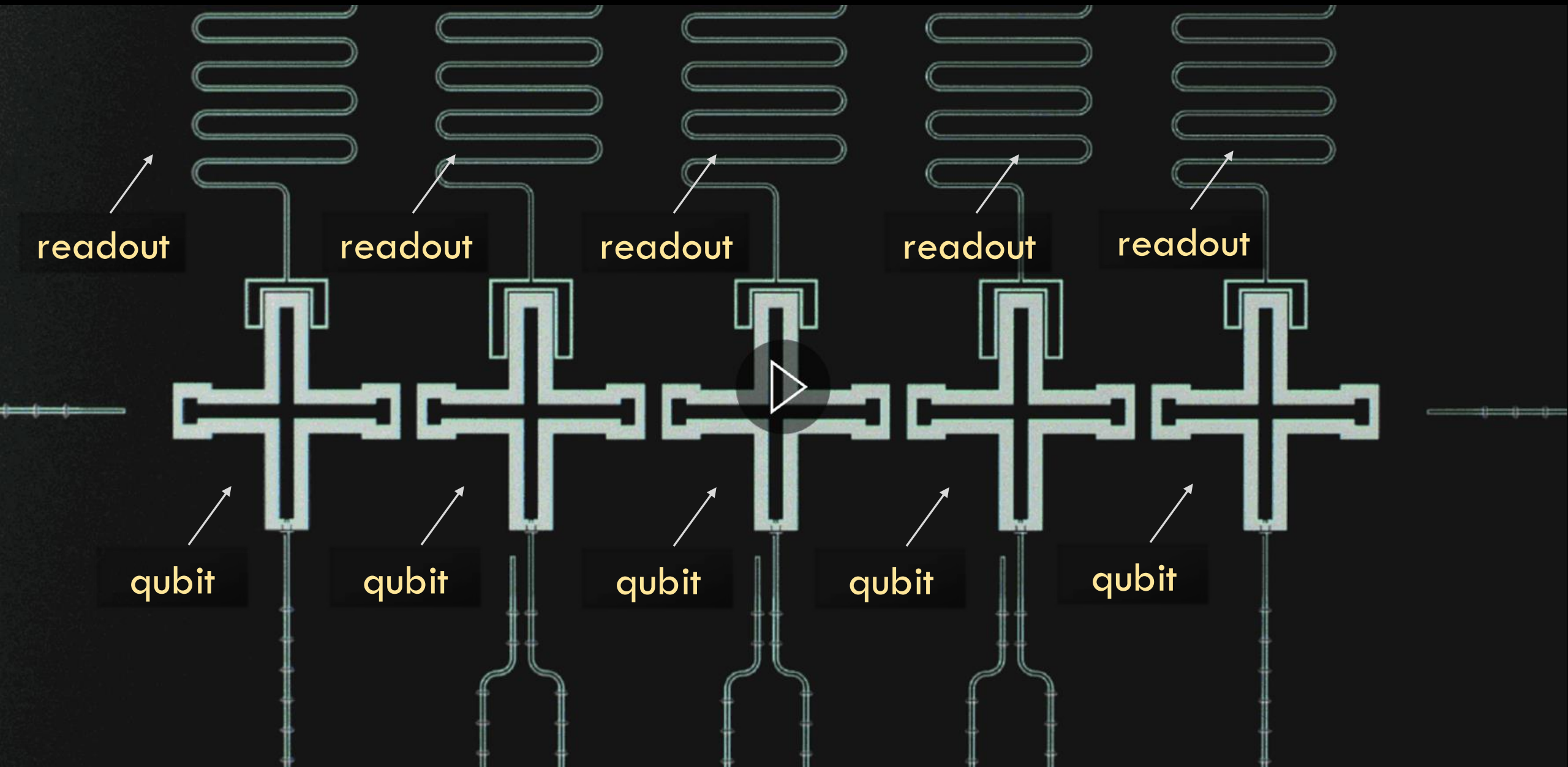




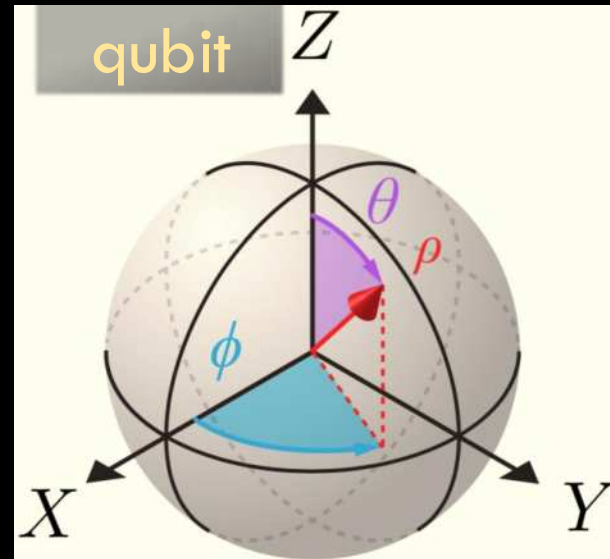
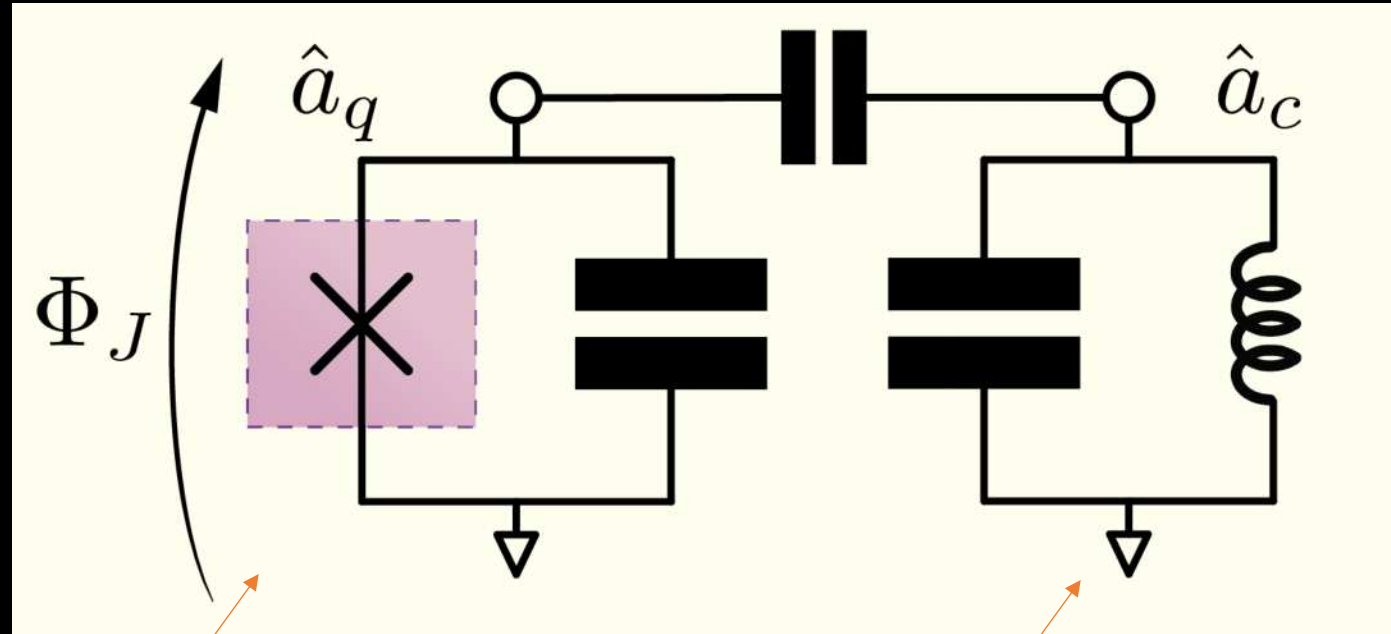
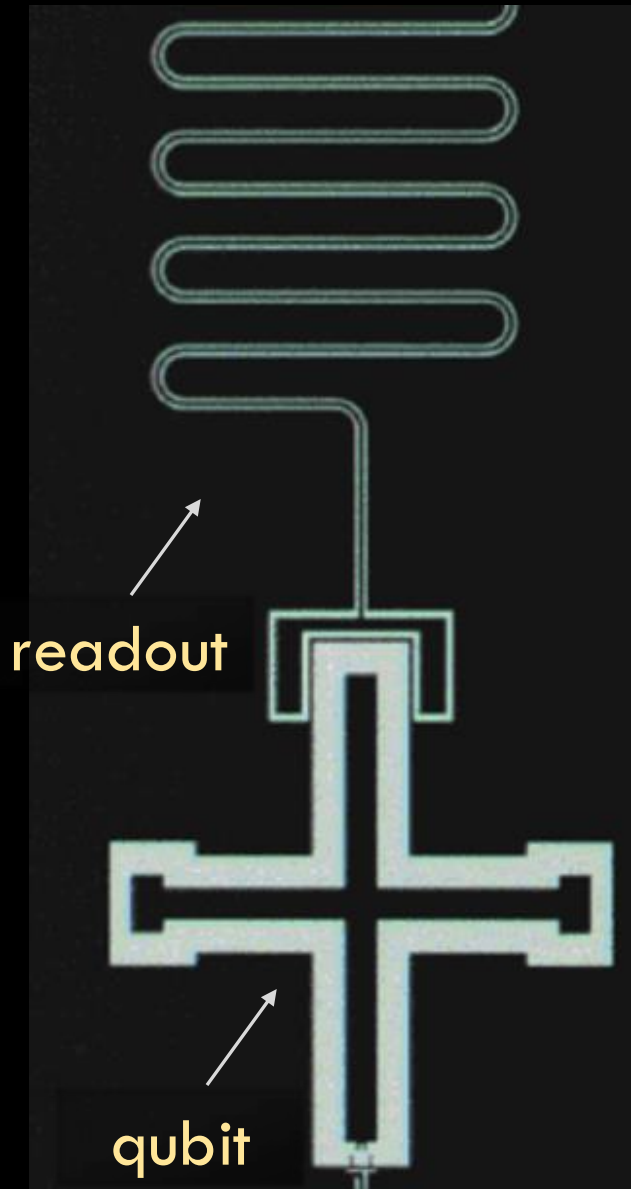
Designing qubits

THE BIG PICTURE
before calculations

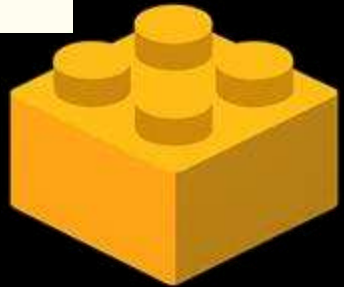
Example Google Architecture



Physical and abstract circuits and Hilbert space



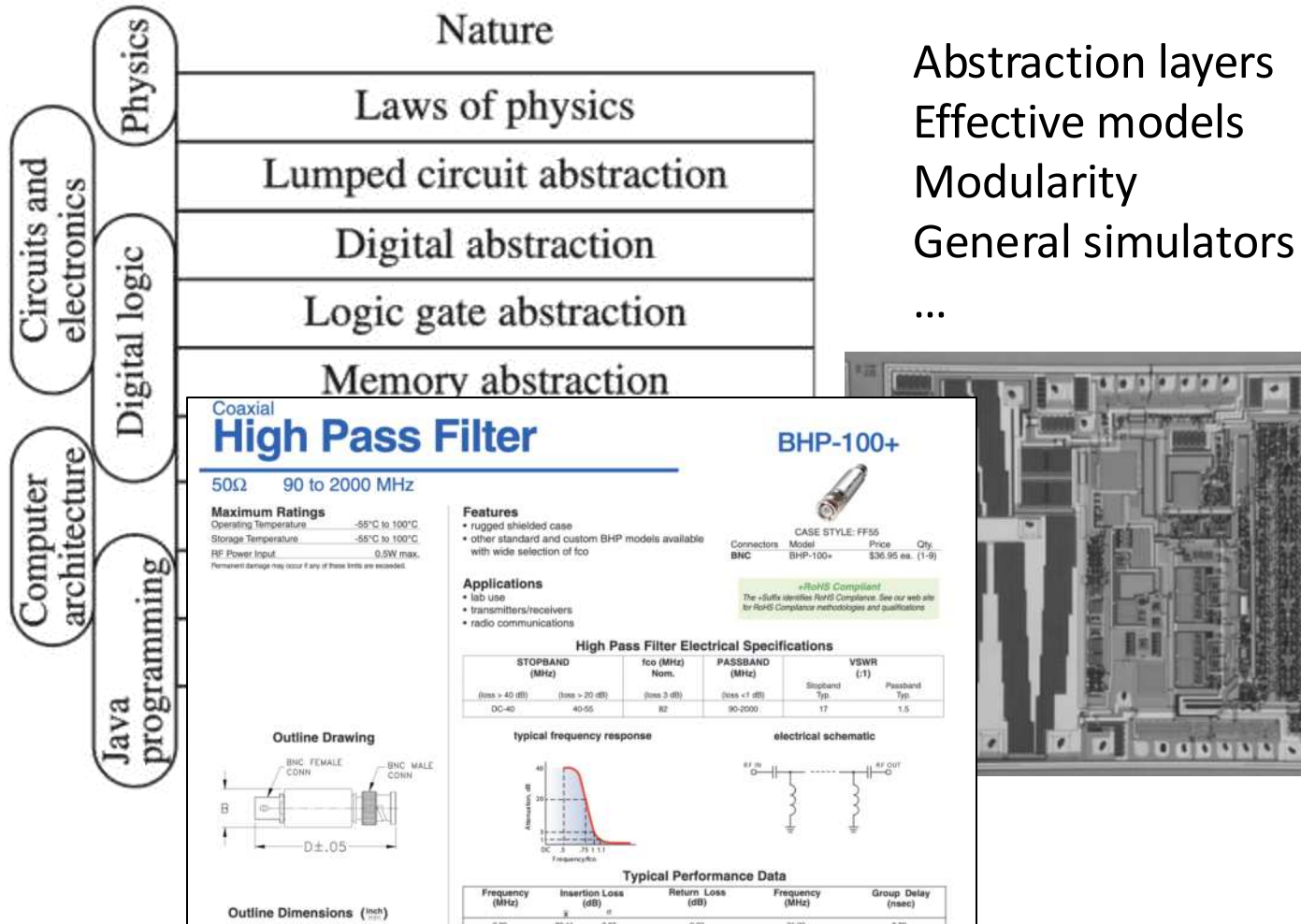
readout



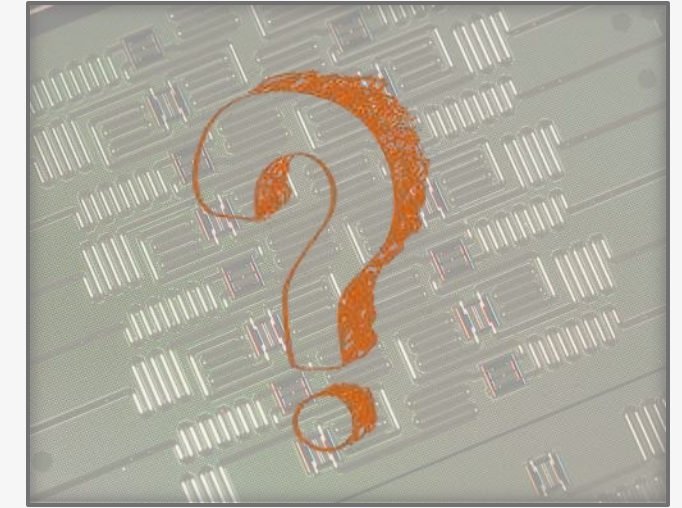
$$\hat{H} \approx \omega_0 \hat{a}^\dagger \hat{a} - \frac{\alpha}{2} \hat{a}^{\dagger 2} \hat{a}^2$$

How to design a classical processor

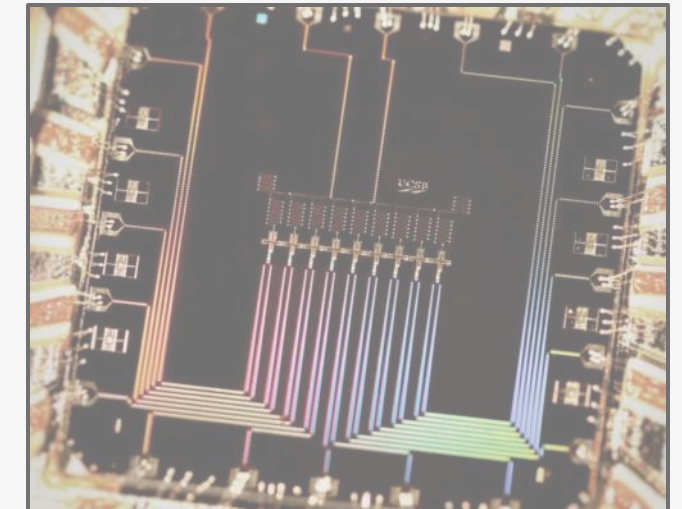
Classical hierarchy of abstractions



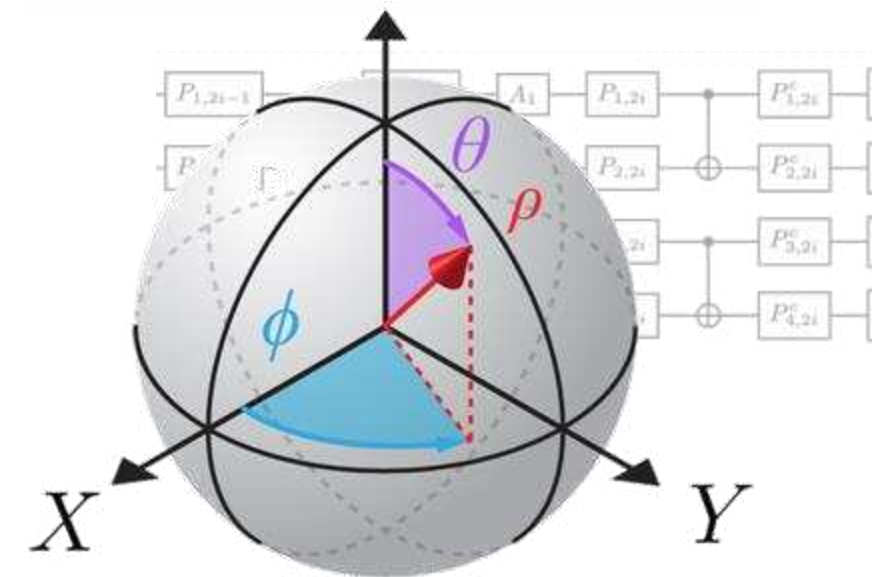
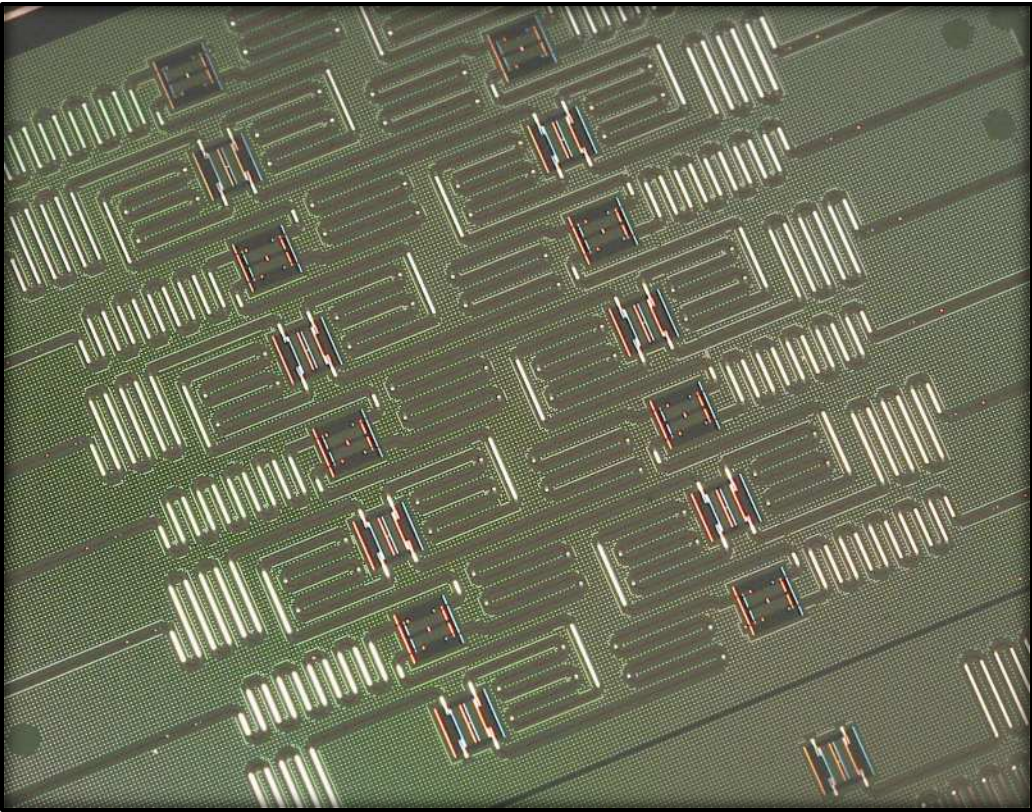
How to design a quantum processor?



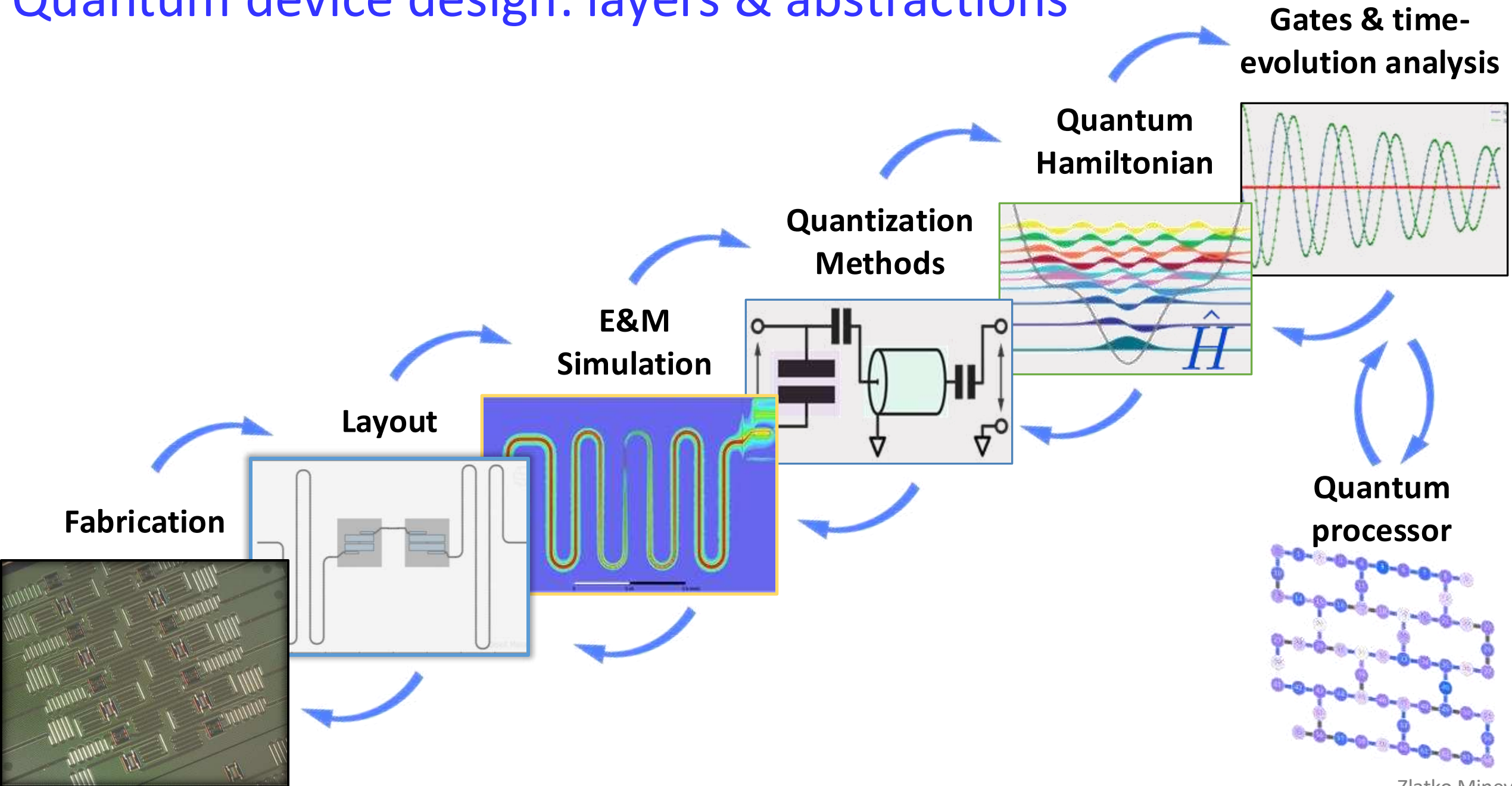
IBM 16-qubit processor (2018)



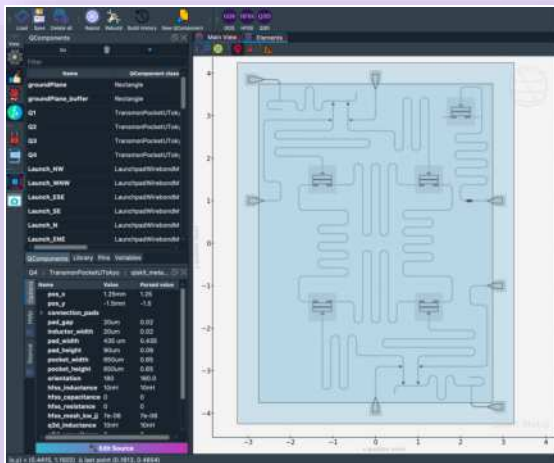
John Martinis group (2015)



Quantum device design: layers & abstractions



Qiskit | quantum device
design

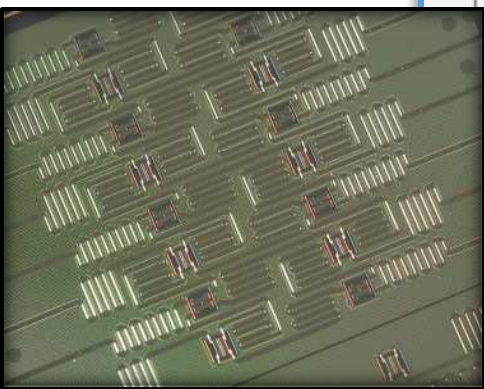


+ pyEP
+ SCQ
+ QuT
+ ⋮

Open source

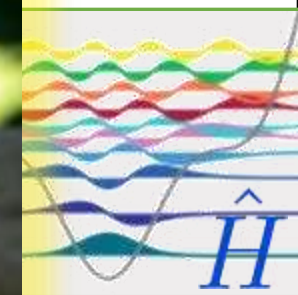
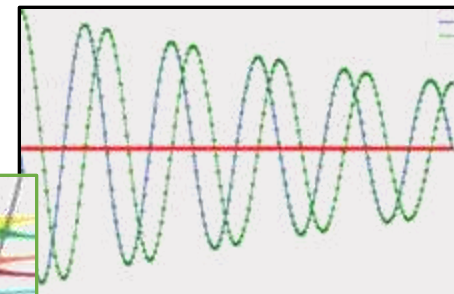
Layout

Fabrication

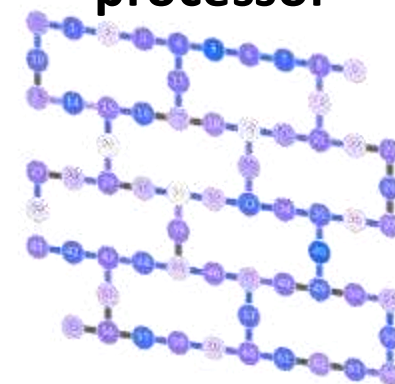


Gates & time-
evolution analysis

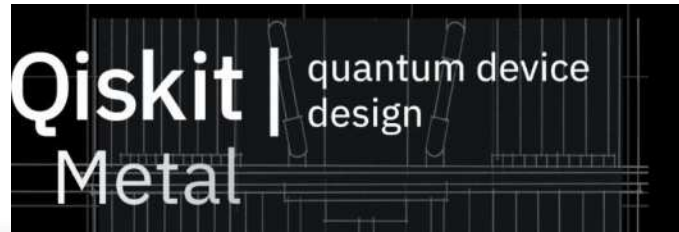
quantum
miltonian



Quantum
processor

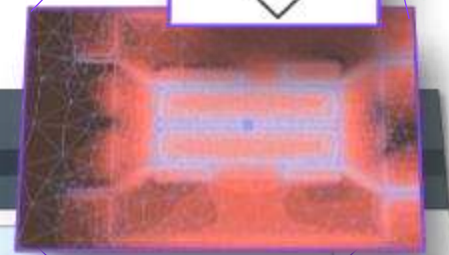
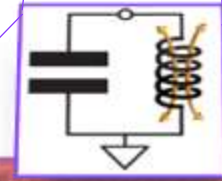
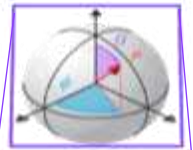
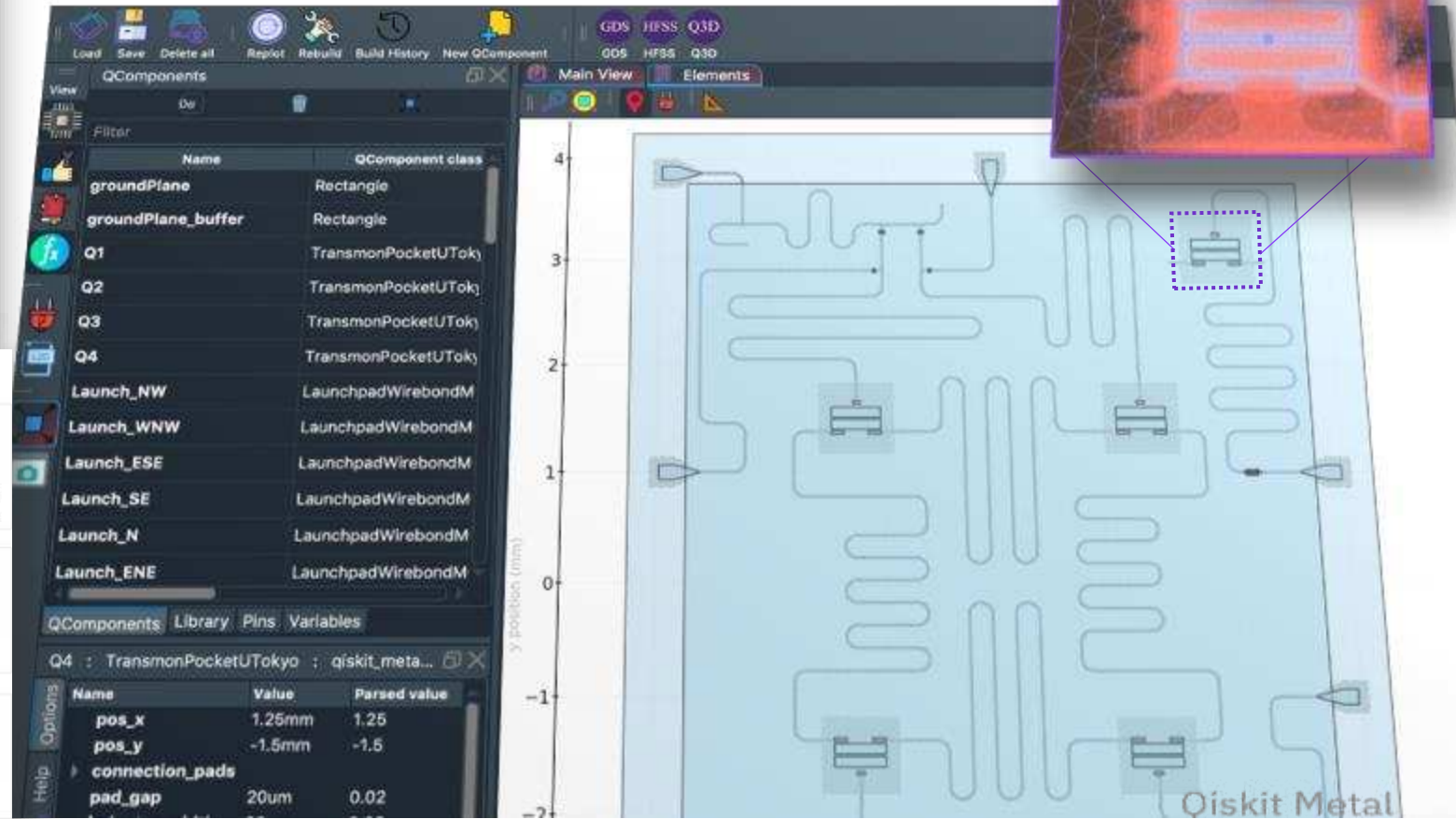
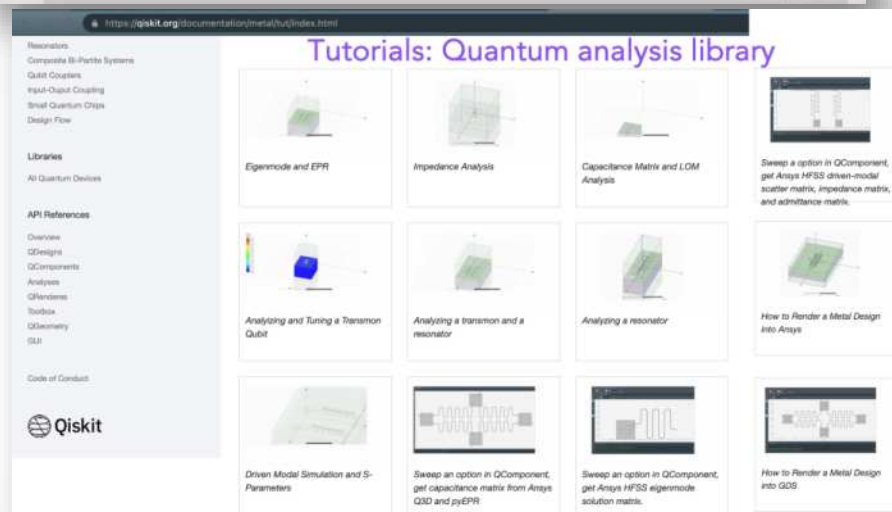
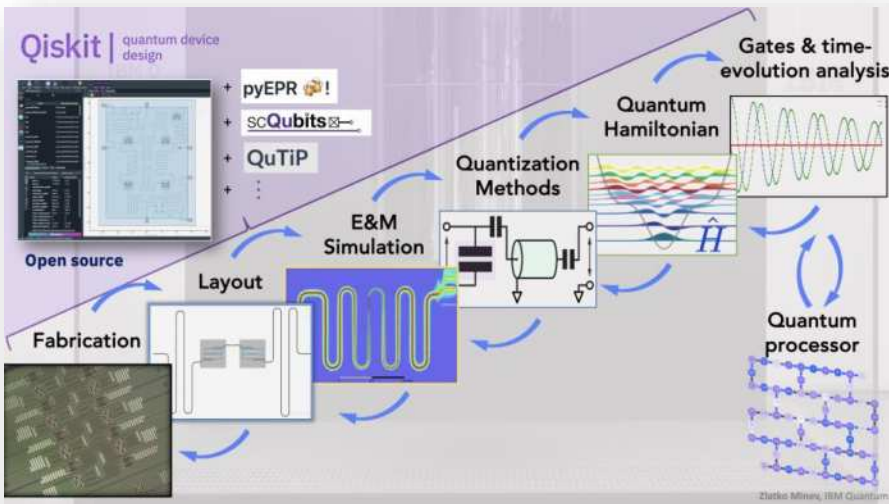


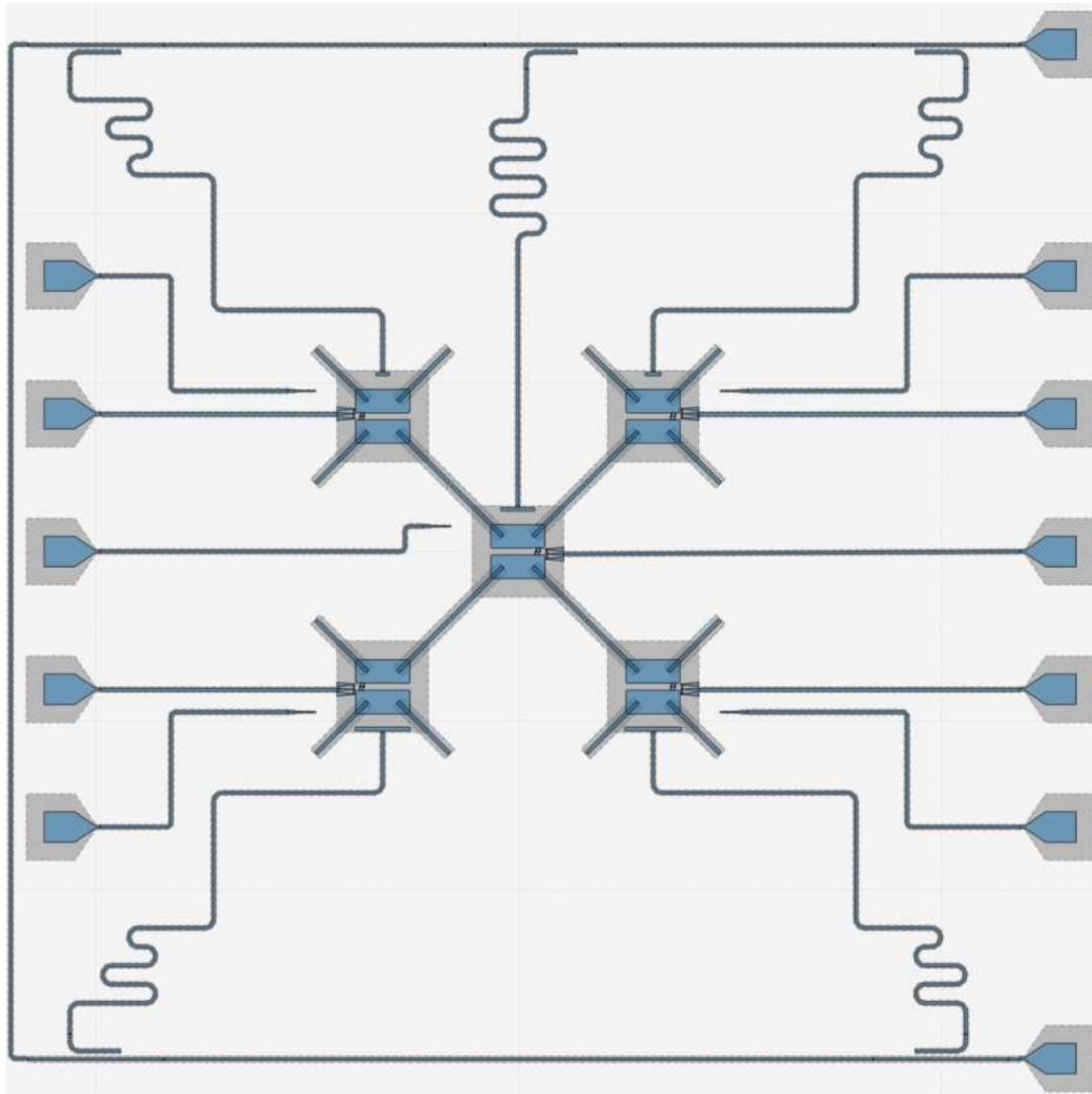
Automation in Practice: Qiskit Metal + Community



- + advanced talk (today)
- + workshops (tomorrow)
- + maintainer's meeting discussion

Community ecosystem driven (Apache-2.0)
github.com/qiskit-community/qiskit-metal





Qiskit | quantum device
Metal design

Example designs from the
community, folks like you!

NQCP
NIELS BOHR INSTITUTE



5 tunable transmon weight-4 devices
(direct coupling and Delft/ETH style)



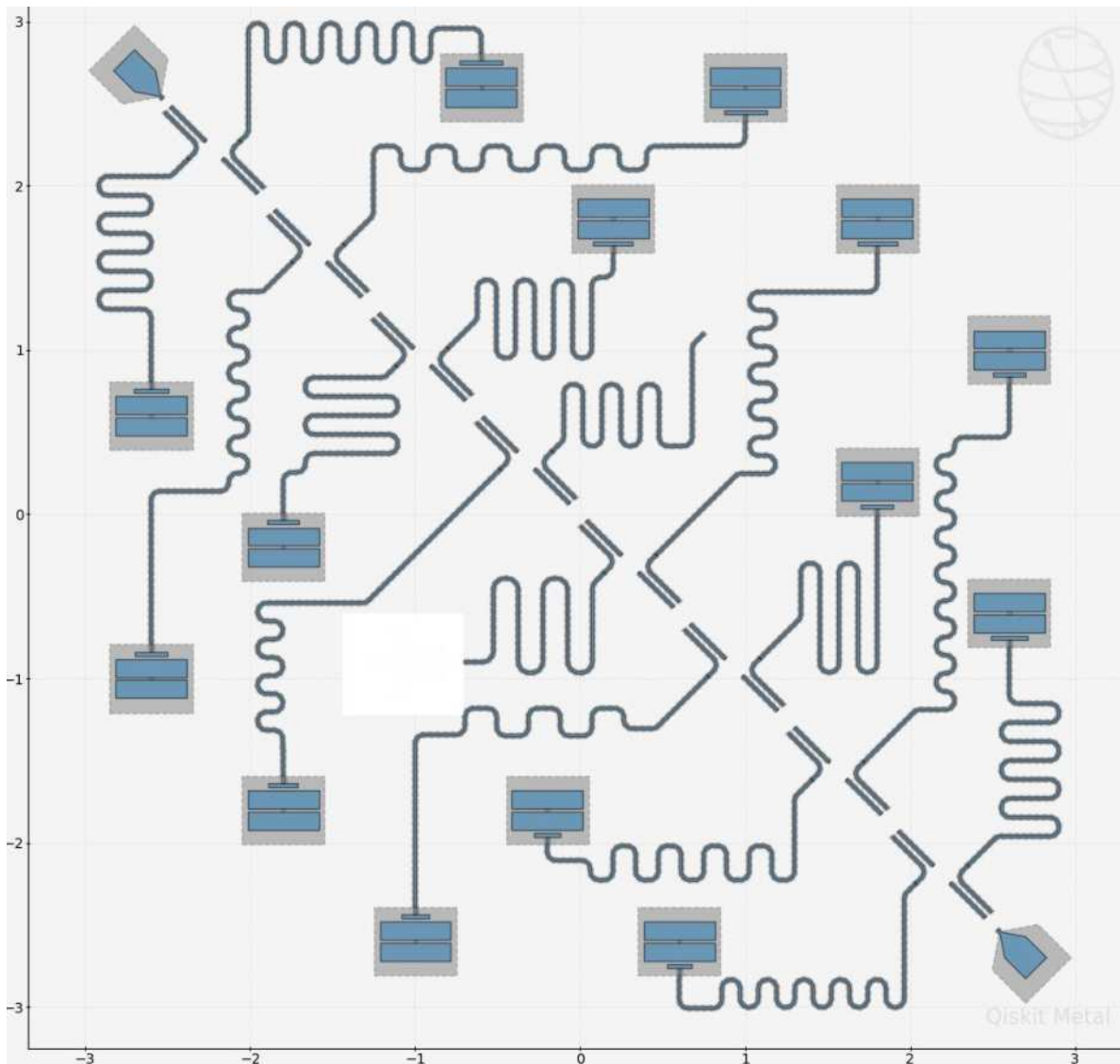
Designs:

- Benjamin Joecker, PhD
- Thue Christian Thann, PhD Candidate



Affiliations:

- Niels Bohr Institute, University of Copenhagen
- Novo Nordisk Quantum Computing Programme (NQCP)



Example designs from the community, folks like you!

NQCP
NIELS BOHR INSTITUTE



Standard candle devices for tracing reproducibility

Designs:

-Thue Christian Thann, PhD Candidate



Affiliations:

- Niels Bohr Institute, University of Copenhagen
- Novo Nordisk Quantum Computing Programme (NQCP)



Example designs from the community, folks like you!

NQCP
NIELS BOHR INSTITUTE



Christopher. W. Warren

Niels Bohr Institute
Verified email at nbi.ku.dk

quantum computing quantum simulation
quantum information

12 Tunable qubit chip; directly coupled (dubbed “Adamite”)

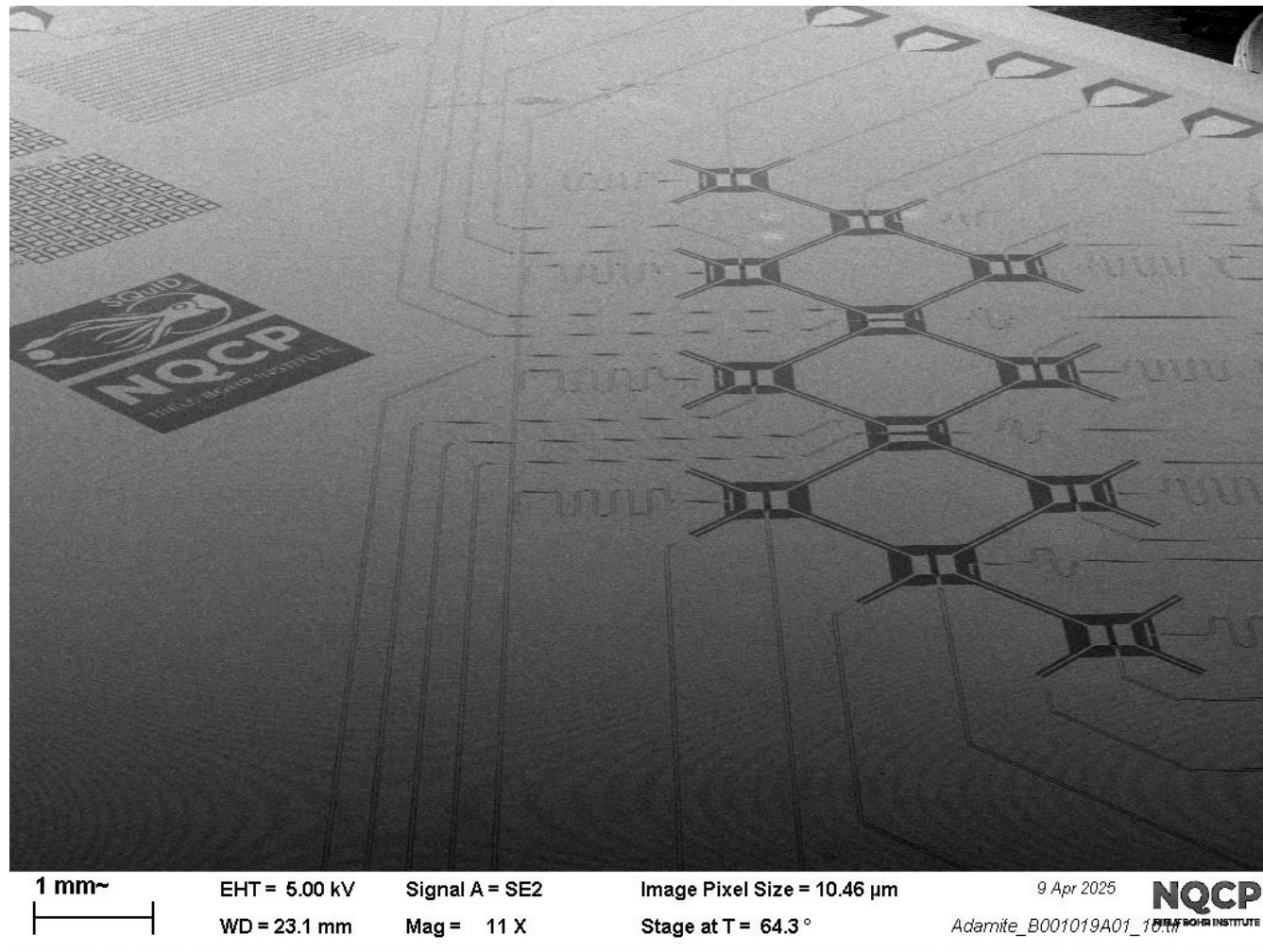
Design:

- Christopher W. Warren, PhD

Affiliations:

- SQuID Lab, Niels Bohr Institute, University of Copenhagen
- Novo Nordisk Quantum Computing Programme (NQCP)

Zlatko Minev



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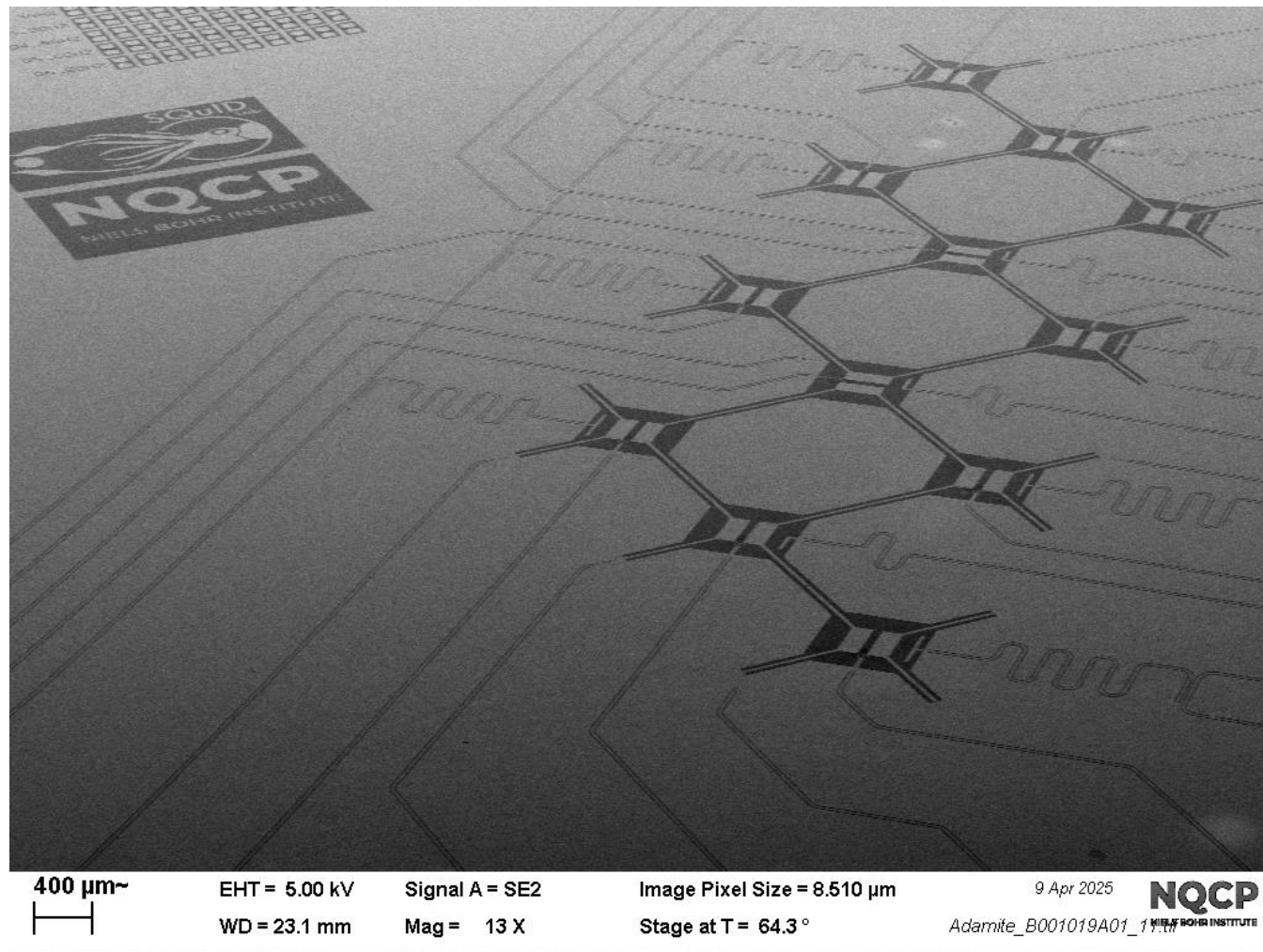
Design:

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Affiliations:

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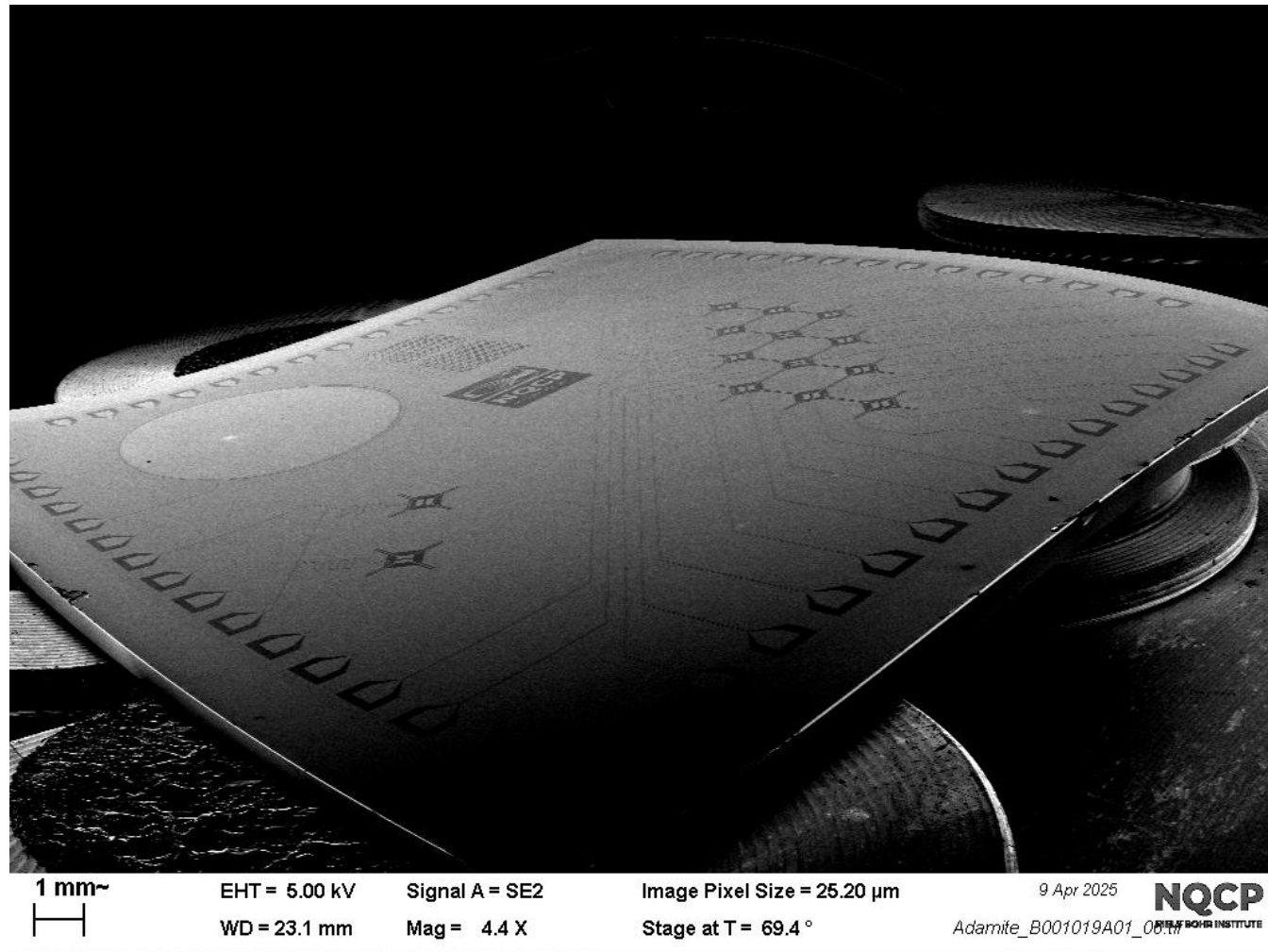
Design:

- Christopher W. Warren, PhD

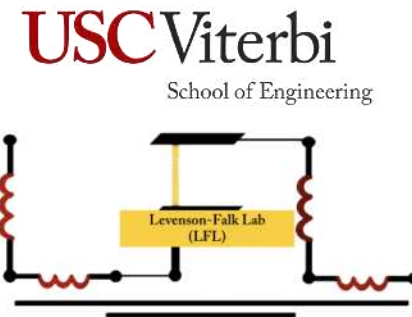
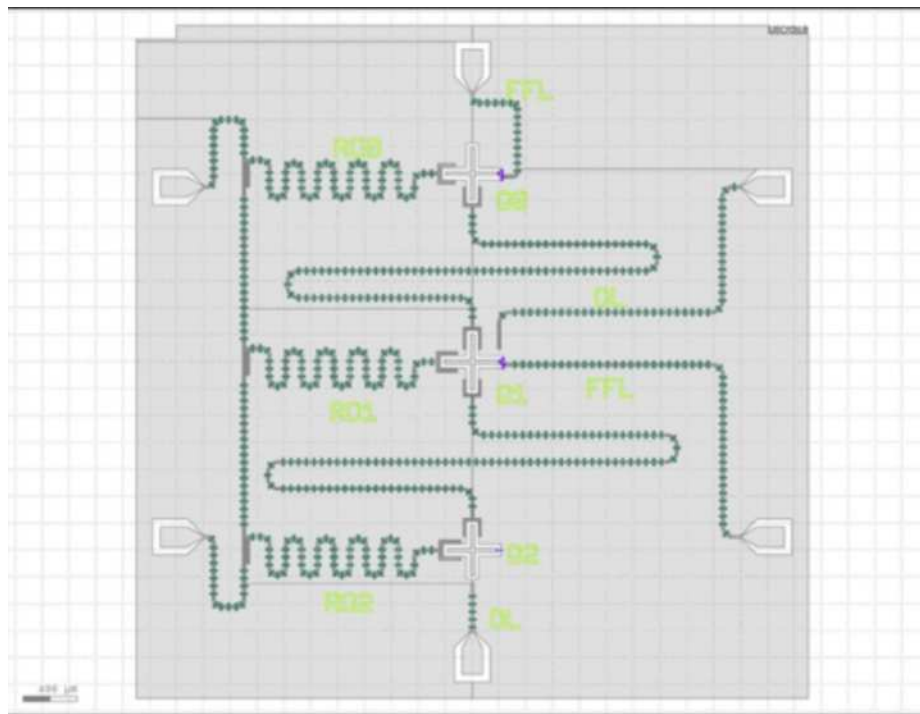
Affiliations:

- SQUID Lab, Niels Bohr Institute, University of Copenhagen
- Novo Nordisk Quantum Computing Programme (NQCP)

Zlatko Minev



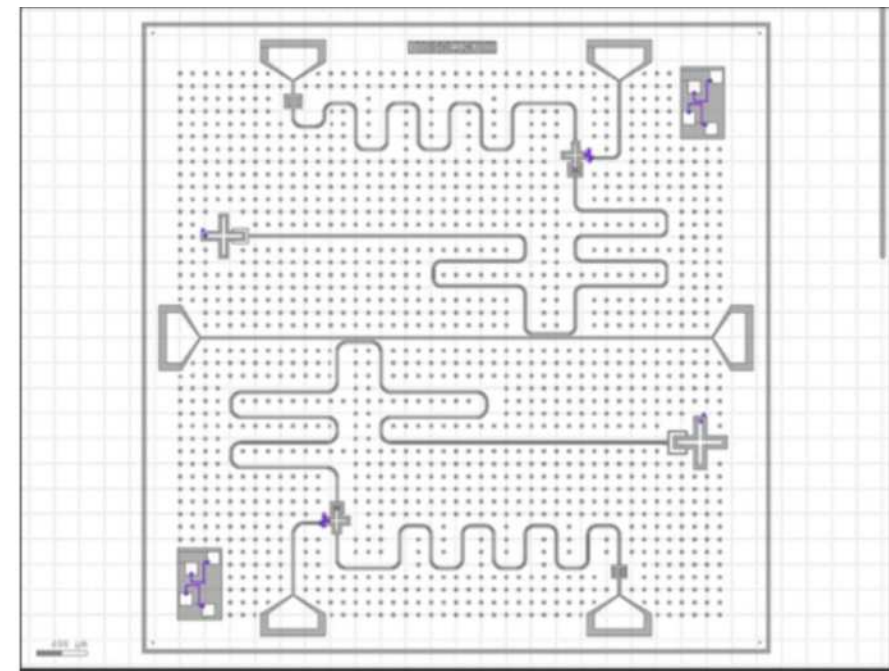
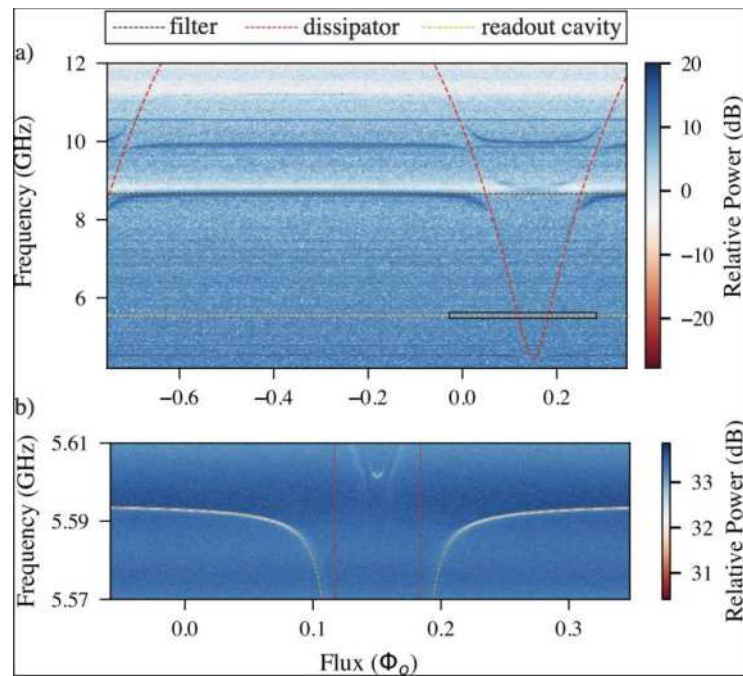
Example designs from the community, folks like you!



Sadman Ahmed Shanto



Eli Levenson-Falk



Zlatko Minev

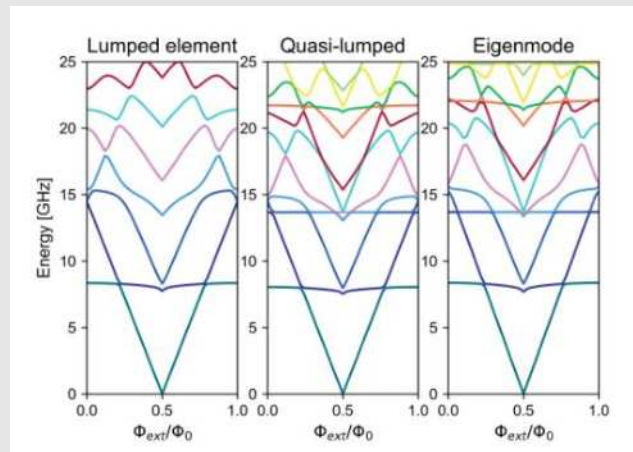
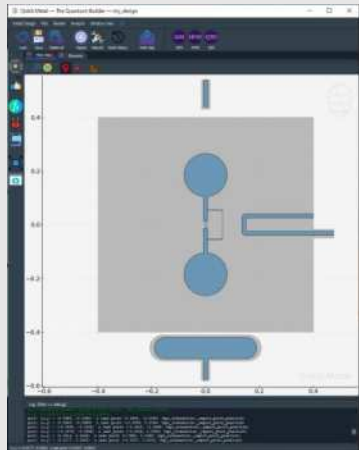
Example designs from the community, folks like you!

QuTech / Technische Universiteit Delft

Figen Yilmaz, C. Andersen Lab: Fluxonium chip design

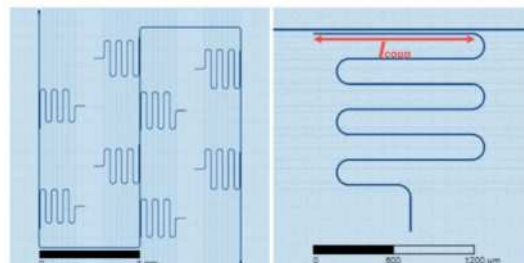
Roald van den Boogaart-Andersen Lab: Fluxonium spectrum (eigenmode = pyEPR) to show how they extended the analysis methods to be able to handle fluxonium qubits

Sara Buhktari-Andersen Lab: Fabricated resonator design being measured



3.1. Final design

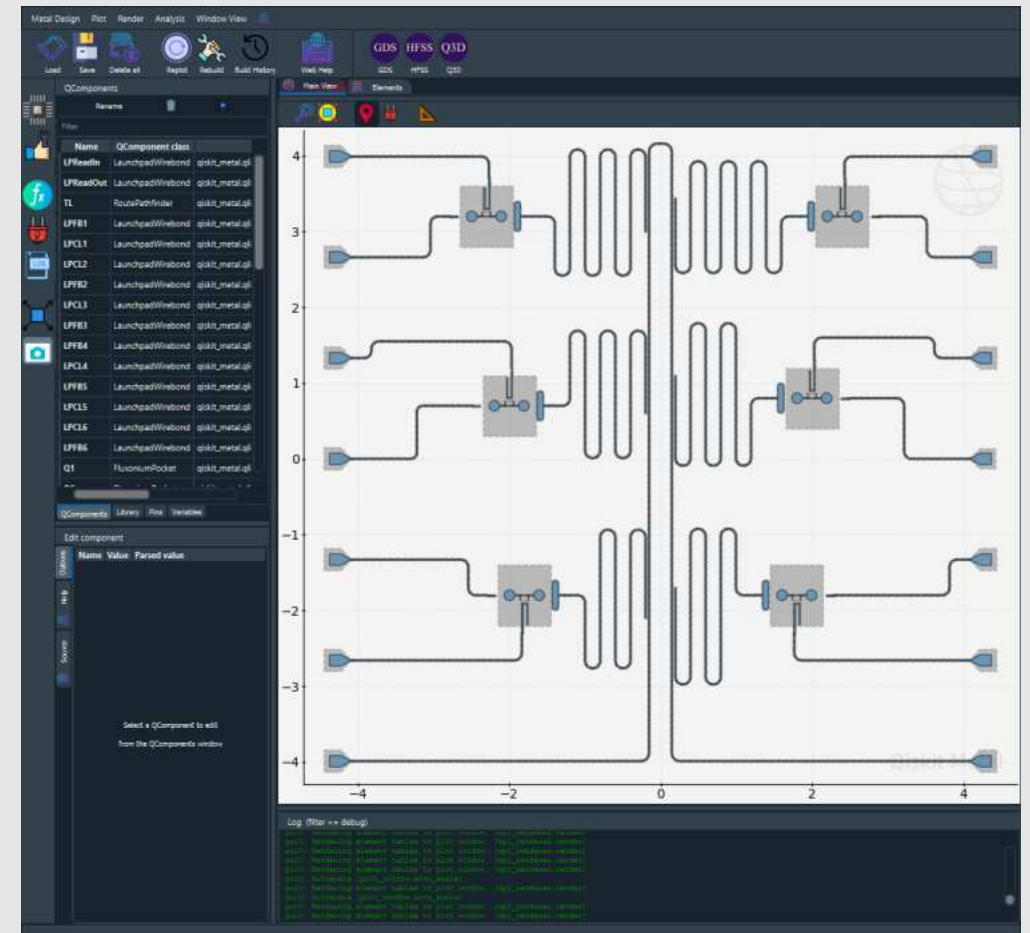
The final design consists of an S-shaped feedline routed between two launchpads. Eight resonators are capacitively coupled to this feedline. Every component on this 9 x 9 mm device is implemented by a coplanar waveguide transmission line. The schematic representations of the full device and single resonator geometry are depicted in Figure 6.



“Qiskit-Metal is the best tool if one studies superconducting qubits” -- Figen Yilmaz

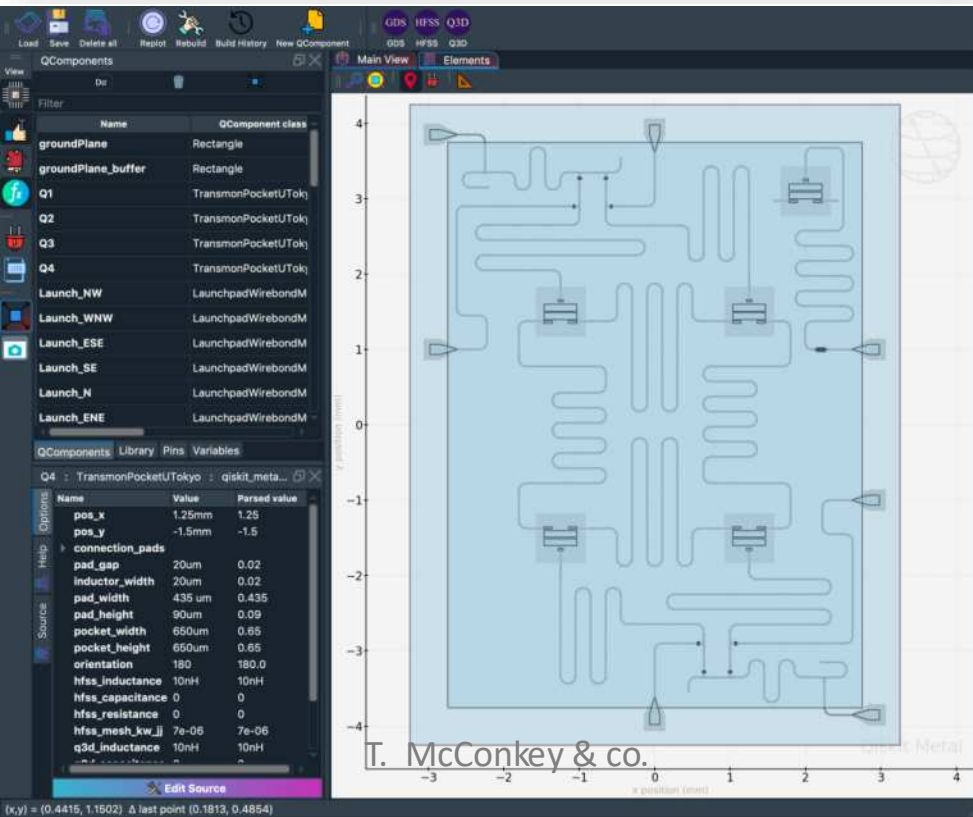
QuTech / Technische Universiteit Delft

Intermediate chip design.

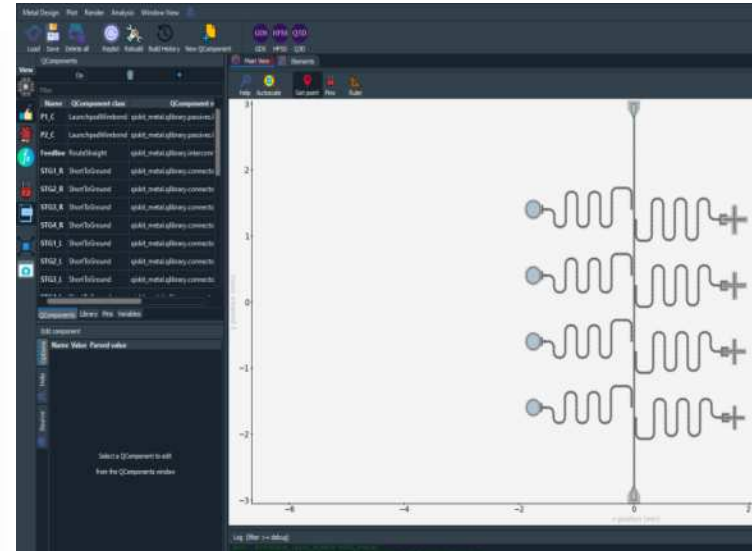


Example designs from the community, folks like you!

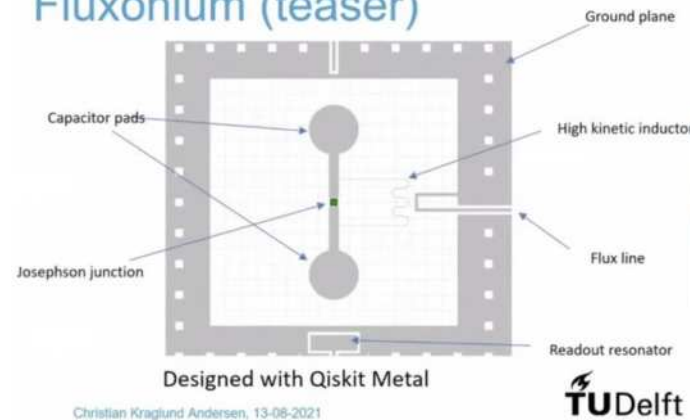
IBM 5Q Tsuru U Tokyo



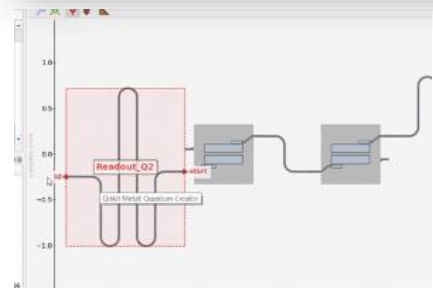
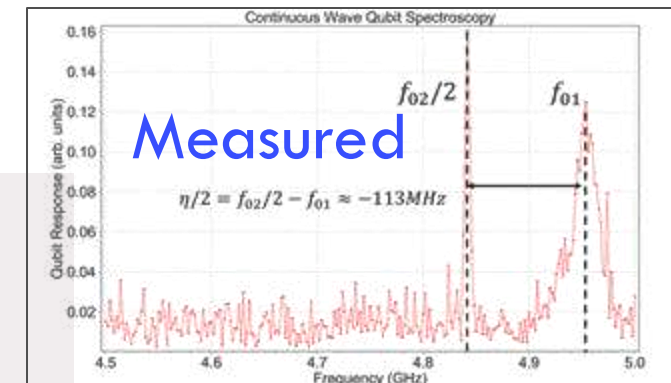
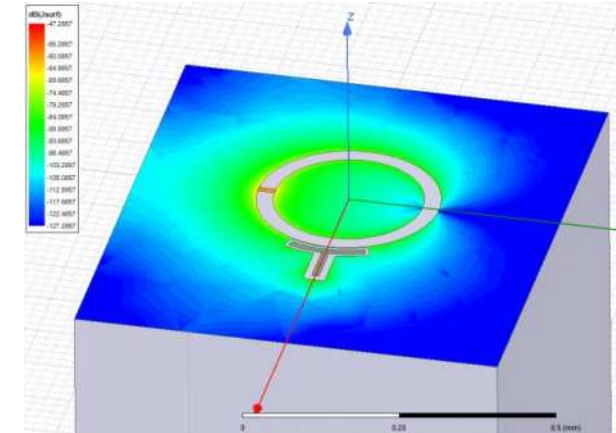
Chalmers 8Q



TU Delft Fluxonium
Fluxonium (teaser)



Simulated



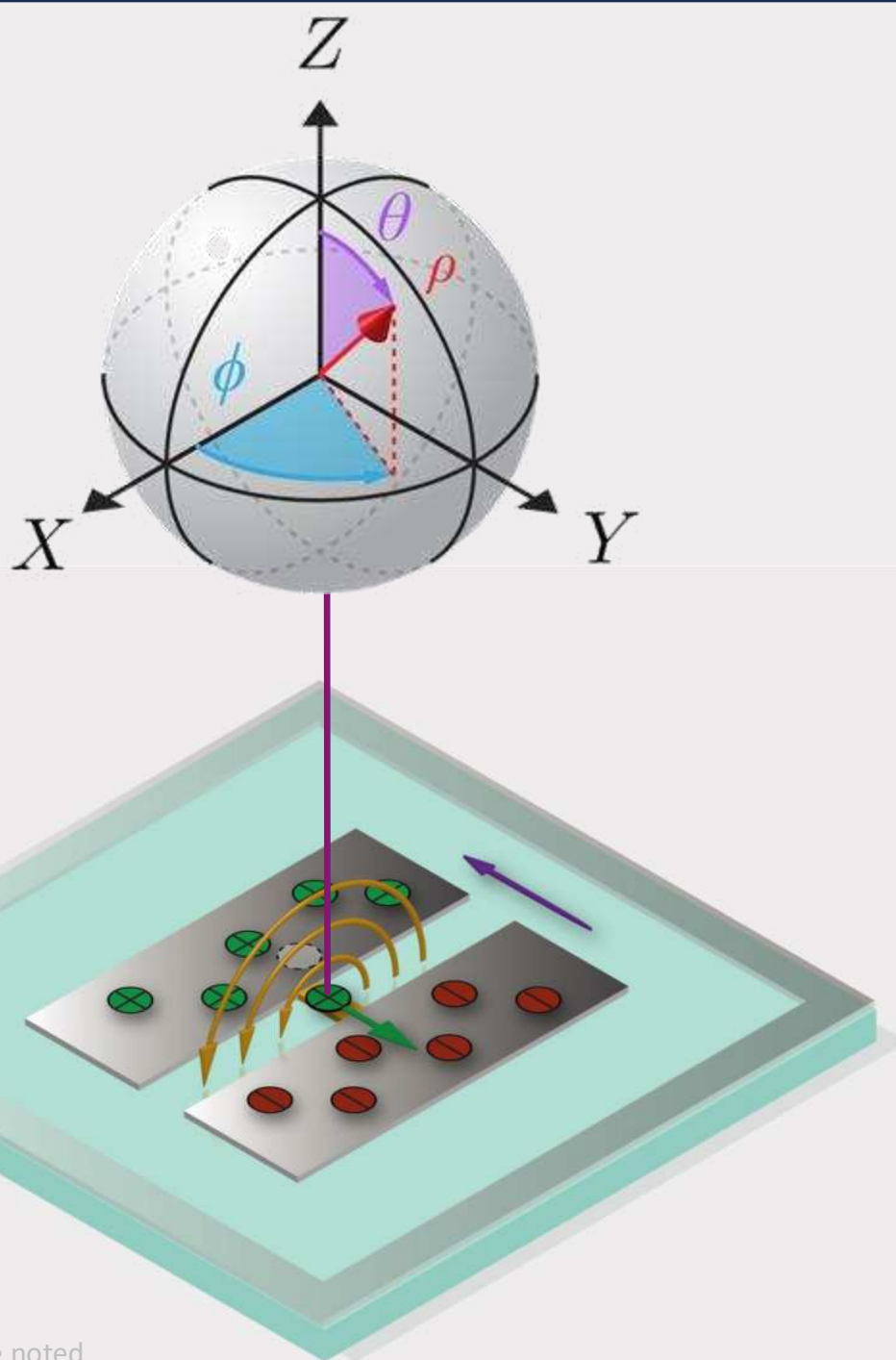
APS physics **MARCH MEETING 2021**
MARCH 15-19 ONLINE

CEC/ICMC **21**
VIRTUAL CONFERENCE JULY 19-23
Cryogenics • Through • Collaboration

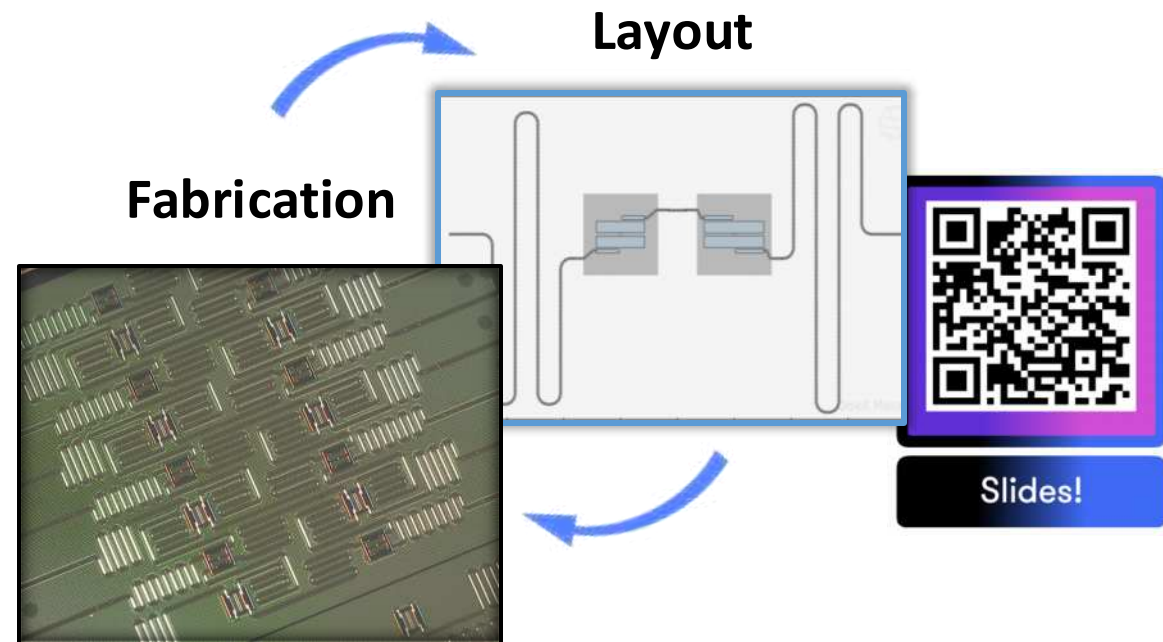


IEEE **QUANTUM WEEK** IEEE International Conference on Quantum Computing and Engineering — QCE21

Zlatko Mineev



Stage 1: Layout+ (Classical)



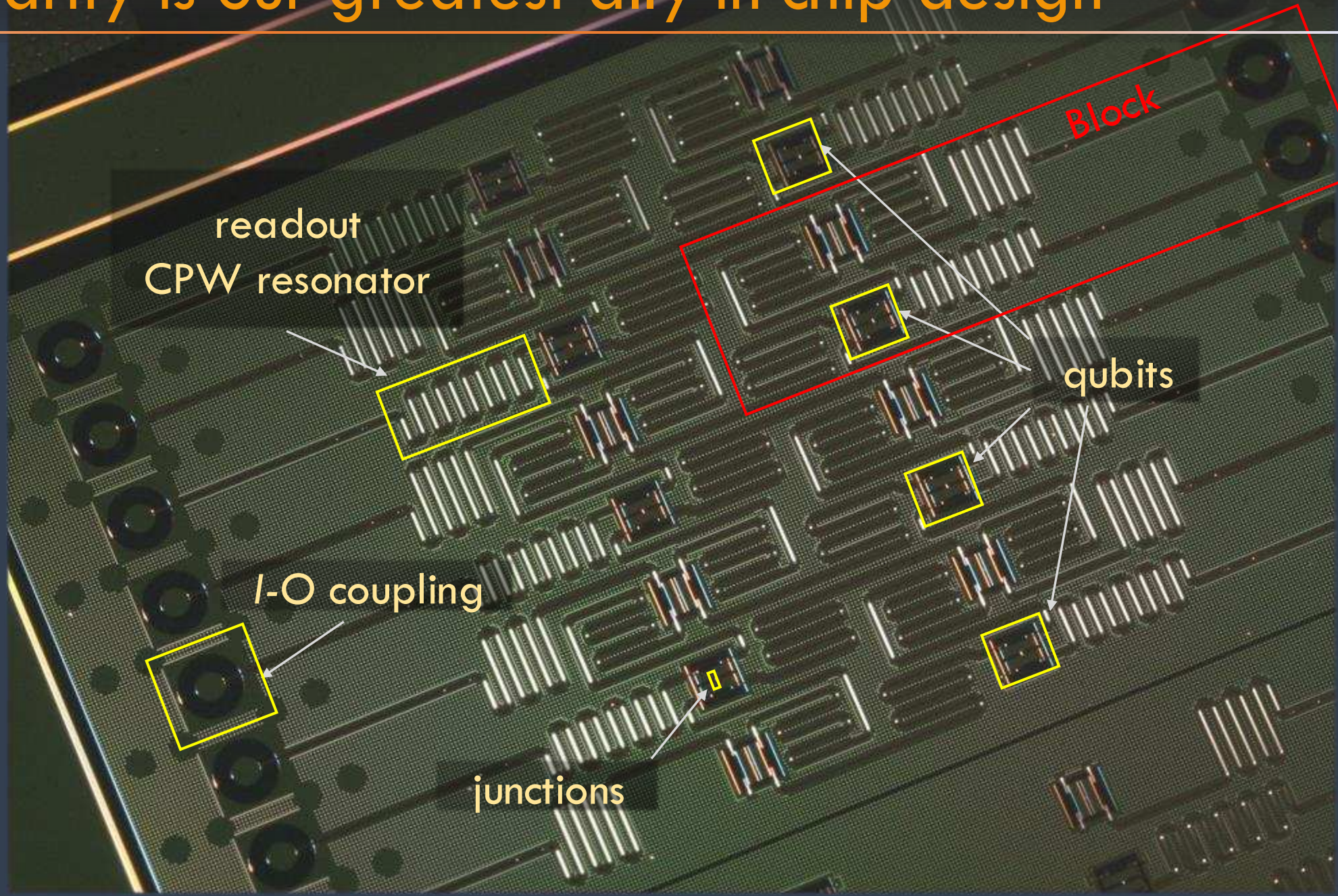
Regularity is our greatest ally in chip design

Reuse

Fine-tune

Automate

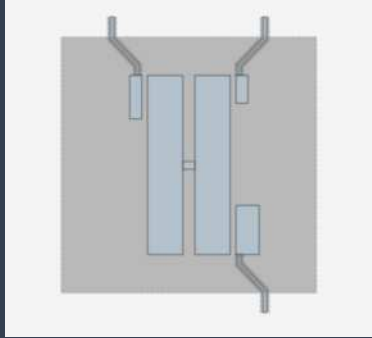
Extend



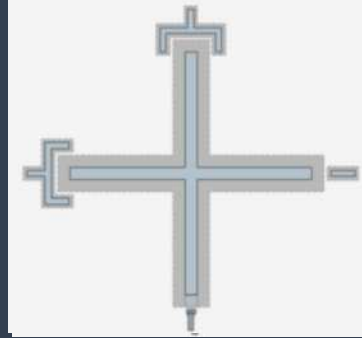
Device Library

Qiskit | quantum device
design

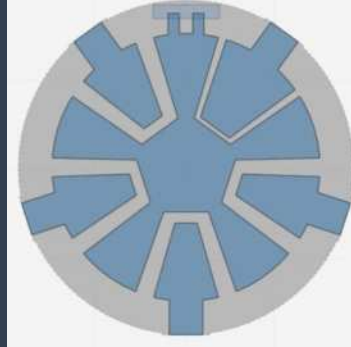
Qubits



Single transmon-
floating



Single transmon-
grounded flux lines



Star-shaped qubit

qiskit-community.github.io/qiskit-metal

qiskit.org/documentation/metal/circuit-examples/index.html

Qiskit Documentation Learning Resources Slack Support

0.0.4

Search Docs

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- Installing Qiskit Metal
- Frequently Asked Questions
- Roadmap
- Qiskit Metal Workflow
- Quantization Methods Overview

Contributor Guide

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- Reporting Bugs and Requesting Enhancements
- Contributing Code
- Contributing to Documentation

Tutorials

- Overview
- Components
- Renderers
- Analysis
- Quick Topics
- Video Recordings

Circuit Example Library

- Qubits

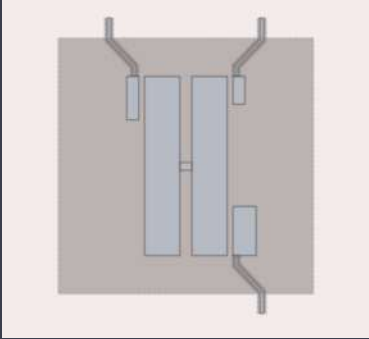
Qubits

 Single Transmon - Grounded (xmon)	 Single Transmon - Floating	 Concentric Transmon
 Interdigitated Transmon Qubits	 Single Transmon - Grounded (xmon) flux lines	 Single Transmon with 6 coils
 Single Transmon - Grounded (xmon)	 Single Transmon - Floating	 Concentric Transmon

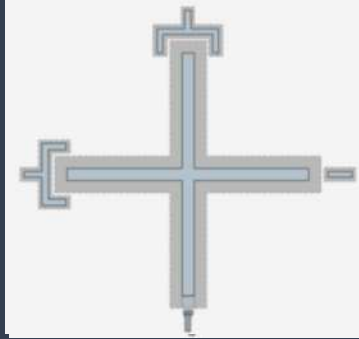
Device Library

Qubits

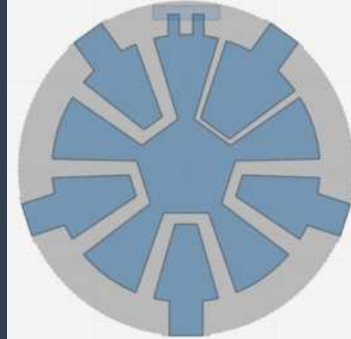
Resonators



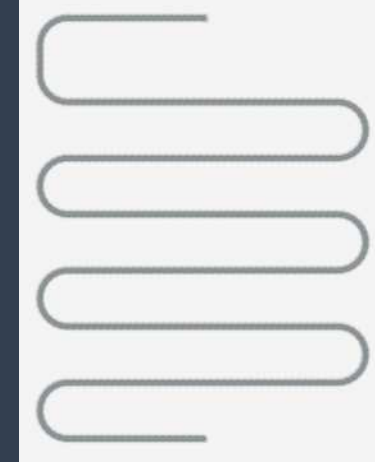
Single transmon-floating



Single transmon-grounded flux lines



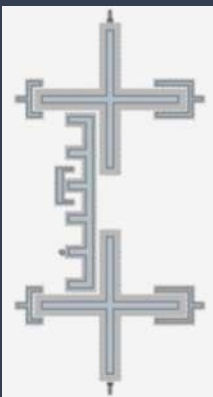
Star-shaped qubit



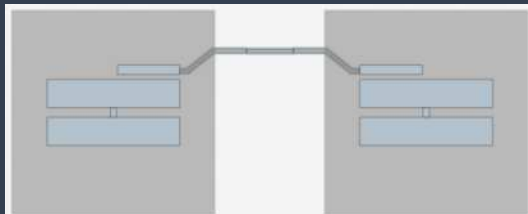
Simple coplanar waveguide (CPW) meander

Qubit Couplers

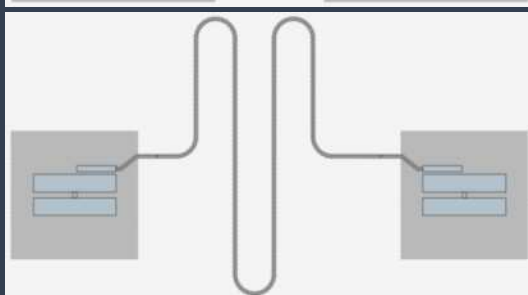
Input – Output Coupling



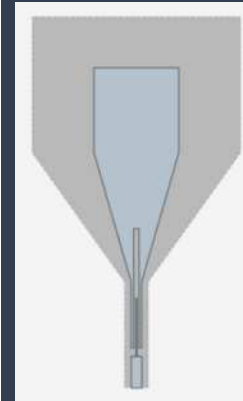
Tunable coupler (MIT)



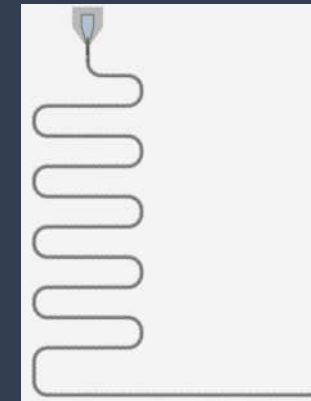
Direct coupler



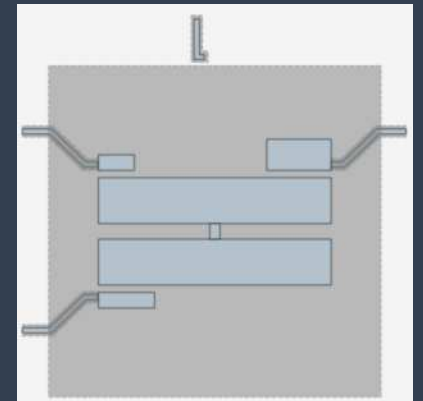
Bus resonator coupler



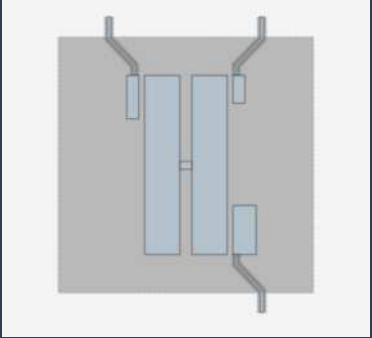
CPW launch pad



Readout line



Charge line



Single transmon-
floating

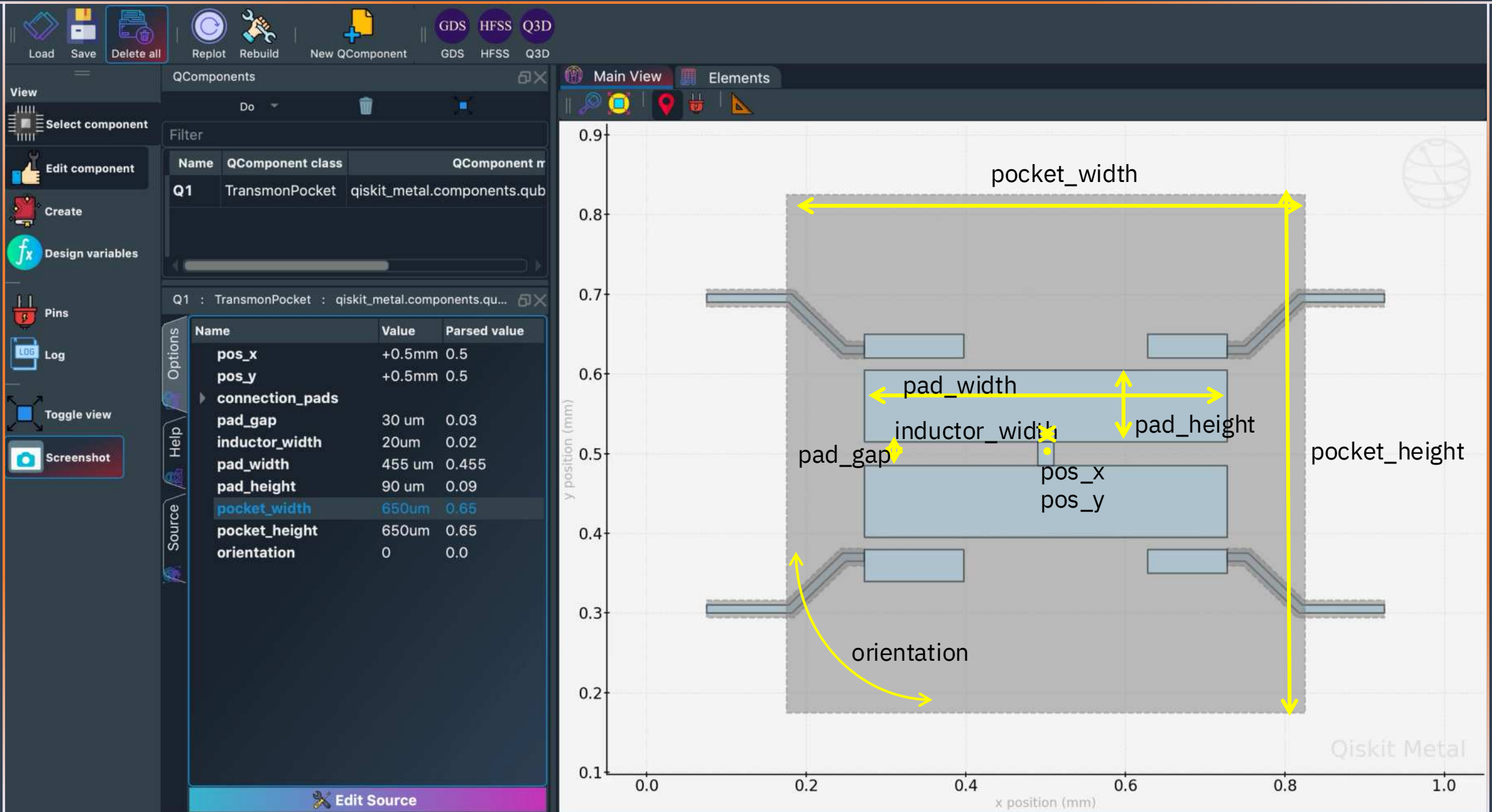
Code

```
from qiskit_metal.qlibrary import qubits  
q1 = qubits.TransmonPocket('Q1', options=dict(...))
```

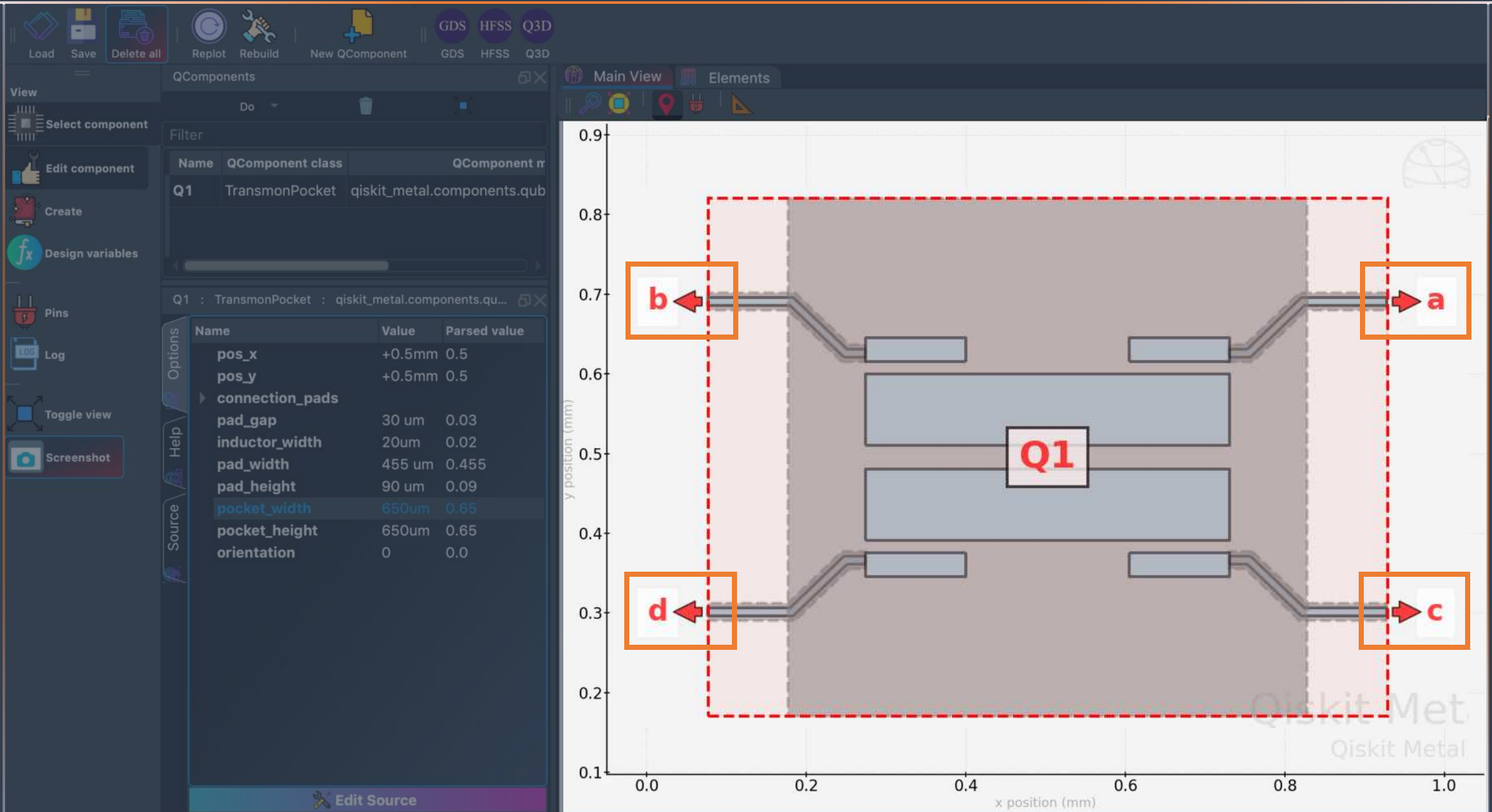
Synced GUI

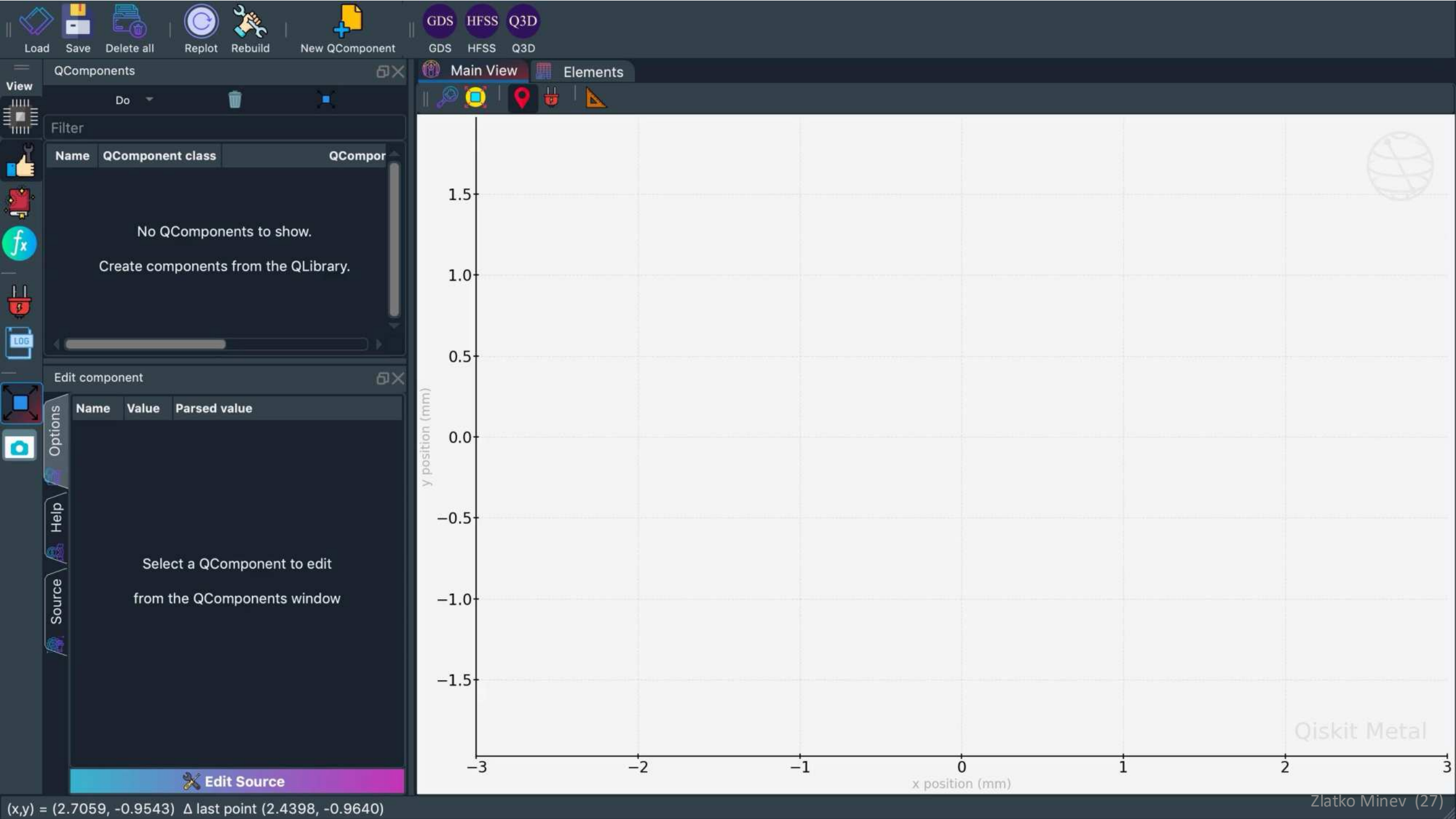
```
from qiskit_metal import MetalGUI  
MetalGUI()
```

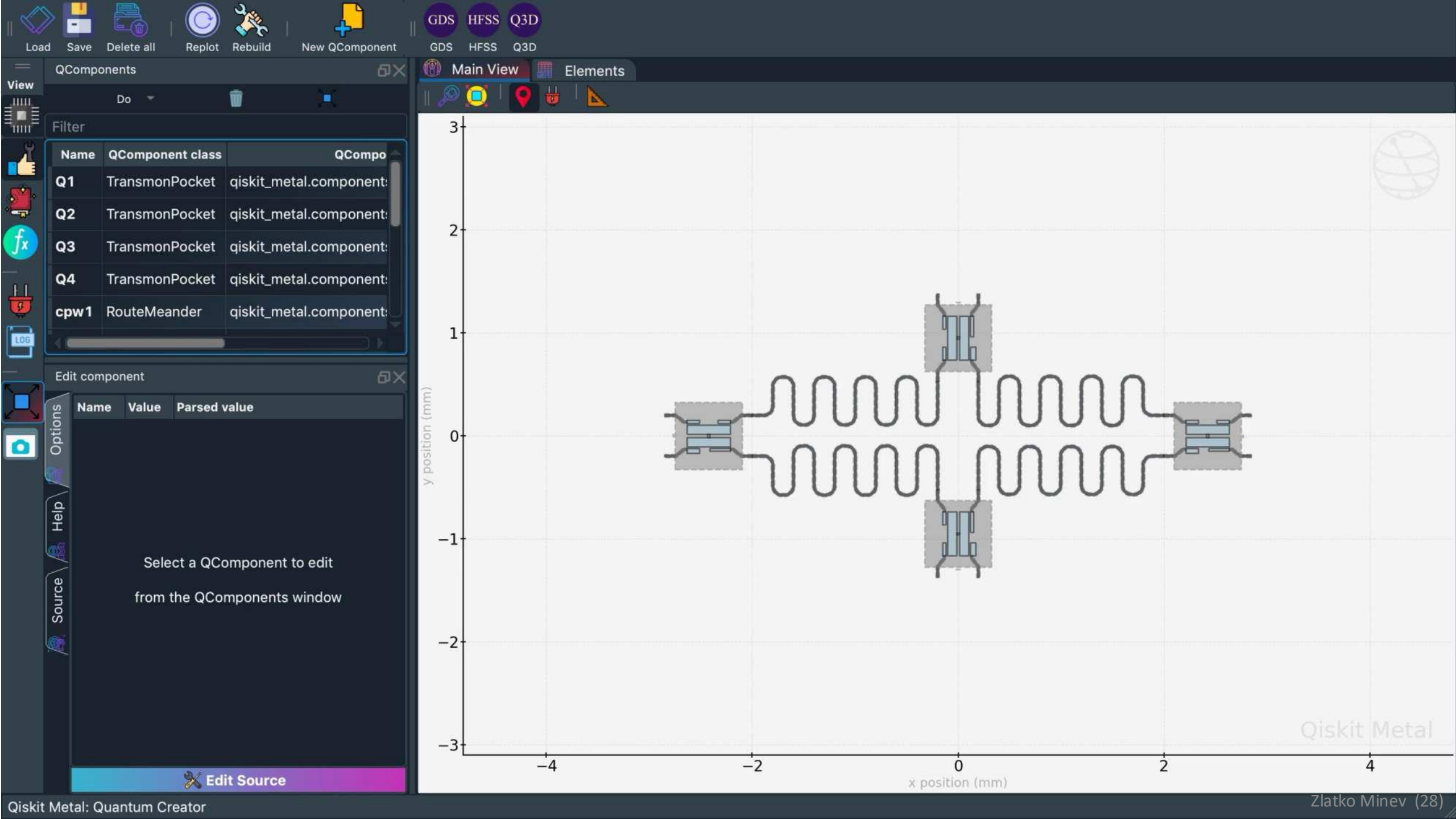
Fine-tune and automate parameters



Connecting quantum components dynamically





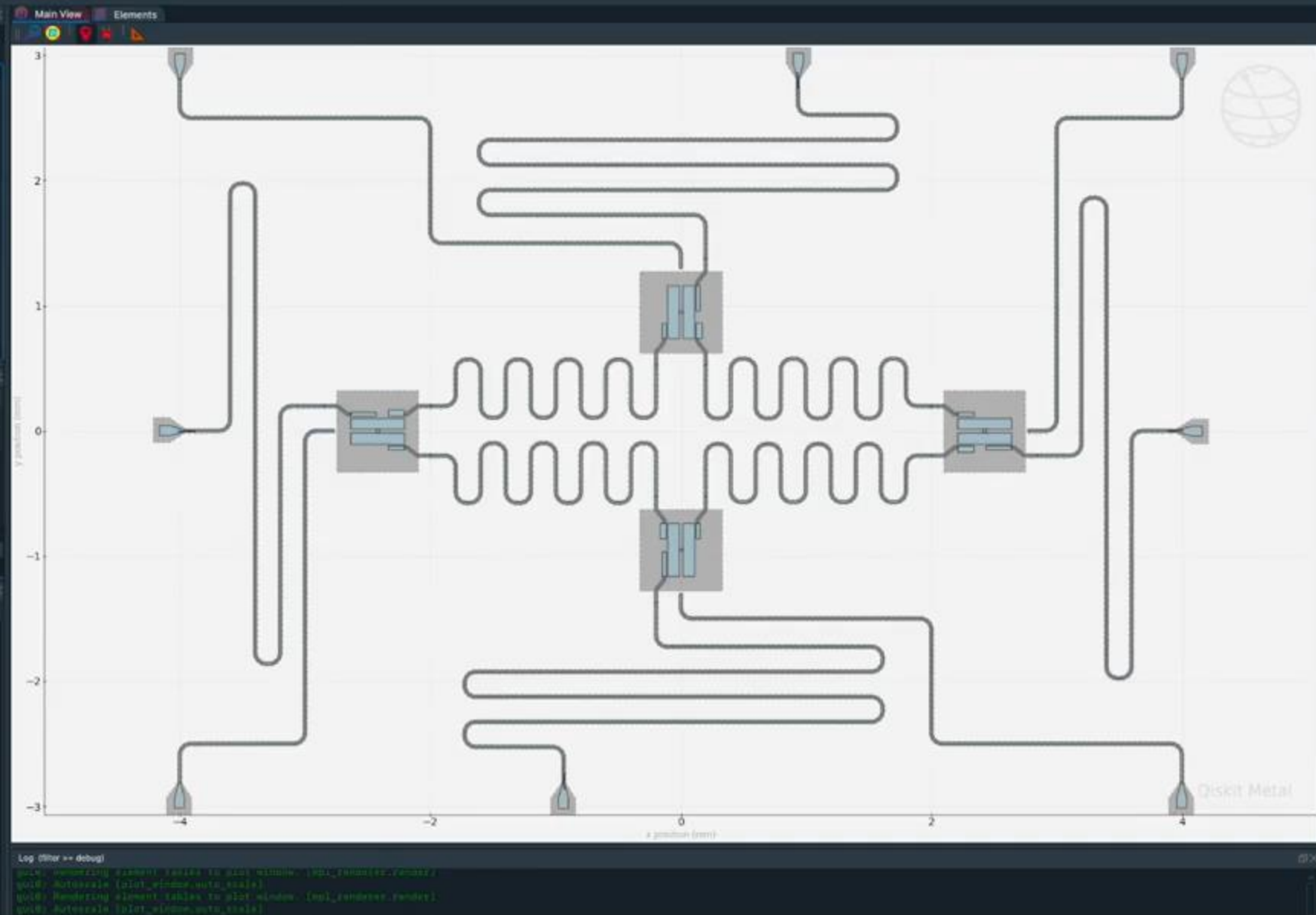


The screenshot shows the Qiskit Aqua GUI's QComponents window. At the top, there's a toolbar with icons for various actions. Below it is a table of components. The table has five columns: Name, QComponent class, QComponent module, Build status, and id. The components listed are TransmonPocket (ids 01-04), RouteMeander (ids cpw1-cpw4), and OpenToGround (ids OT01-OT03). Below the table is a 'Variables' section with a table for variable names, values, and parsed values. The variables shown are 'cpw_width' (10 um, 0.01) and 'cpw_gap' (6 um, 0.006). At the bottom, there's a message: 'Select a QComponent to edit' from the QComponents window.

Name	QComponent class	QComponent module	Build status	id
Q1	TransmonPocket	qiskit_metal.components.qubits.transmon_pocket	good	1
Q2	TransmonPocket	qiskit_metal.components.qubits.transmon_pocket	good	2
Q3	TransmonPocket	qiskit_metal.components.qubits.transmon_pocket	good	3
Q4	TransmonPocket	qiskit_metal.components.qubits.transmon_pocket	good	4
cpw1	RouteMeander	qiskit_metal.components.interconnects.meandered	good	5
cpw2	RouteMeander	qiskit_metal.components.interconnects.meandered	good	6
cpw3	RouteMeander	qiskit_metal.components.interconnects.meandered	good	7
cpw4	RouteMeander	qiskit_metal.components.interconnects.meandered	good	8
OT01	OpenToGround	qiskit_metal.components.connectors.open_to_ground	good	9
OT02	OpenToGround	qiskit_metal.components.connectors.open_to_ground	good	10
OT03	OpenToGround	qiskit_metal.components.connectors.open_to_ground	good	11

Variable name	Value	Parsed value (in um)
cpw_width	10 um	0.01
cpw_gap	6 um	0.006

Select a QComponent to edit
from the QComponents window



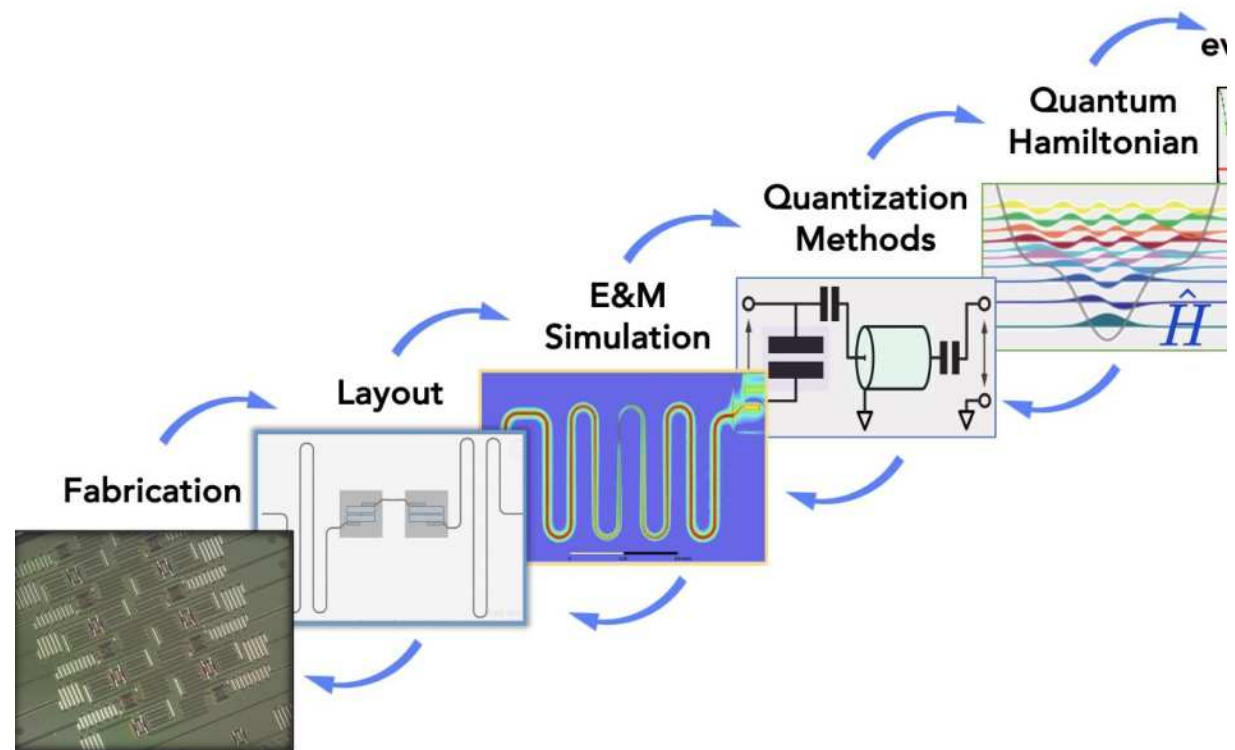
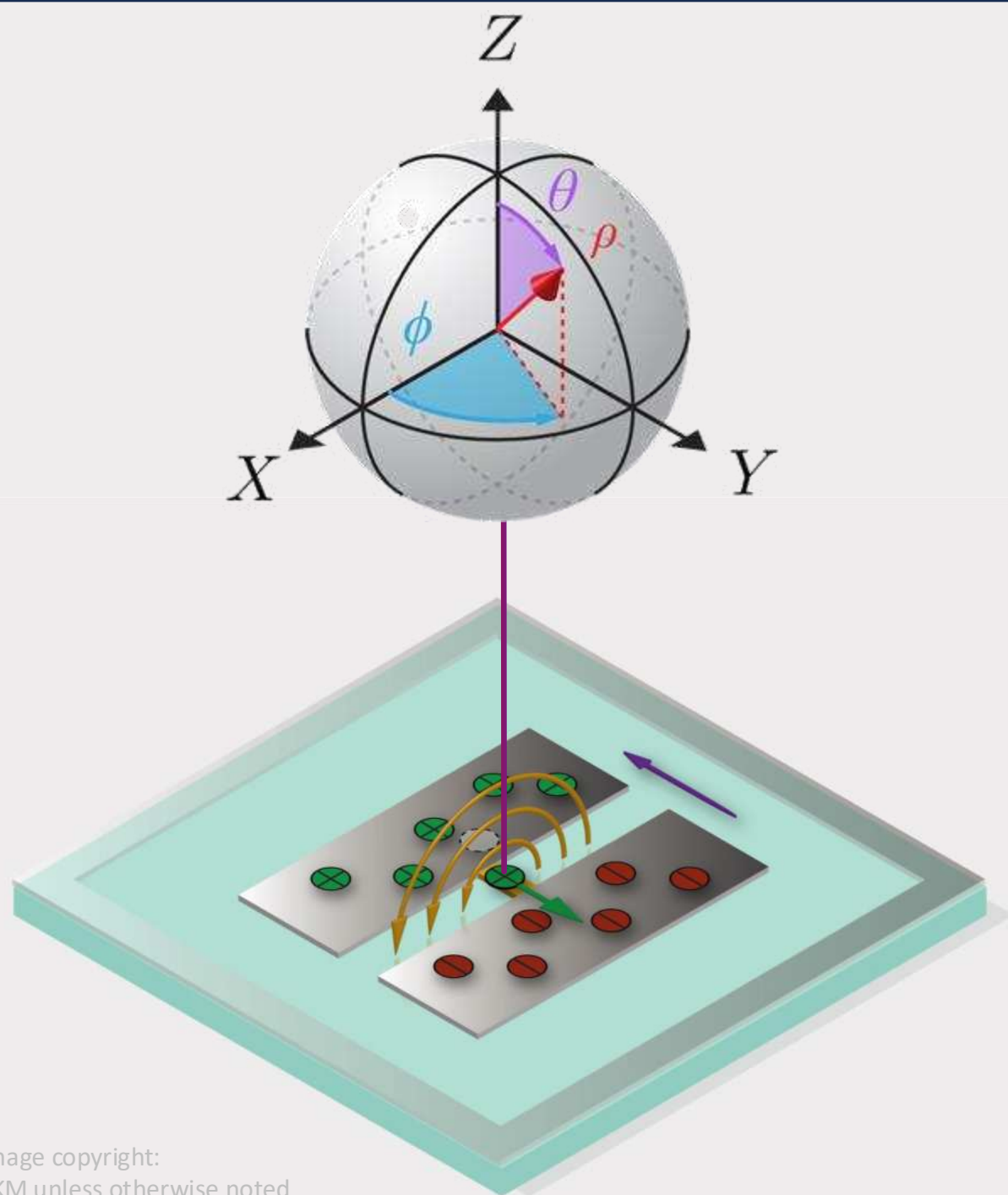
Can I make my own qubit? 😊

```
smile = draw.shapely.geometry.Point(0, 0).buffer(0.8)
cut_sq = draw.shapely.geometry.box(-1, -0.3, 1, 1)
smile = draw.subtract(smile, cut_sq)
face = draw.subtract(face, smile)
face = draw.subtract(face, eye_r)
face = draw.subtract(face, eye_l)
face
```

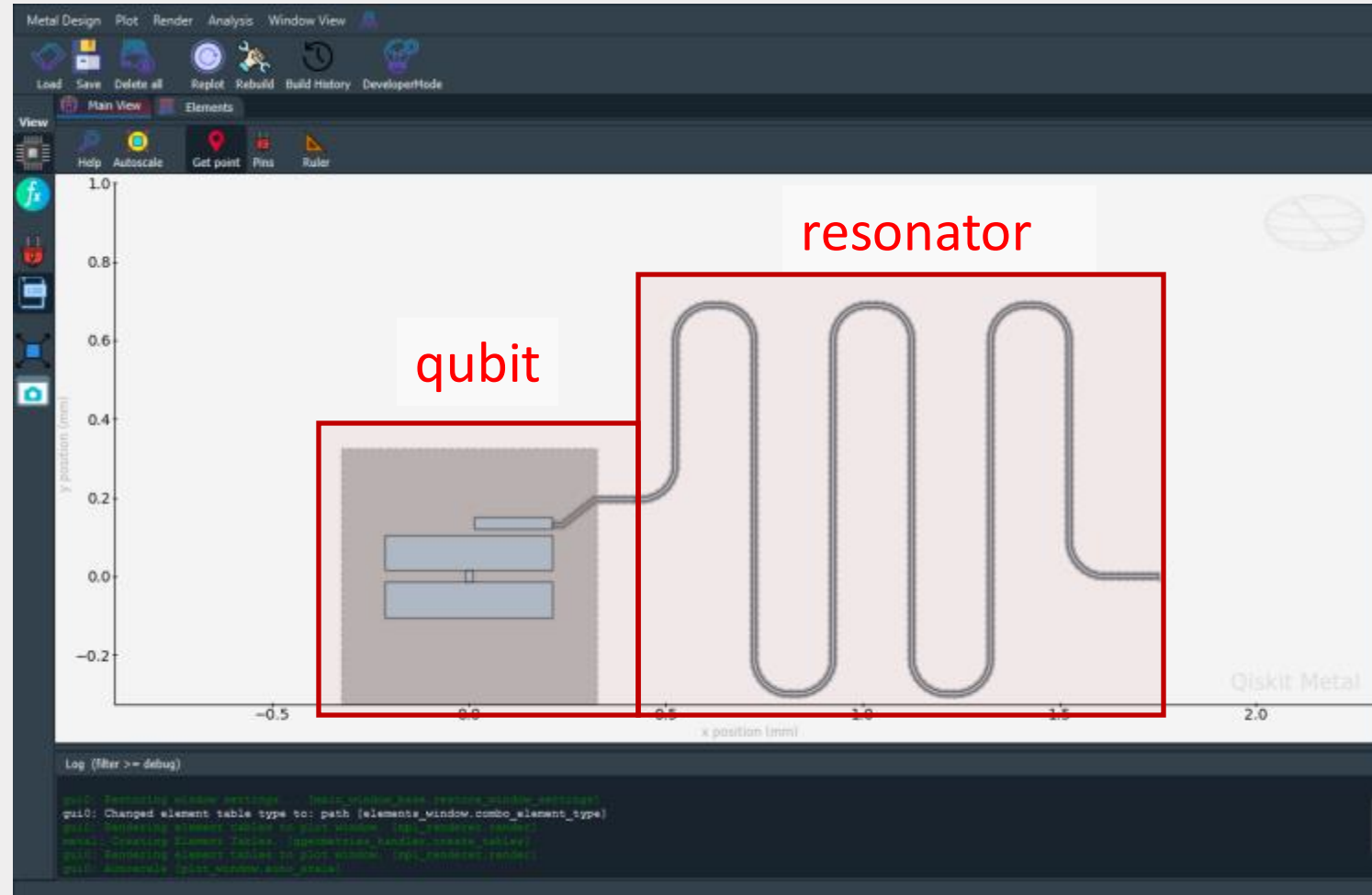
[7]:



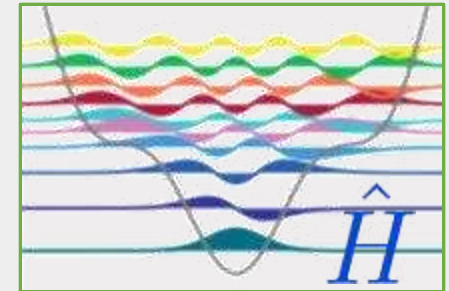
State of the Art Methods



Simple example



?

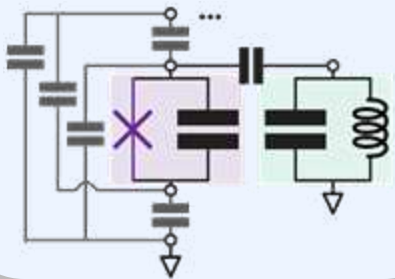


Step 2: Landscape of quantization methods

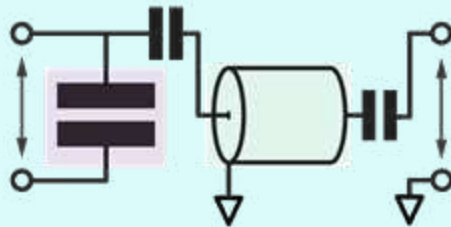
quasi-static

full-wave

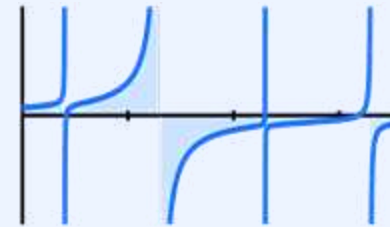
lumped



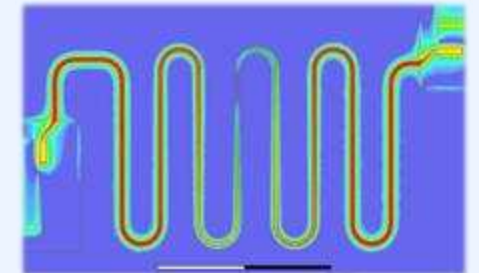
quasi-lumped



impedance



energy



fast

more information, complexity, accuracy

SCQubits

QuCAT

pyEPR

...

See Friday Workshop!

QuTiP

KQC

Qiskit | quantum device design
Metal

Yurke & Denker (1984),
Devoret (1997), Burkard *et al.* (2004), Koch *et al.*...

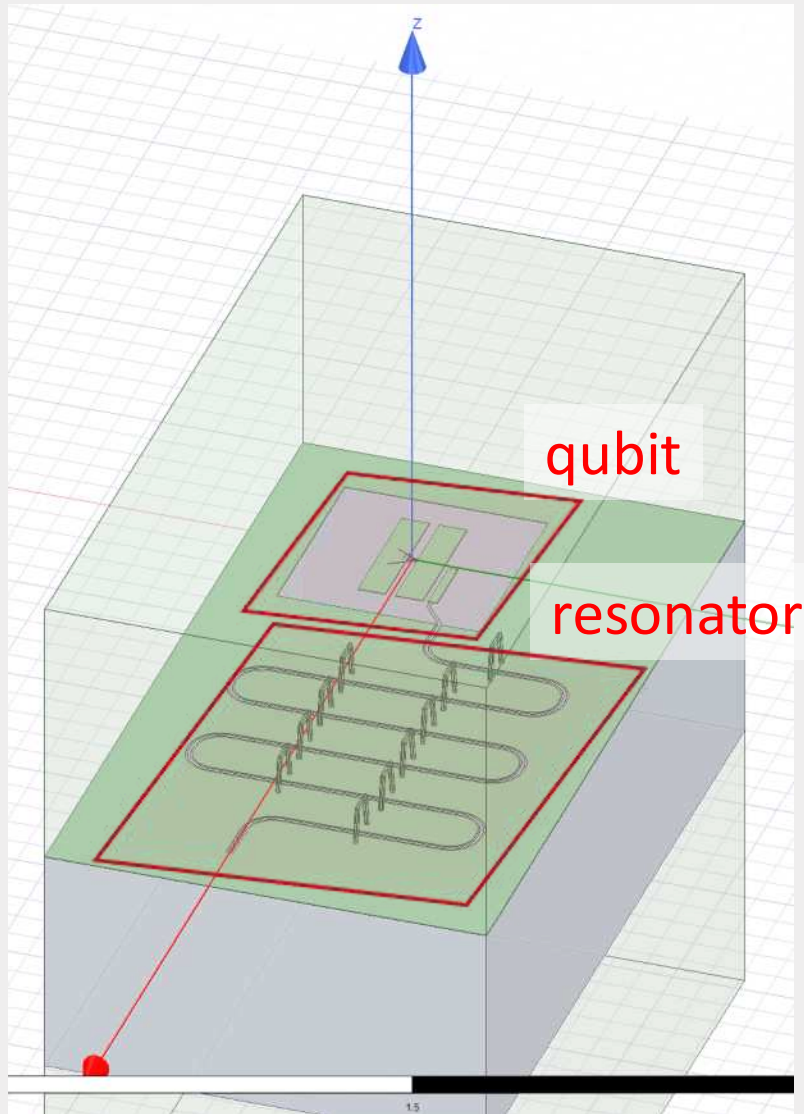
Malekakhlagh *et al.* (2017, 2019),
Gely *et al.* (2019), Parra-Rodriguez *et al.* (2019), Minev *et al.* (2021), ...

Nigg *et al.* (2012), Bourassa *et al.* (2012), Solgun *et al.* (2014, 2015, 2017) ...

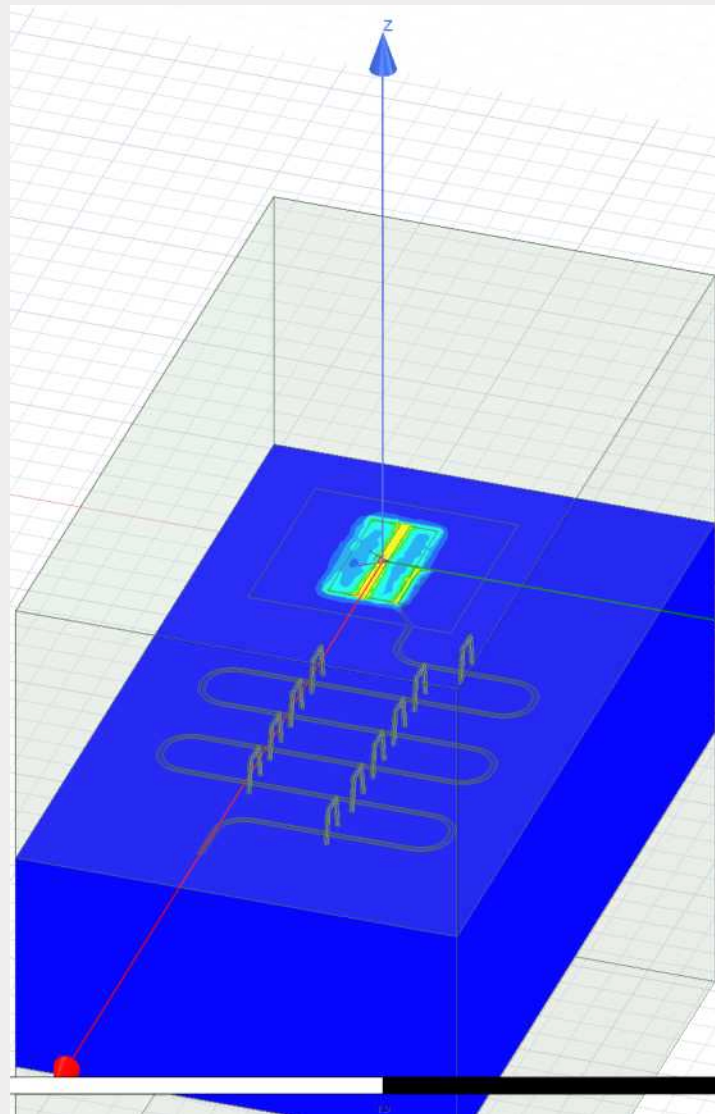
Minev (2018)
Minev *et al.* (2020)

Modeling step A: Classical E&M approximations

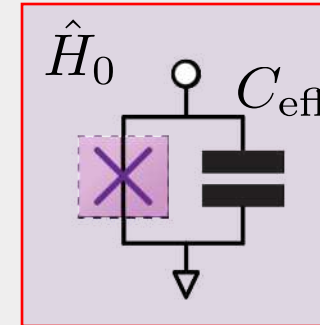
Physical (linear) model



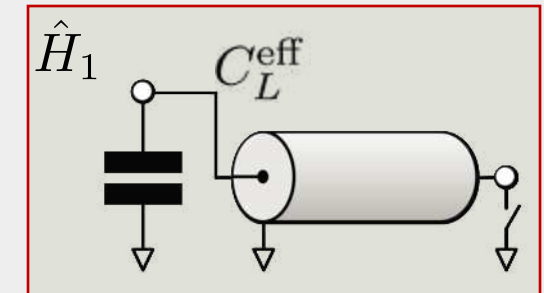
Eigenmodes



Quasi-lumped models

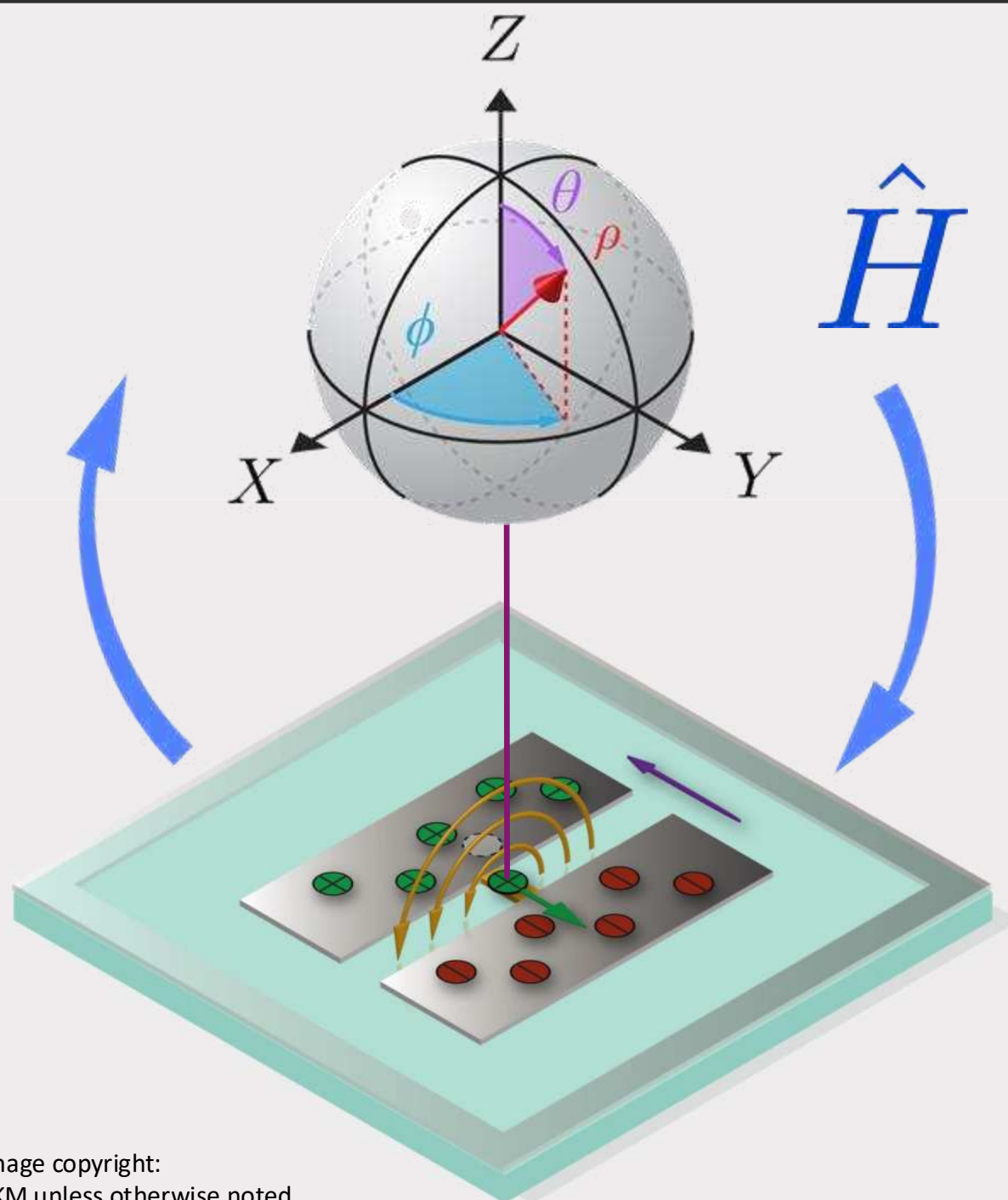


$$\begin{matrix} n_0 & n_1 & n_2 & n_3 & n_4 & n_5 \\ n_0 & \begin{pmatrix} C_{0\Sigma} & -C_{01} & -C_{02} & -C_{03} & -C_{04} & -C_{05} \\ C_{1\Sigma} & C_{11} & -C_{12} & -C_{13} & -C_{14} & -C_{15} \\ & C_{2\Sigma} & -C_{23} & -C_{24} & -C_{25} & \\ & & C_{3\Sigma} & -C_{34} & -C_{35} & \\ & & & C_{4\Sigma} & -C_{45} & \\ & & & & C_{5\Sigma} & \end{pmatrix} \end{matrix}$$



Impedance





Quasi-Lumped Element Oscillator Models (LOM)

arXiv > quant-ph > arXiv:2103.10344

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Quantum Physics

[Submitted on 18 Mar 2021]

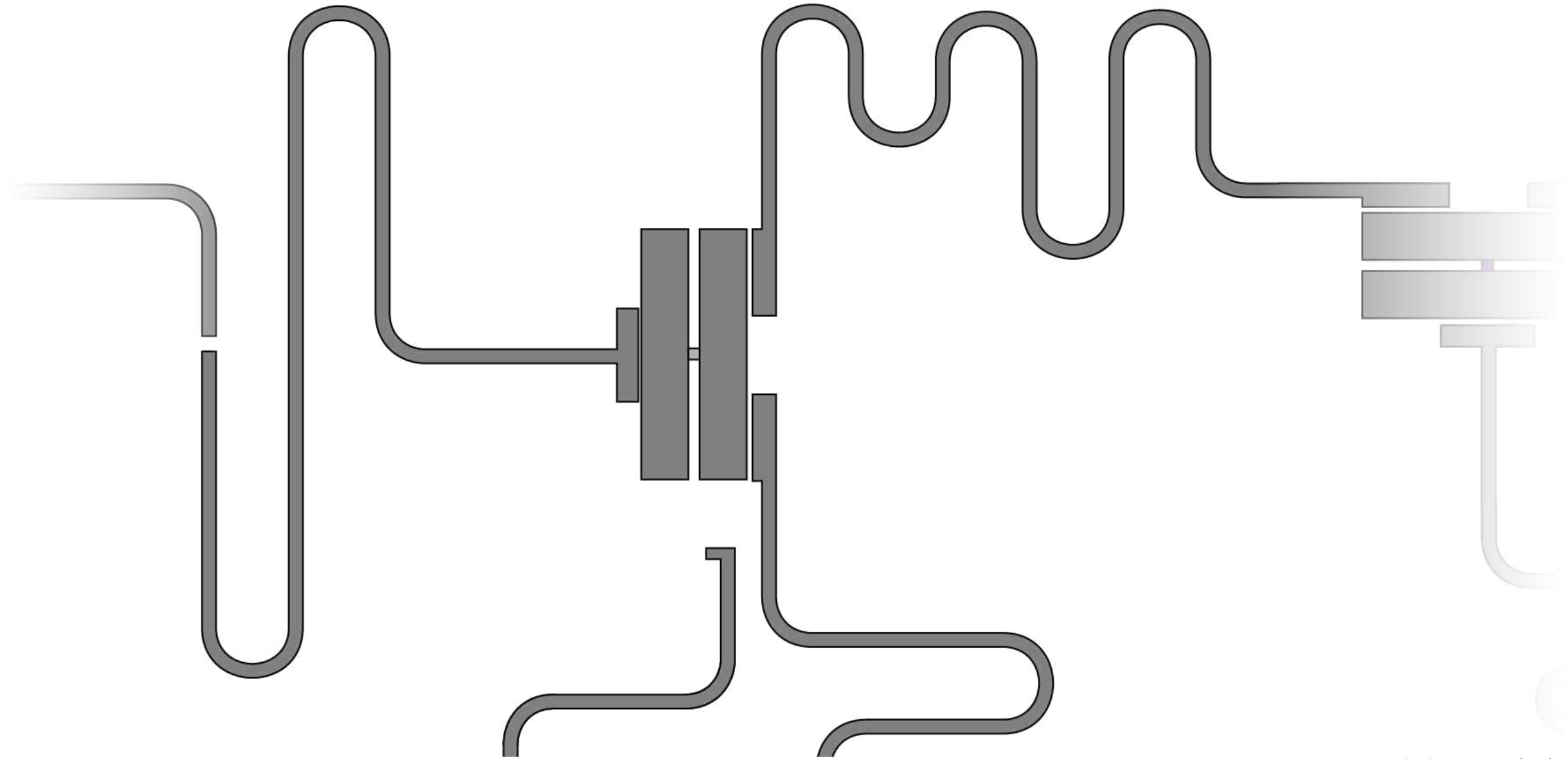
Circuit quantum electrodynamics (cQED) with modular quasi-lumped models

Zlatko K. Mineev, Thomas G. McConkey, Maika Takita, Antonio D. Corcoles, Jay M. Gambetta

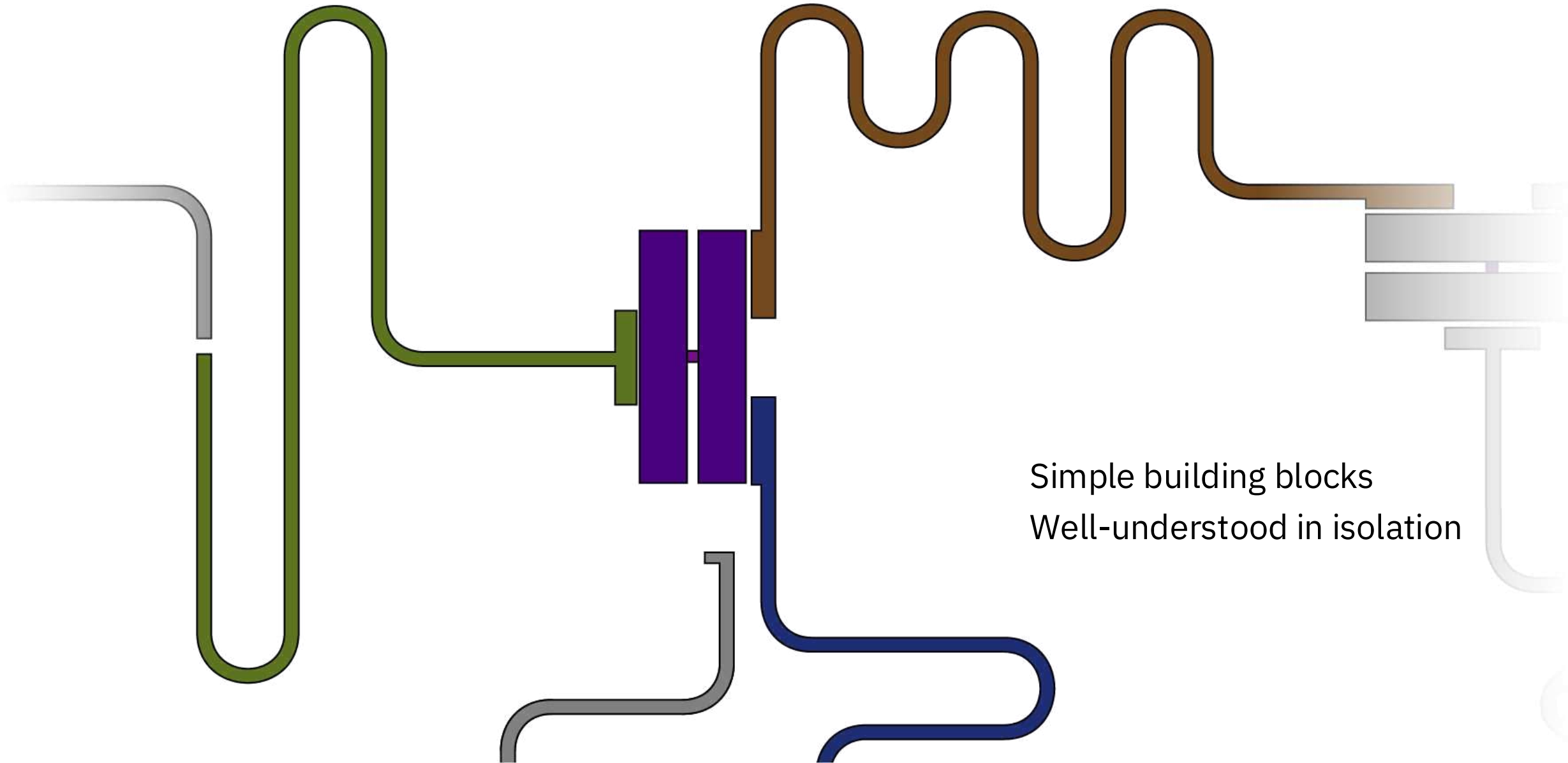
arXiv:2103.10344

Example

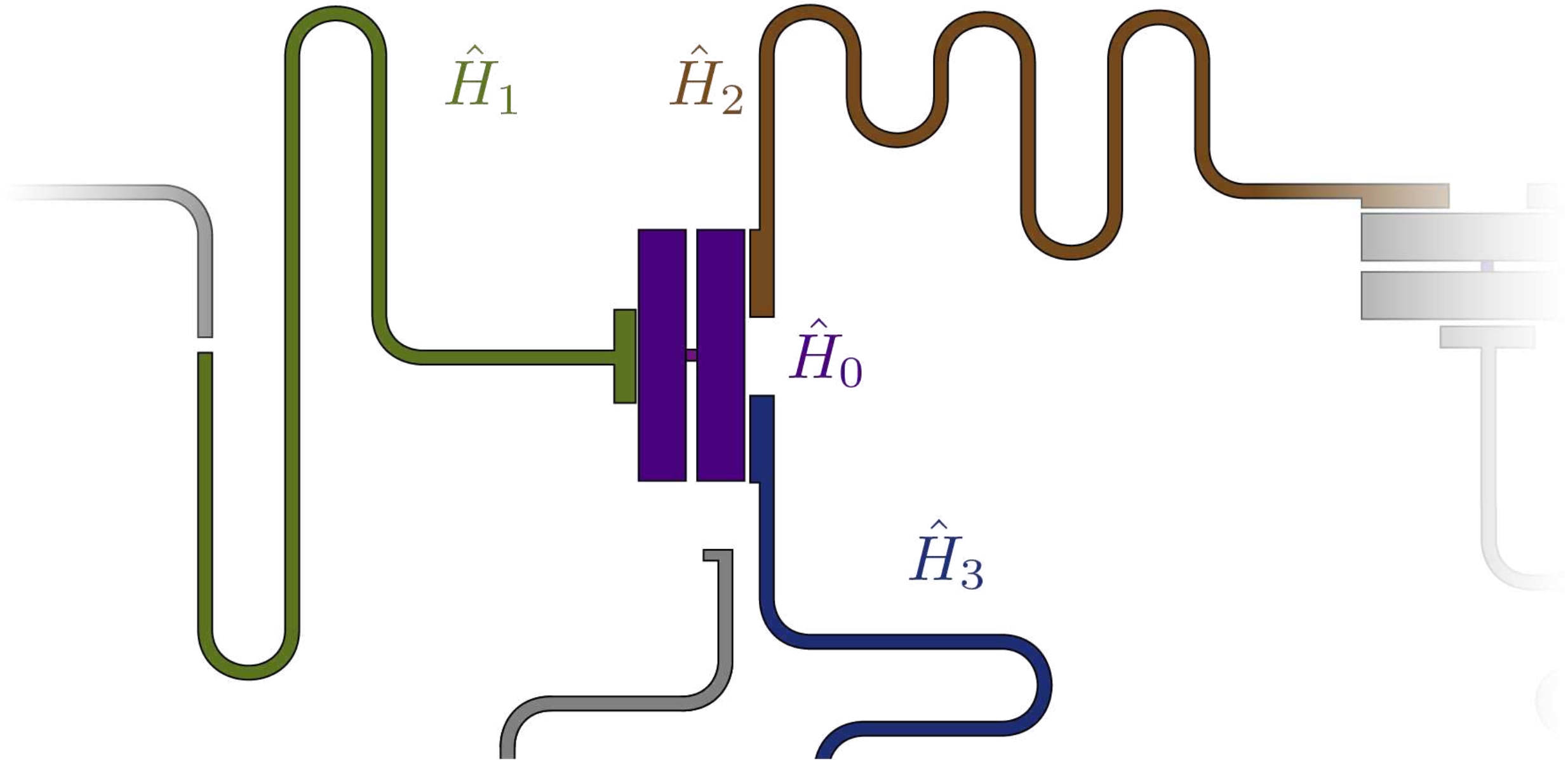
Physical layout zoom in



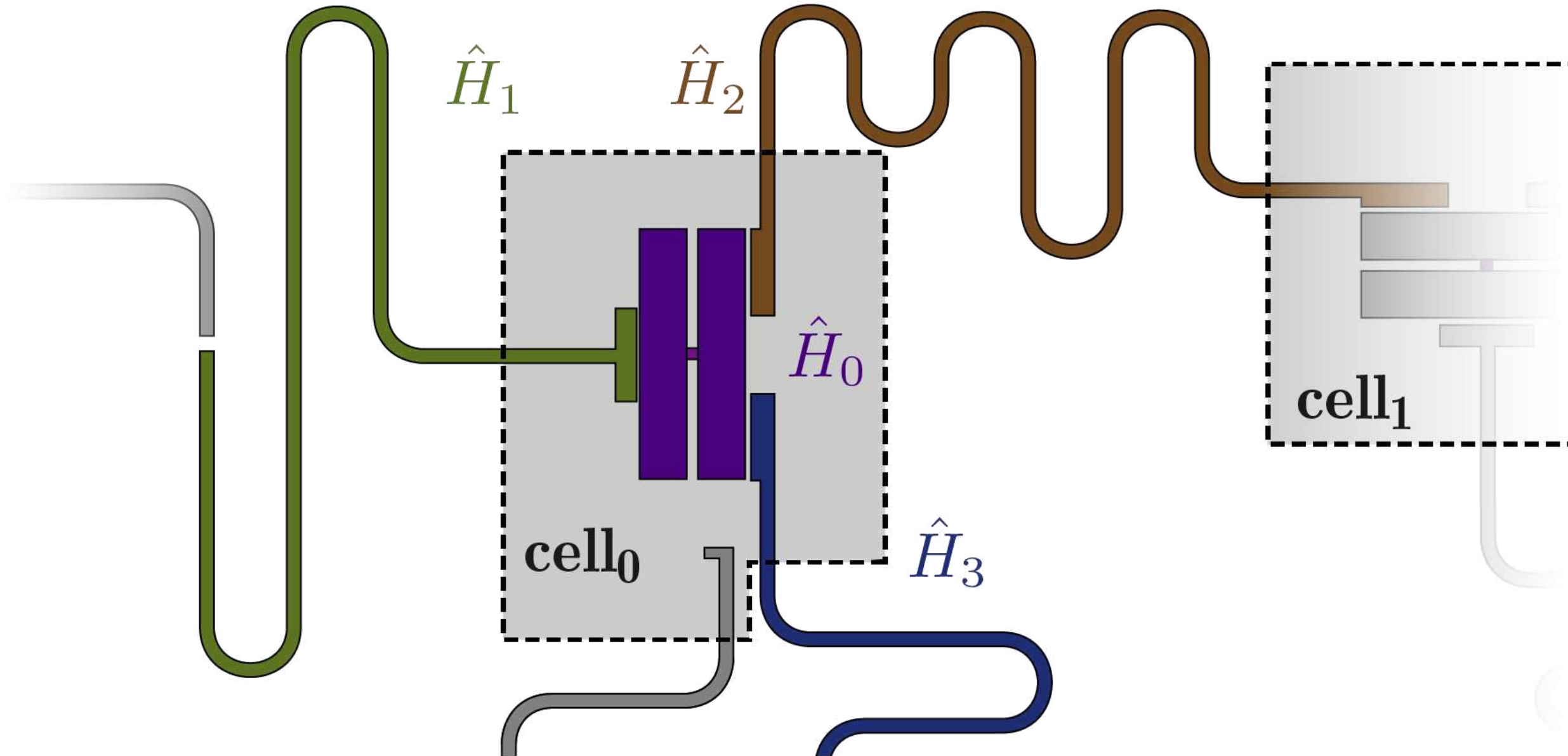
Identifying systems



Dressed subsystem Hamiltonians

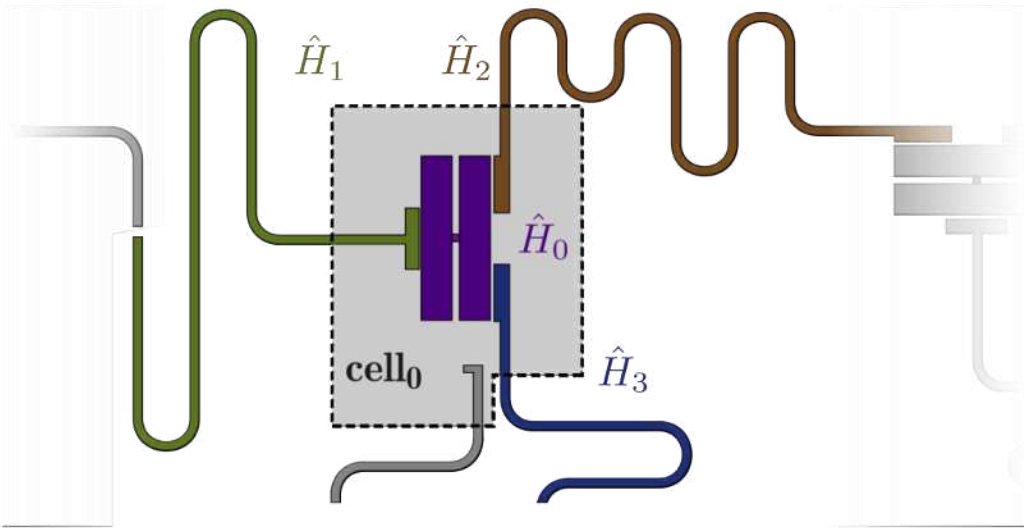


Subsystem & cell partitions



Simulating a cell

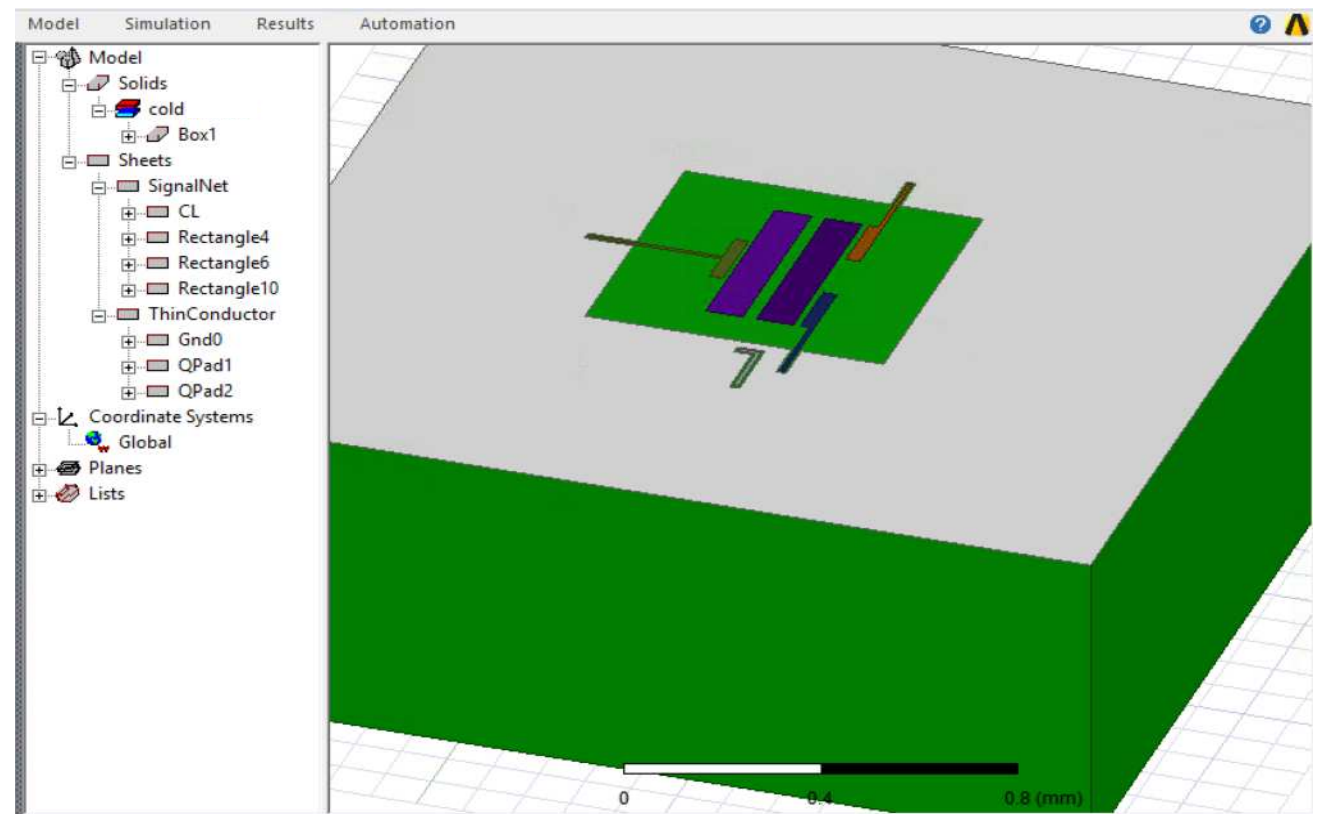
device layout



Automated with
Qiskit | quantum device
design



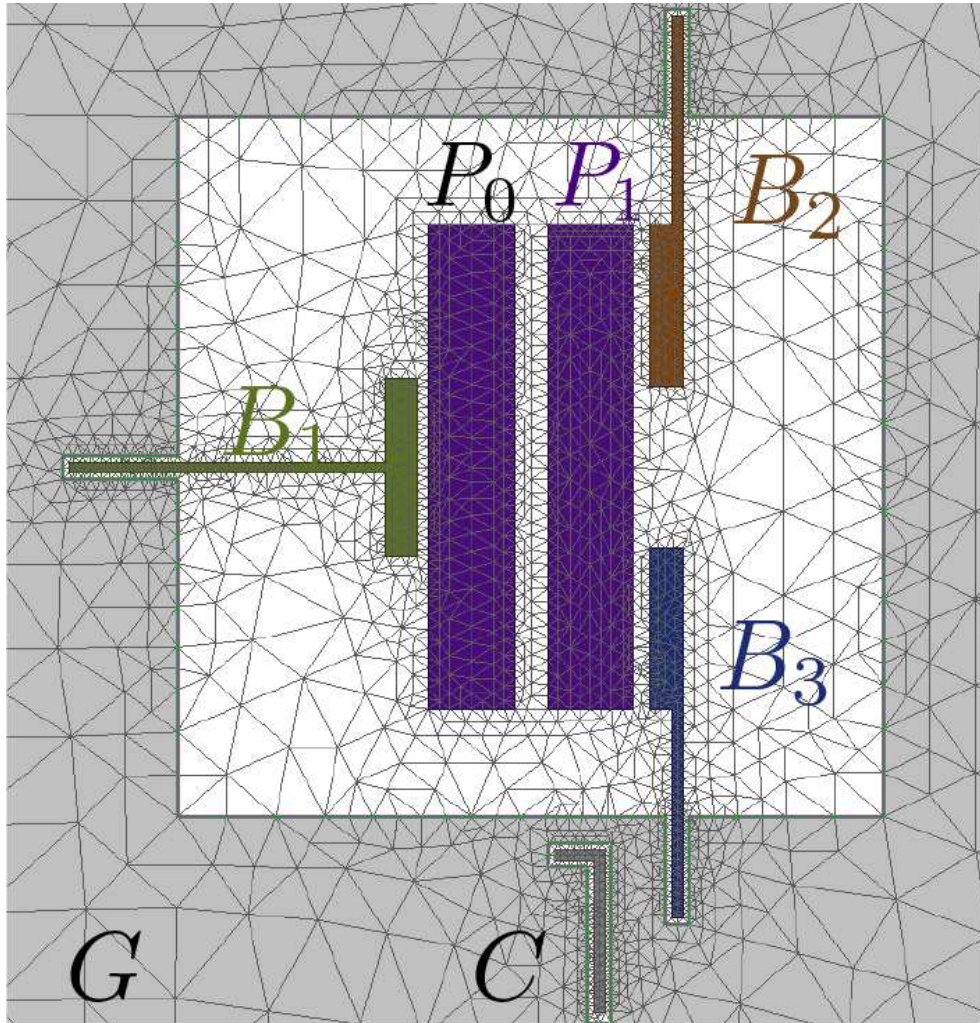
extractor model



Extracting cell parameters

netlist & simulation

Maxwell capacitance matrix

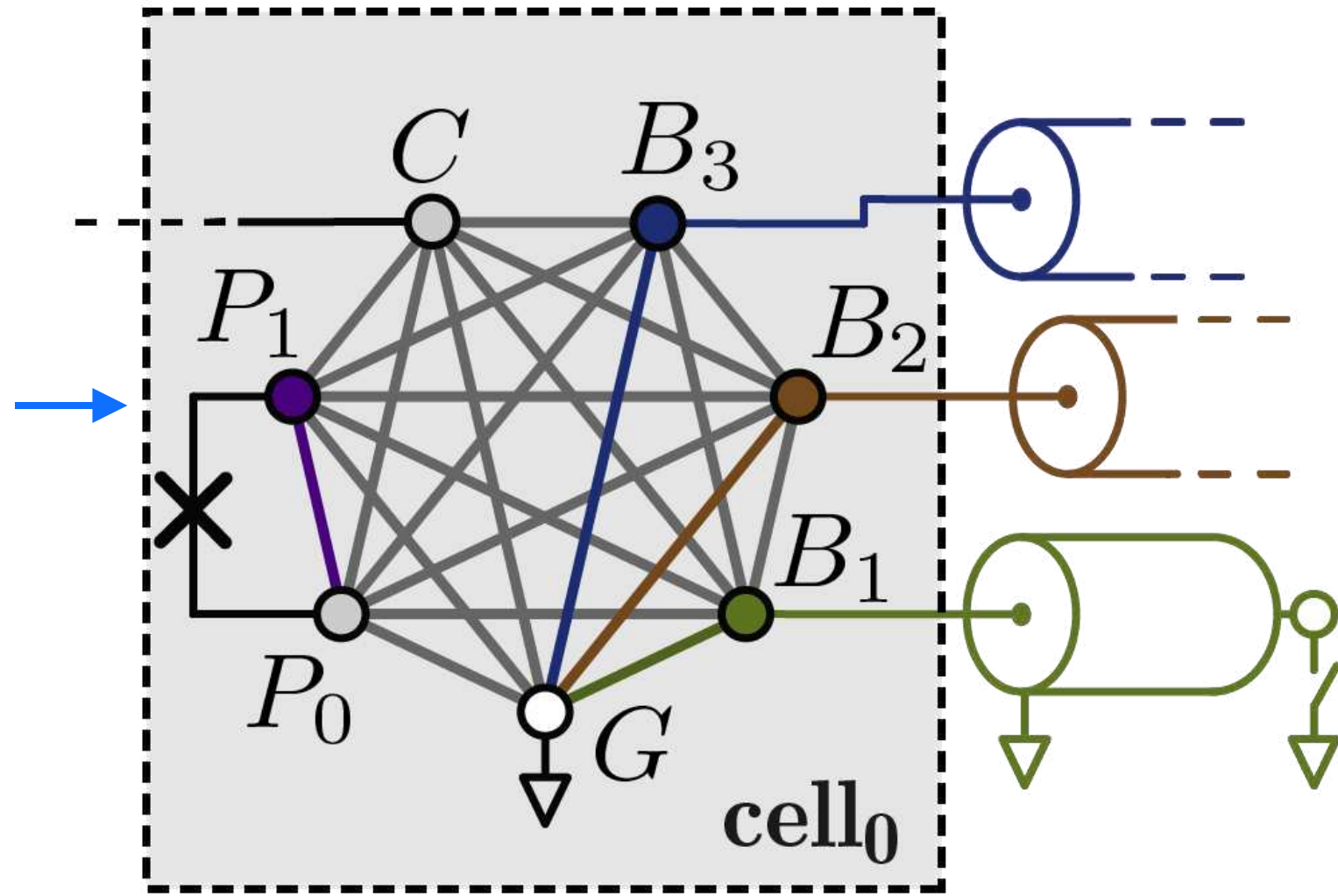
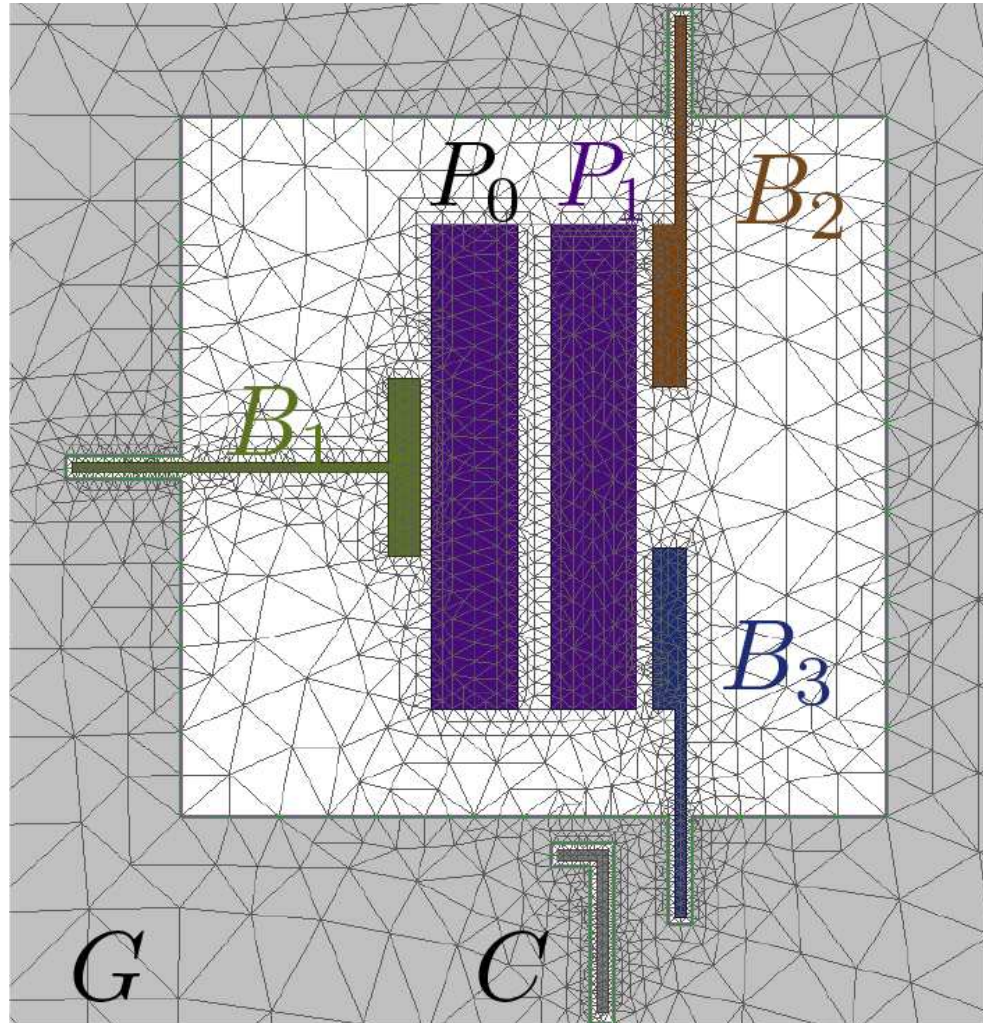


$$\begin{matrix} n_0 \\ n_1 \\ n_2 \\ n_3 \\ n_4 \\ n_5 \end{matrix} \begin{pmatrix} n_0 & n_1 & n_2 & n_3 & n_4 & n_5 \\ C_{0\Sigma} & -C_{01} & -C_{02} & -C_{03} & -C_{04} & -C_{05} \\ & C_{1\Sigma} & -C_{12} & -C_{13} & -C_{14} & -C_{15} \\ & & C_{2\Sigma} & -C_{23} & -C_{24} & -C_{25} \\ & & & C_{3\Sigma} & -C_{34} & -C_{35} \\ & & & & C_{4\Sigma} & -C_{45} \\ & & & & & C_{5\Sigma} \end{pmatrix}$$

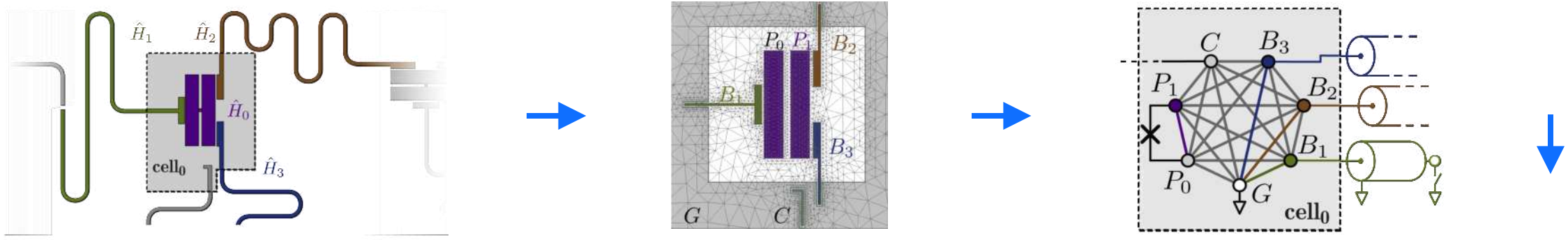
Automated with
Qiskit | quantum device
 design

Cell & composite-system network

full network model



Reduction to dressed subsystems



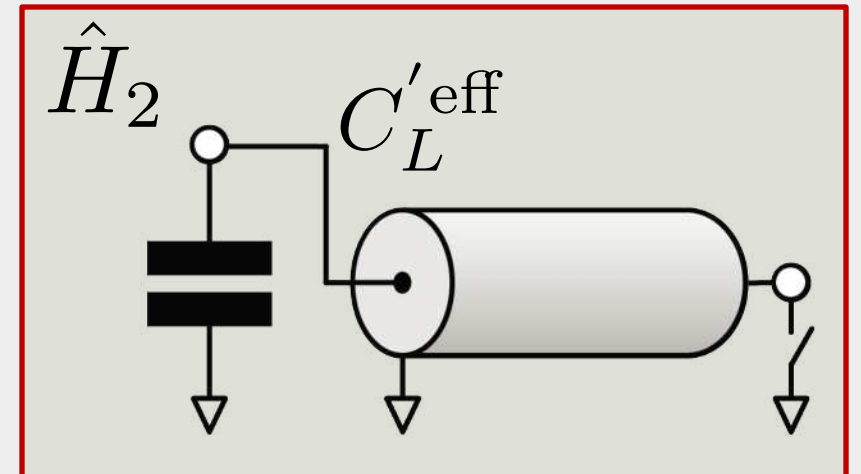
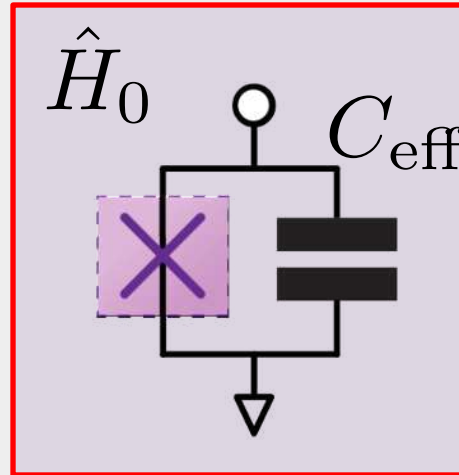
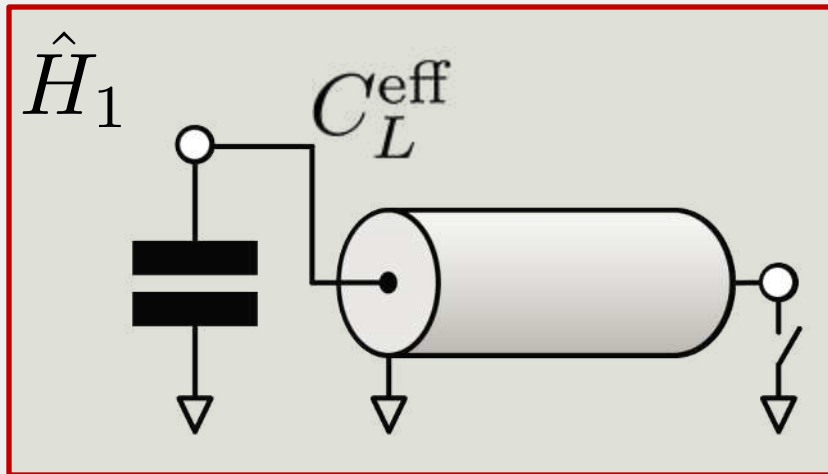
...

$$\hat{H}_{01}$$

$$\hat{H}_{12}$$

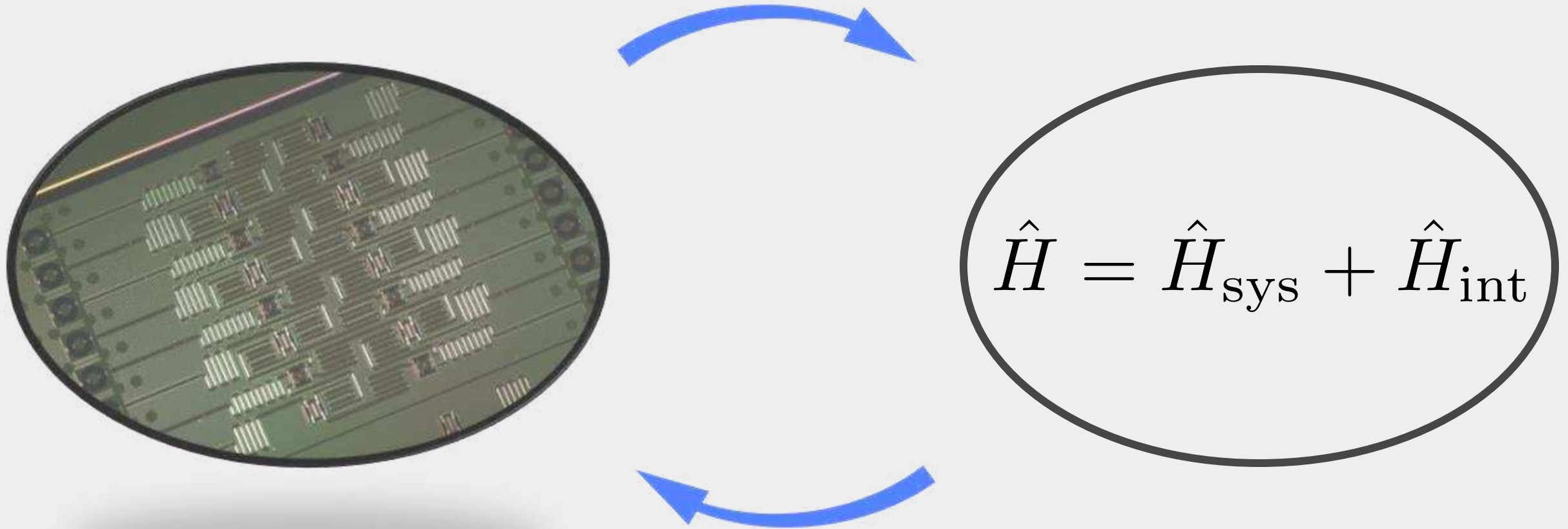
$$\hat{H}_{02}$$

...

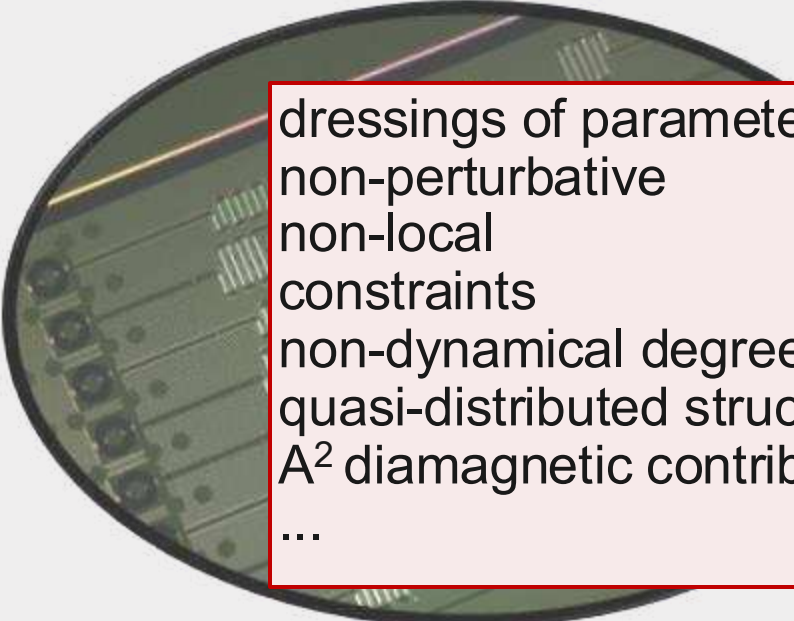


...

Physical Hardware $\leftrightarrow$ Modular Systems



Summary: Physical Hardware ↔ Modular Systems



dressings of parameters due to couplings
non-perturbative
non-local
constraints
non-dynamical degrees of freedom (Ding et al.)
quasi-distributed structures
 A^2 diamagnetic contributions (Malekakhlagh et al.)
...

$$\hat{H}_{\text{sys}} = \sum_n \hat{H}_n$$

small,
known
building blocks

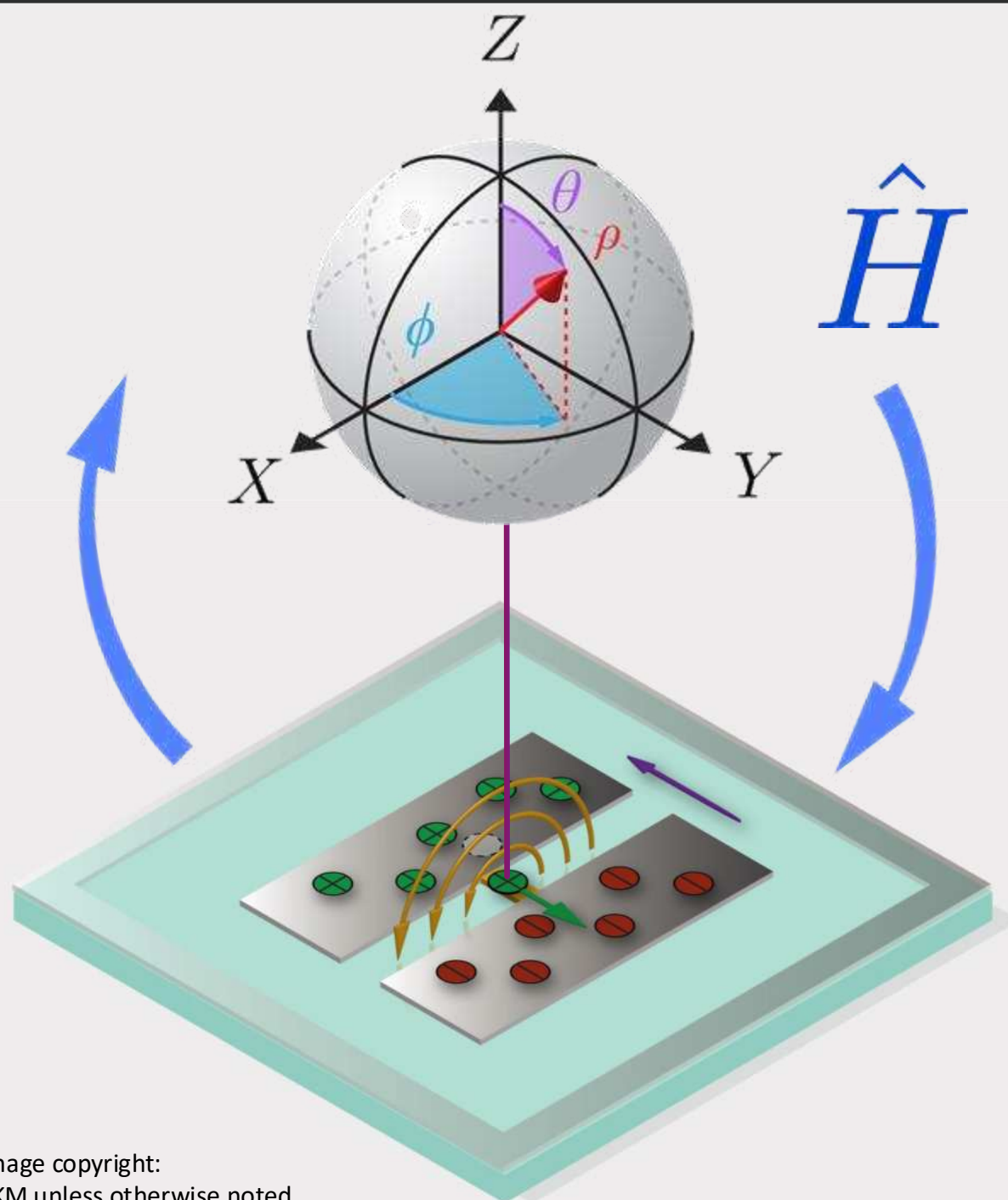
$$\hat{H}_{\text{int}} = \sum_n \sum_m \hbar g_{nm} \hat{A}_n \hat{B}_m$$

$$[\hat{H}_n, \hat{H}_m] = 0$$

$$[\hat{A}_n, \hat{B}_m] = 0$$

Jens Koch
Peter Groszkowski

- transmon (fixed | tunable)
- fluxonium
- 3-jct. flux qubit
- 0- π qubit
- cos(2 φ) qubit ^{NEW}
- oscillator | Kerr oscillator ^{NEW}



The End: Quasi-Lumped Element Oscillator Models (LOM)

arXiv > quant-ph > arXiv:2103.10344

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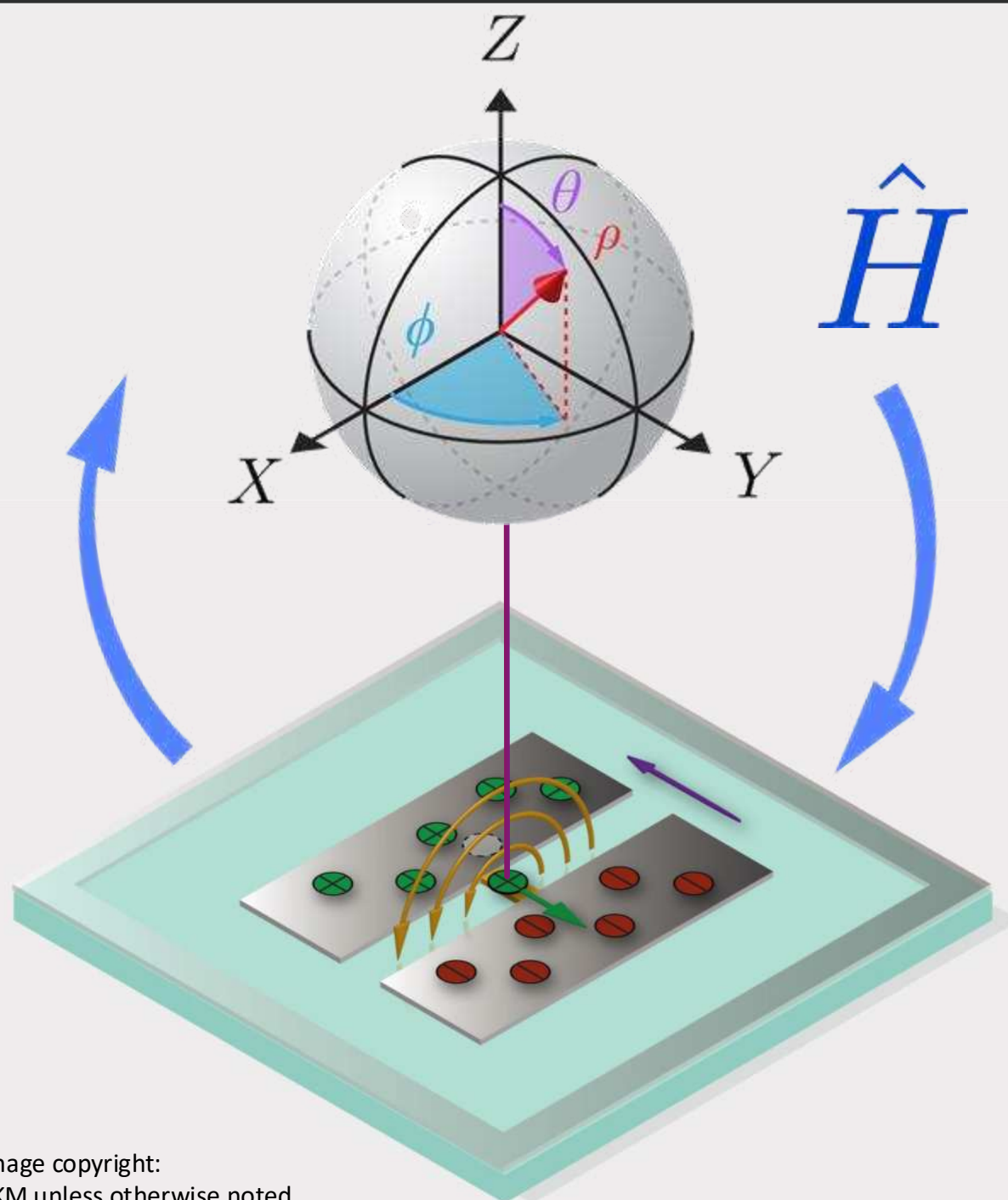
Quantum Physics

[Submitted on 18 Mar 2021]

Circuit quantum electrodynamics (cQED) with modular quasi-lumped models

Zlatko K. Mineev, Thomas G. McConkey, Maika Takita, Antonio D. Corcoles, Jay M. Gambetta

arXiv:2103.10344



Energy- Participation Ratio (EPR)

arXiv.org > quant-ph > arXiv:2010.00620

Quantum Physics

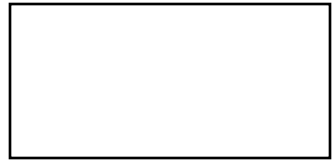
[Submitted on 1 Oct 2020]

Energy-participation quantization of Josephson circuits

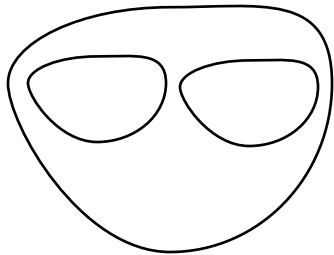
Zlatko K. Minev, Zaki Leghtas, Shantanu O. Mundhada, Lysander Christakis, Ioan M. Pop, Michel H. Devoret

Hamiltonian of a distributed system

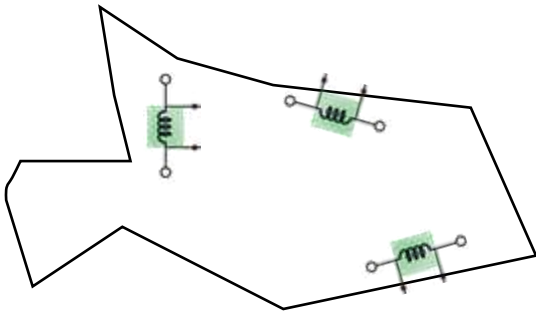
Linear system



$$\sum_{m=1}^M \hbar \omega_m \hat{a}_m^\dagger \hat{a}_m$$

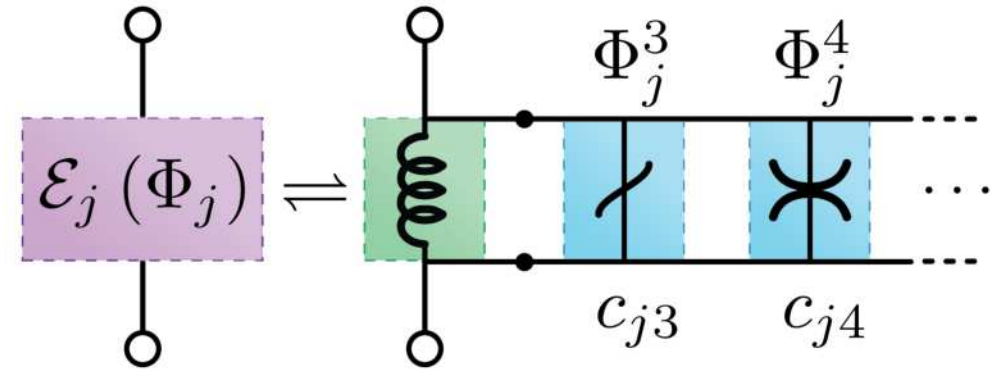


$$\sum_{m=1}^M \hbar \omega_m \hat{a}_m^\dagger \hat{a}_m$$

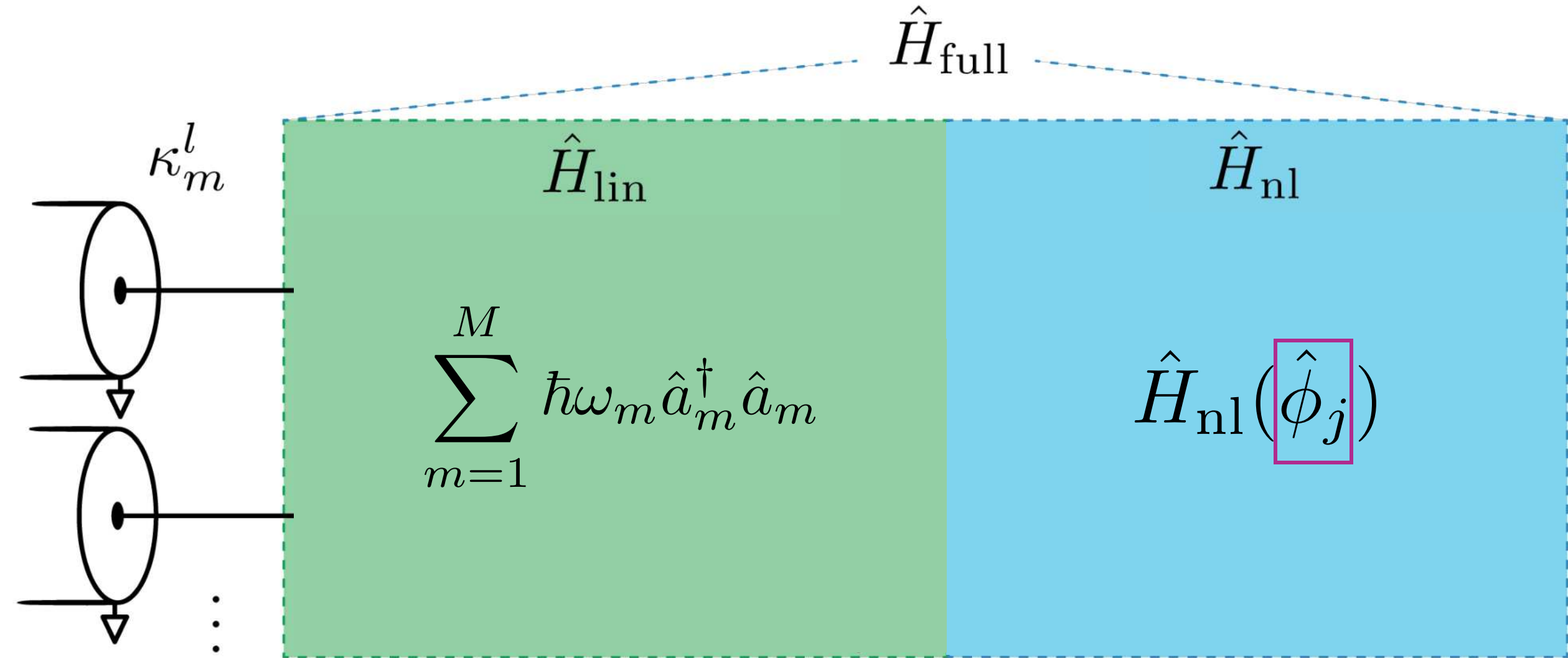


$$\sum_{m=1}^M \hbar \omega_m \hat{a}_m^\dagger \hat{a}_m$$

Non-linear dipole



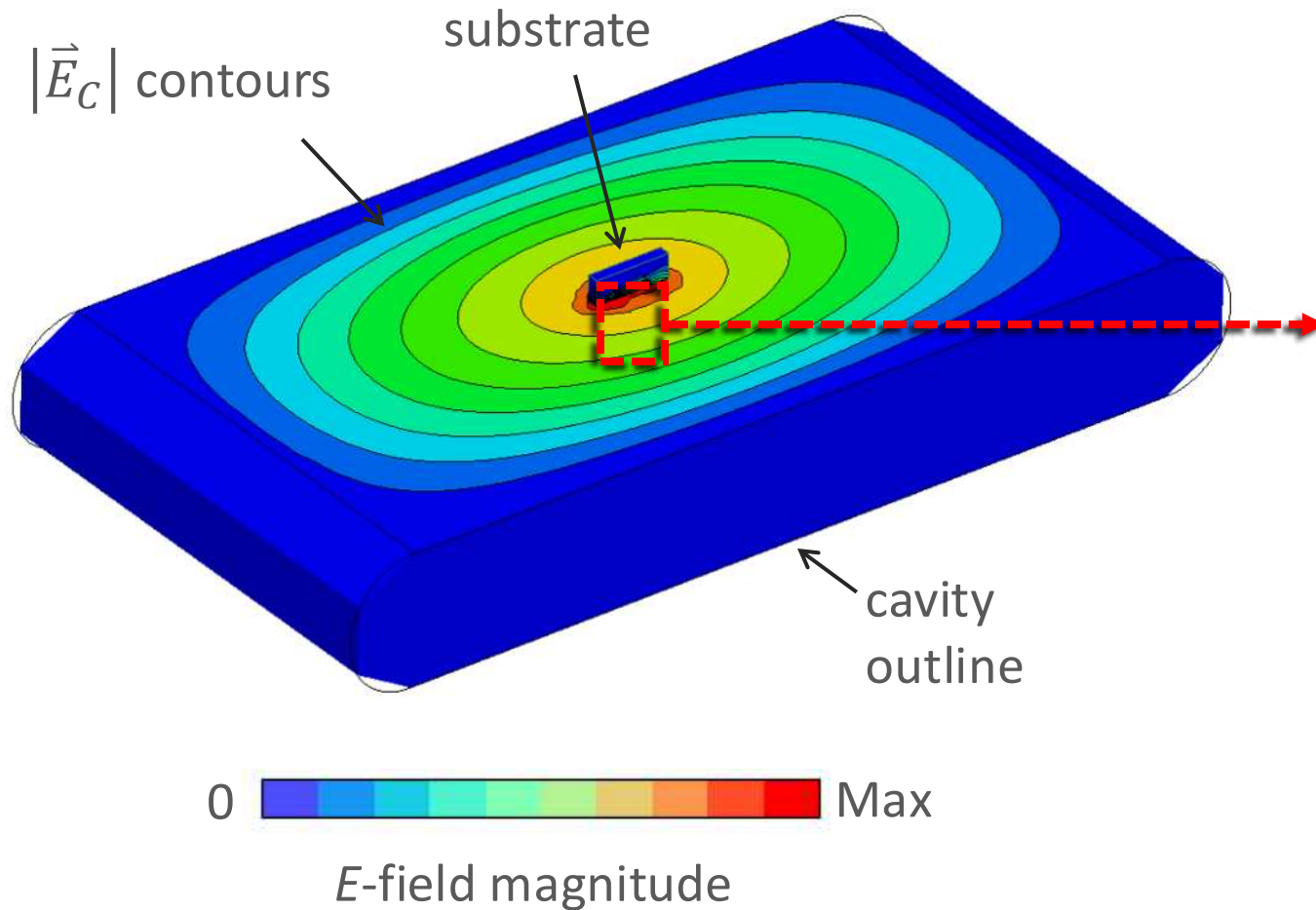
Decomposition of a general circuit



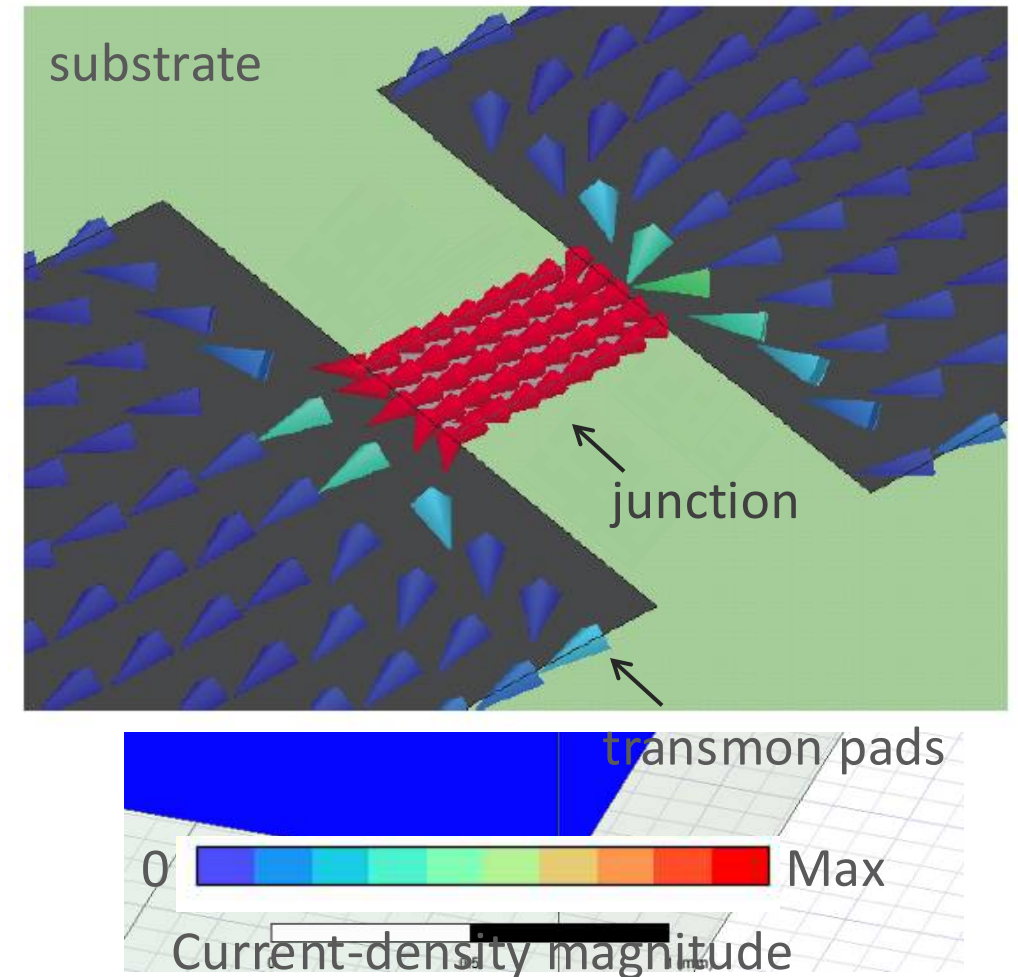
second quantization in eigen basis of linearized circuit

$\mathcal{H}_{\text{lin}}$ eigen modes

Cavity mode (7.0 GHz)



Qubit mode (linearized, 5 GHz)

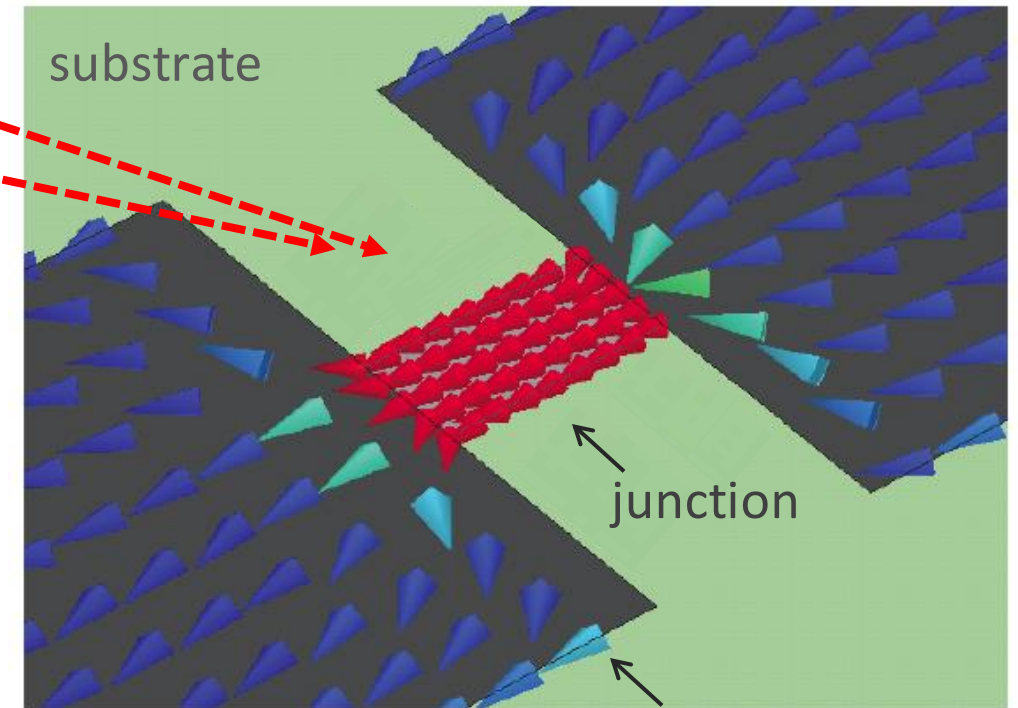


Energy participation of the junction

$$p_m = \frac{\text{Energy stored in junction}}{\text{Inductive energy stored in mode } m}$$

$p_m = \frac{\frac{1}{2} L_j I_j^2}{E_{ind,m}}$

Qubit mode (linearized, 5 GHz)

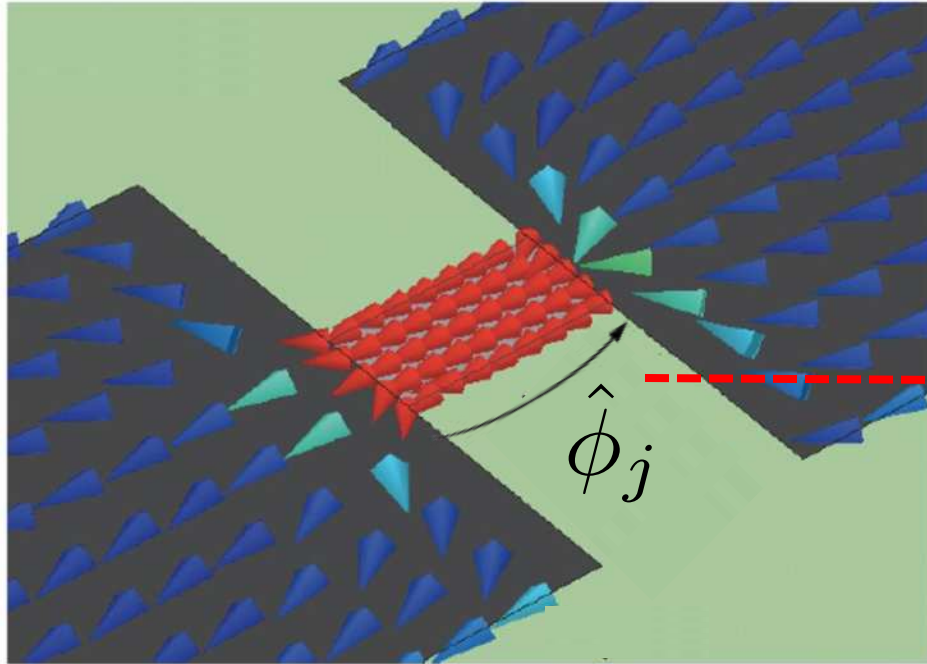


transmon pads



Current-density magnitude

Quantum operator

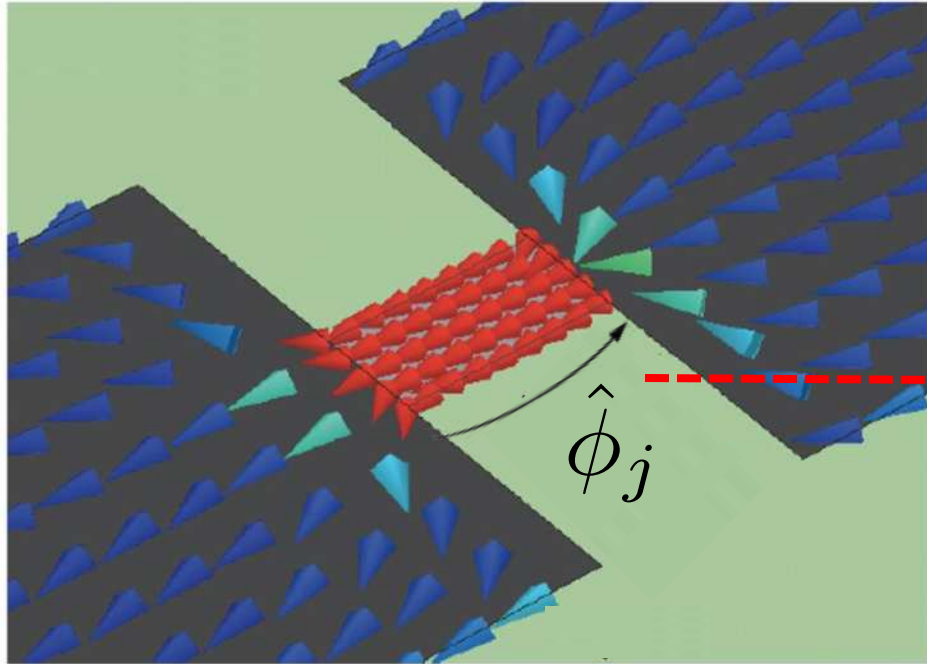


What is zero-point fluctuation of the phase of junction j due to mode m ?

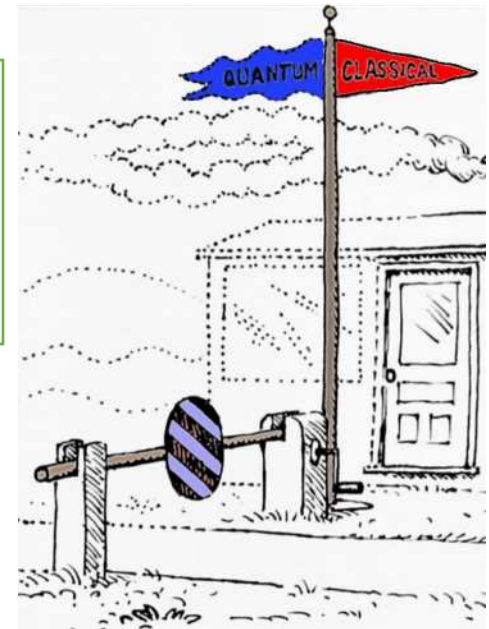
$$\hat{\phi}_j = \sum_{m=1}^M \phi_{mj} (\hat{a}_m^\dagger + \hat{a}_m)$$

EPR bridge from classical to quantum

What fraction of the energy of mode m is stored in junction j ?



$$\frac{1}{\hbar} \phi_{mj}^2 = p_{mj} \frac{\omega_m}{2E_j}$$



for $j > 1$, root requires sign bit $s_{mj} = \pm 1$

Drawing: Zurek, Physics Today (1991)

Step 3: Hamiltonian

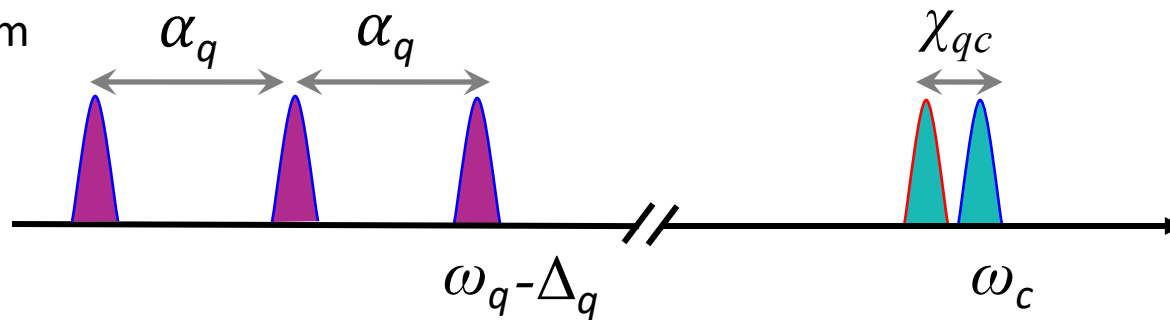
Qubit/cavity anharmonicity

$$\alpha_{q/c} = p_{q/c}^2 \frac{\hbar \omega_{q/c}^2}{8E_J}$$

Qubit-cavity dispersive shift

$$\chi_{qc} = p_q p_c \frac{\hbar \omega_q \omega_c}{4E_J}$$

Transition spectrum

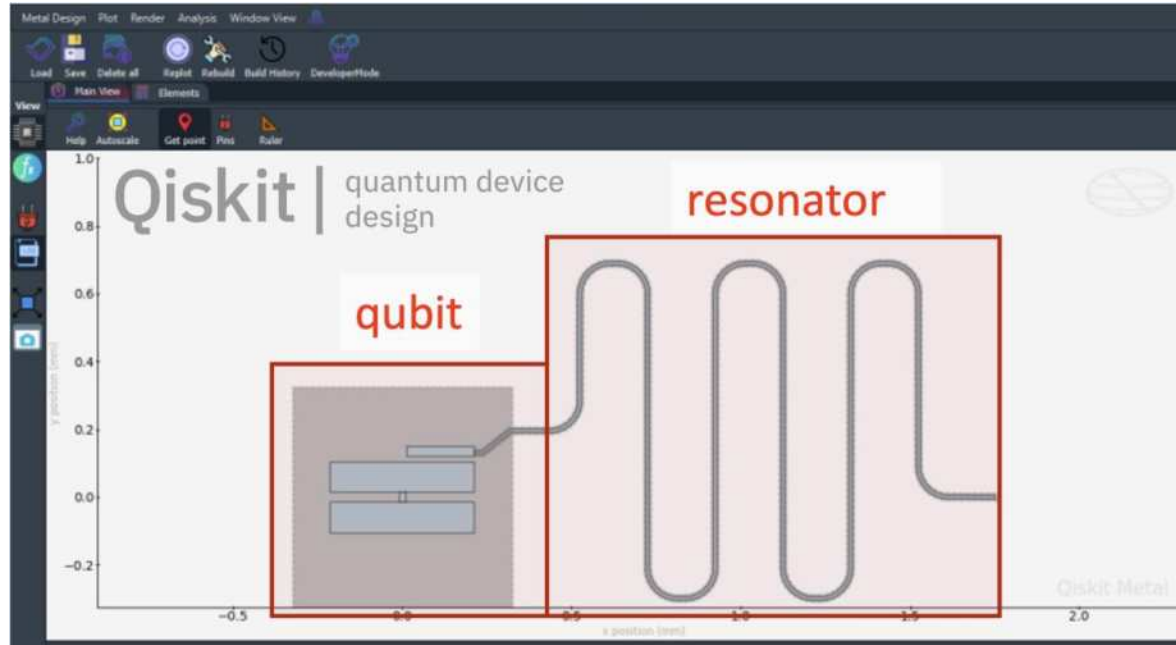


Qubit Lamb shift

$$\Delta_q = \alpha_q - \frac{1}{2}\chi_{qc}$$

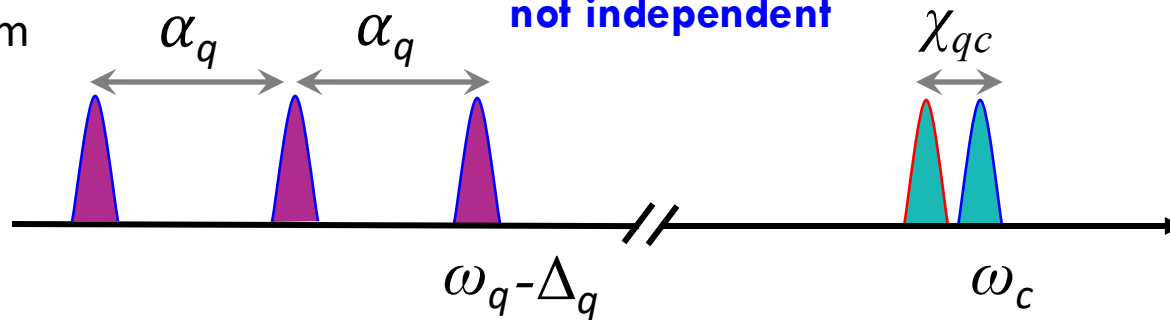
H_{eff} : for simplicity, showing up to $\mathcal{O}(\varphi^6)$ in RWA

Hamiltonian: Universal constraints & EPR monogamy

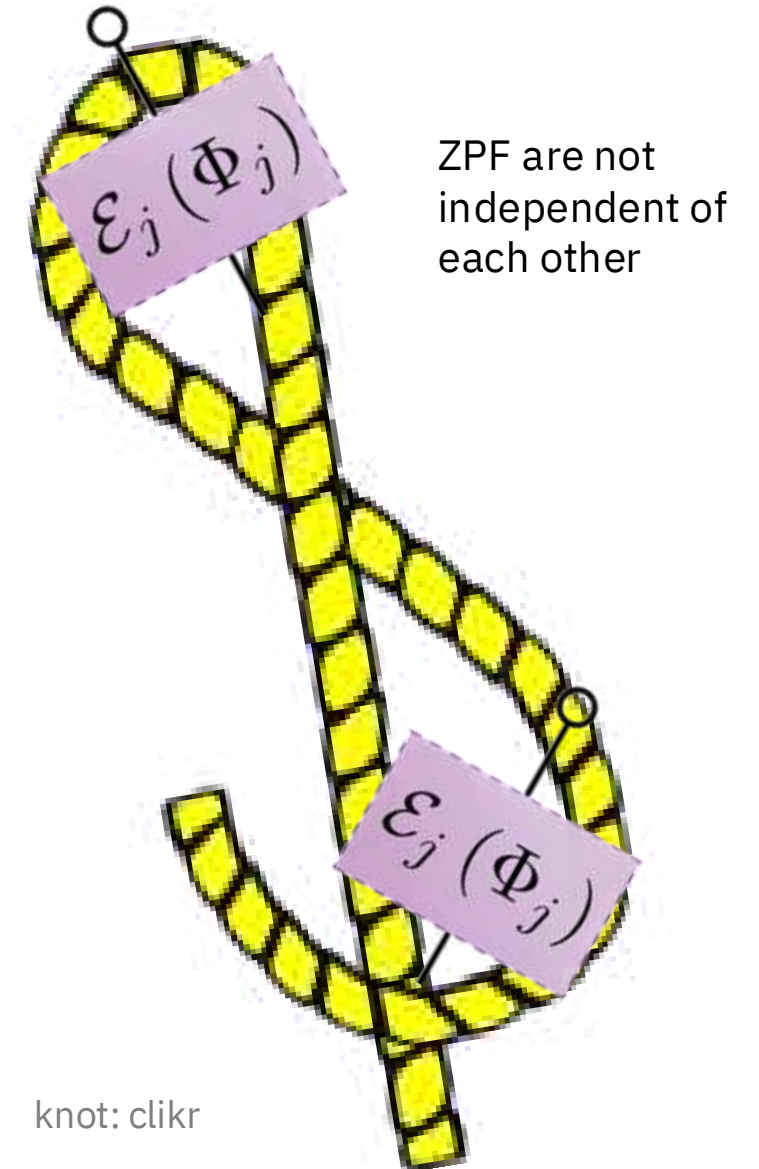


Transition spectrum

Quantum parameters are not independent



H_{eff} : for simplicity, showing up to $\mathcal{O}(\varphi^6)$ in RWA



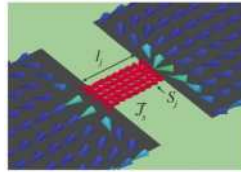
knot: klikr

EPR is monogamous

Commandments of EPR

- I. All modes and all dipole are created equal
- II. Each dipole has exactly 1 EPR it must dilute among modes
- III. Each mode receives exactly 1 EPR units

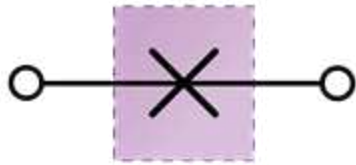
EPR



$$0 \leq p_{mj} \leq 1$$

EPR is bounded

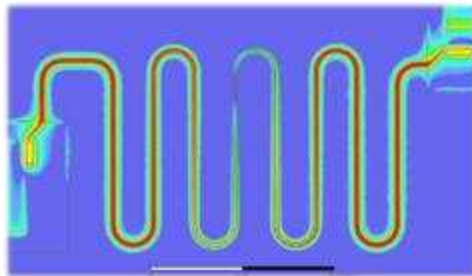
Non-linear dipole



$$\sum_{m=1}^M p_{mj} = 1 \quad \text{for } j \in \{1, \dots, J\}$$

Every dipole comes w/ 1 unit of EPR

Mode



$$0 \leq \sum_{j=1}^J p_{mj} \leq 1 \quad \text{for } m \in \{1, \dots, M\}$$

Every mode accepts at most 1 total unit of EPR

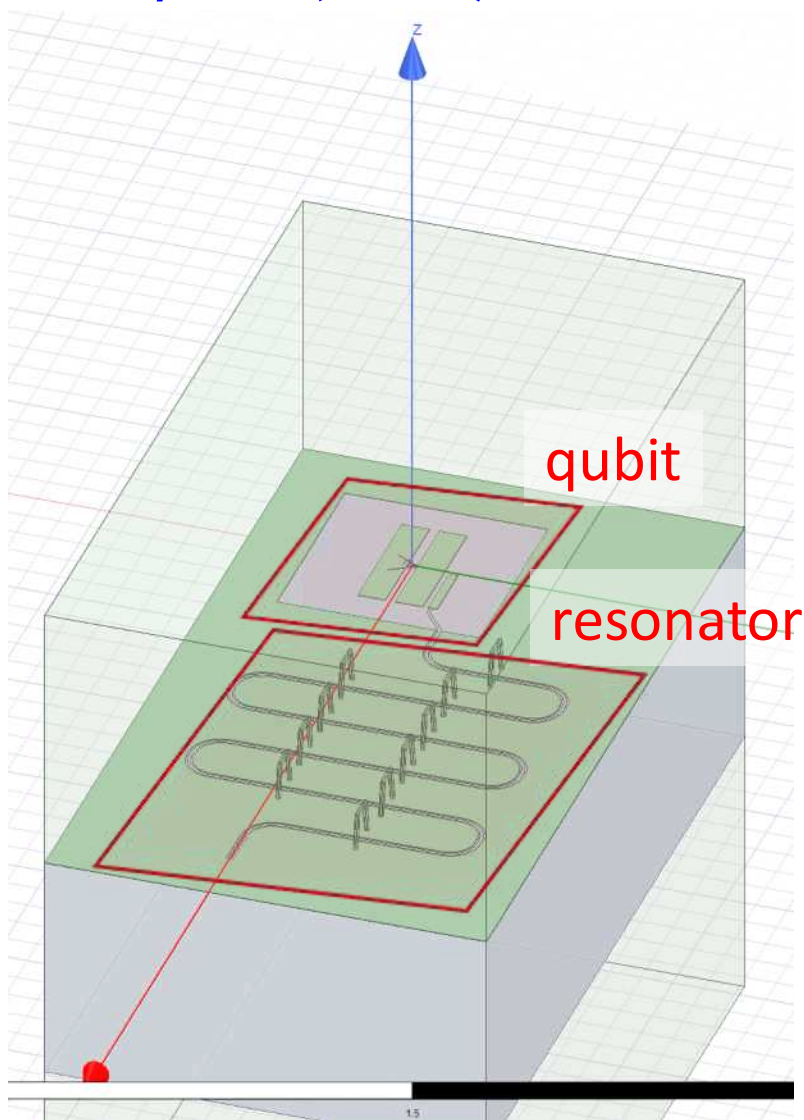
Orthogonality

$$\sum_{m=1}^M s_{mj} s_{mj'} \sqrt{p_{mj} p_{mj'}} = 0 \quad \text{for } j \neq j'$$

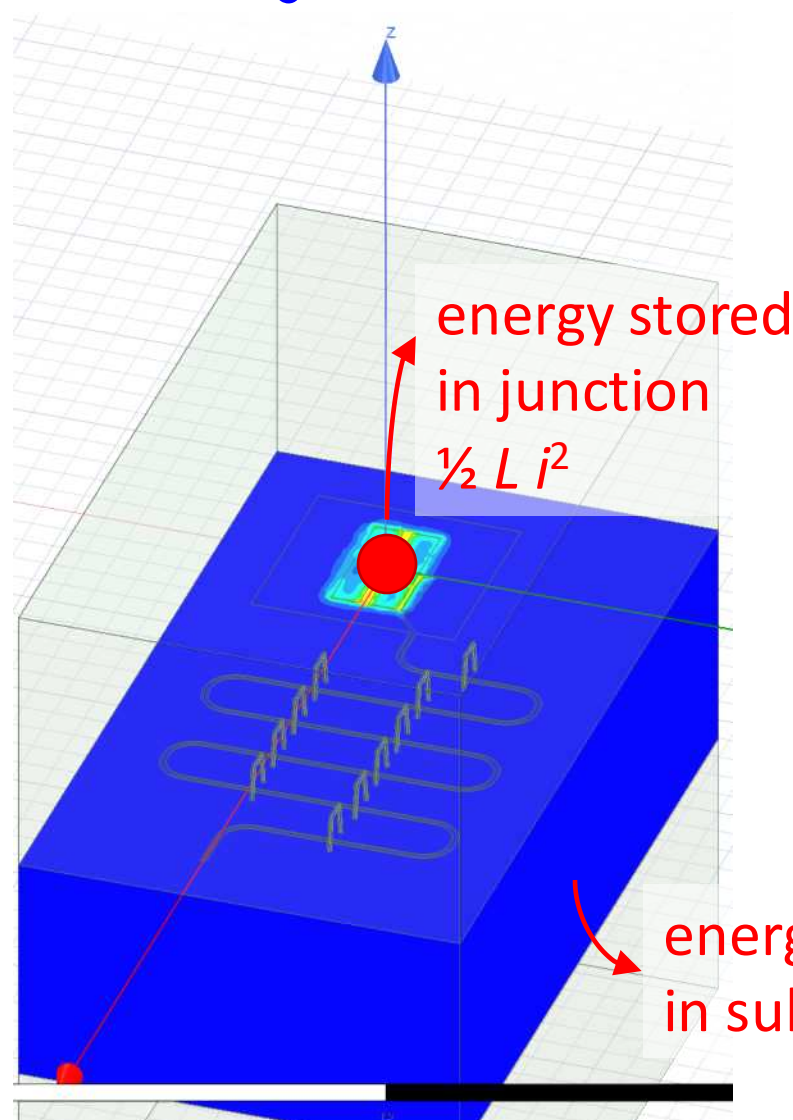
For a dipole, its EPRs across all modes are orthogonal

Example of step 2 with EPR

Physical (linear) model



Eigenmodes



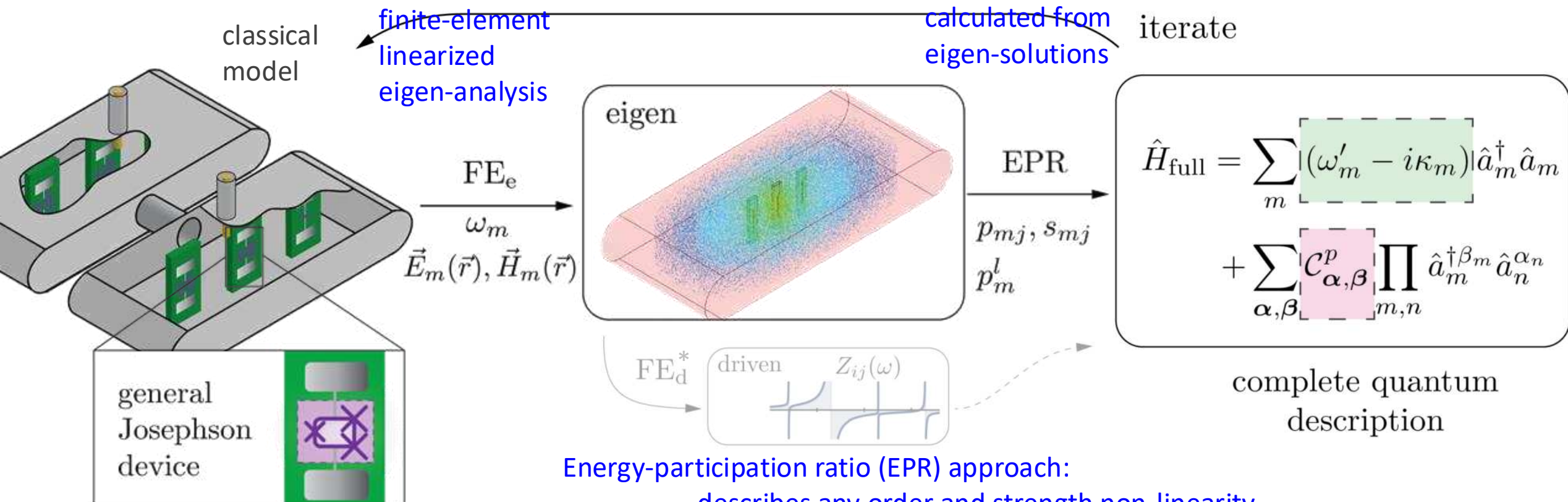
Non-linear device
params

$$p_j \longrightarrow \hat{H}_{\text{full}}$$

Dissipation
budget

$$p_l \longrightarrow \mathcal{D}[\sqrt{\kappa} \hat{a}]$$

Overview of energy-participation approach



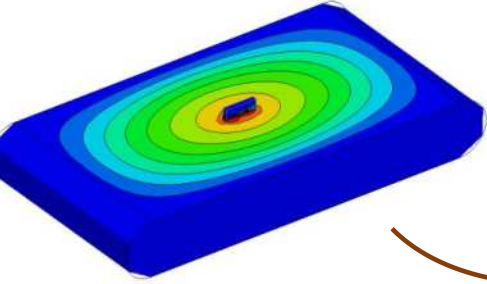
Energy-participation ratio (EPR) approach:

- describes any order and strength non-linearity
- describes arbitrary (composite) non-linear inductive devices
- first-principle derivation
- zero approximations (aside from truncation of modes)
- fully automated in Qiskit Metal & in pyEPR

Practical limits: Fock and mode basis truncation due to computing power

* Nigg, Paik, *et al.*, PRL (2012),
Bourassa *et al.* (2012),
Solgun *et al.* (2014, 2015, 2017), ...

Lossy energy-participation ratios (EPRs)



Capacitive



bulk

$$p_{ml}^{\text{cap}} = \frac{1}{\mathcal{E}_{\text{elec}}} \frac{1}{4} \Re \int_{V_l} \vec{E}_{\text{max}}^* \overleftrightarrow{\epsilon} \vec{E}_{\text{max}} dv ,$$

surface

$$p_{ml}^{\text{cap,surf}} = \frac{1}{\mathcal{E}_{\text{elec}}} \frac{t_l \epsilon_l}{4} \Re \int_{\text{surf}_l} |\vec{E}_{\text{max}}|^2 ds ,$$

Inductive



bulk

$$p_{ml}^{\text{ind,surf}} = \frac{1}{\mathcal{E}_{\text{mag}}} \frac{\lambda_0 \mu_l}{4} \Re \int_{\text{surf}_l} |\vec{H}_{\text{max},\parallel}|^2 ds ,$$

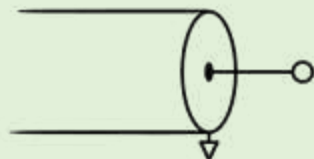
surface

$$p_{ml}^{\text{ind,bulk}} = \frac{1}{\mathcal{E}_{\text{mag}}} \frac{1}{4} \Re \int_{V_l} \vec{H}_{\text{max}}^* \overleftrightarrow{\mu} \vec{H}_{\text{max}} dv .$$

seam

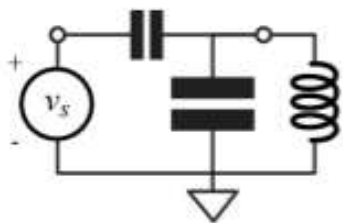
$$p_{ml}^{\text{ind,seam}} = \frac{1}{\mathcal{E}_{\text{mag}}} \frac{\lambda_0 t_l \mu_l}{4} \Re \int_{\text{seam}_l} |\vec{H}_{\text{max},\perp}|^2 dl ,$$

Radiative



external

$$Q_{mp} = \frac{\omega_m \mathcal{E}_m(0)}{\frac{1}{2} R I_{mp}^2} .$$



Charge dispersion with EPR

$$\omega_\theta = 6.50 \text{ GHz}$$

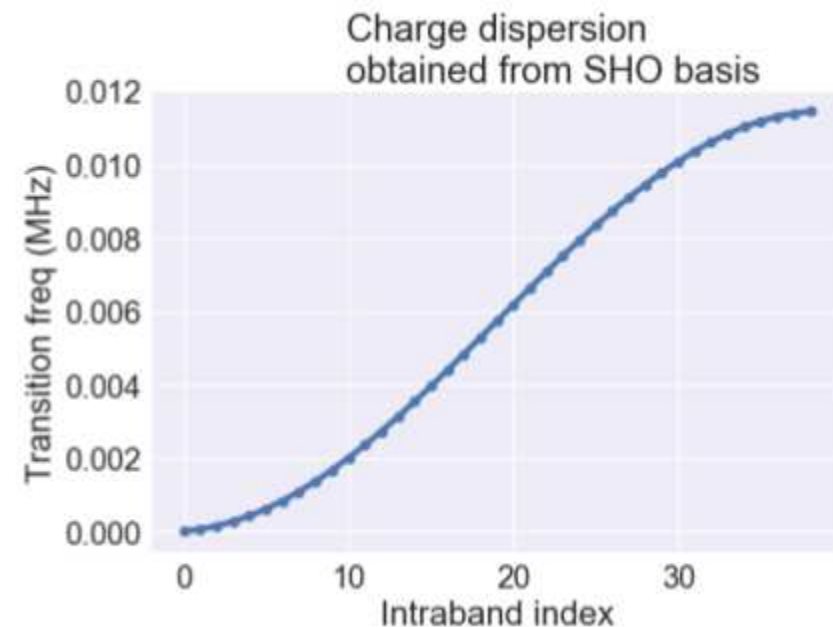
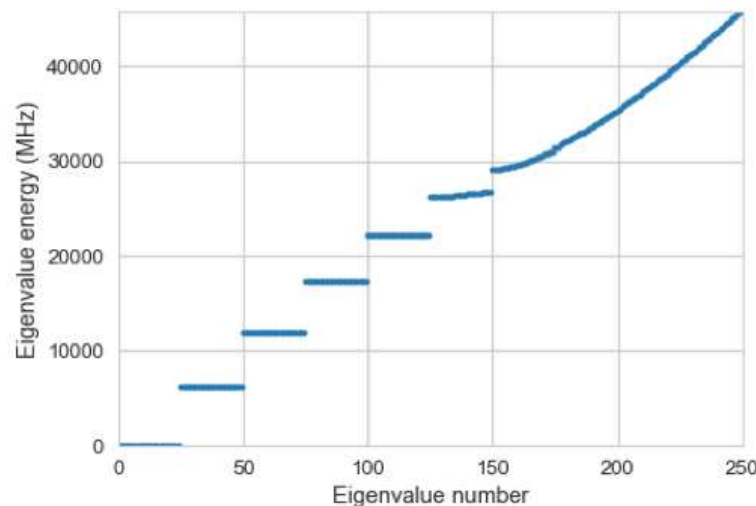
$$\alpha^{(4)} = 322.00 \text{ MHz}$$

$$\phi_{ZPF} = 0.45$$

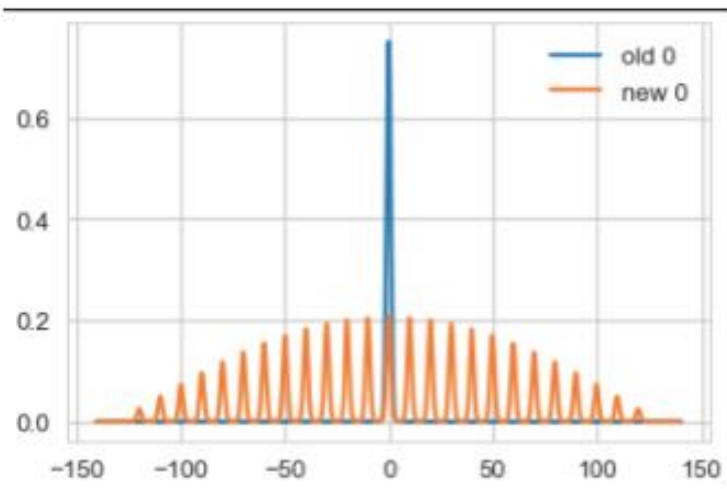
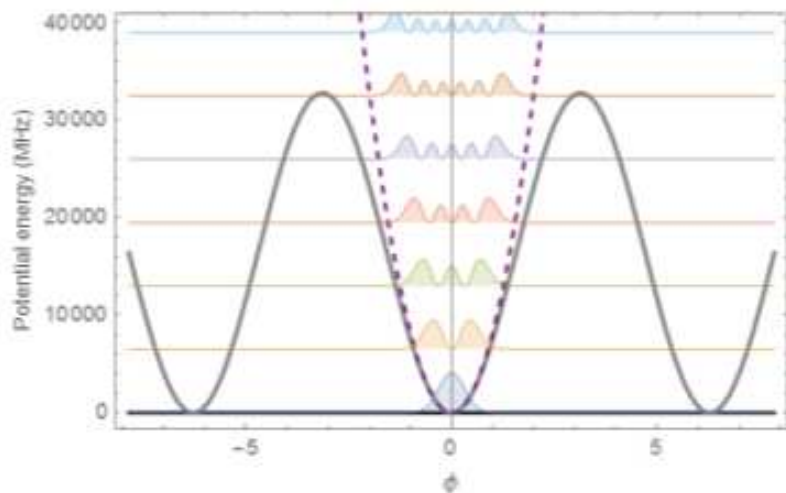
$$E_J = 16.40 \text{ GHz}$$

$$E_C = 0.32 \text{ GHz} \quad (E_J/E_C = 50.90)$$

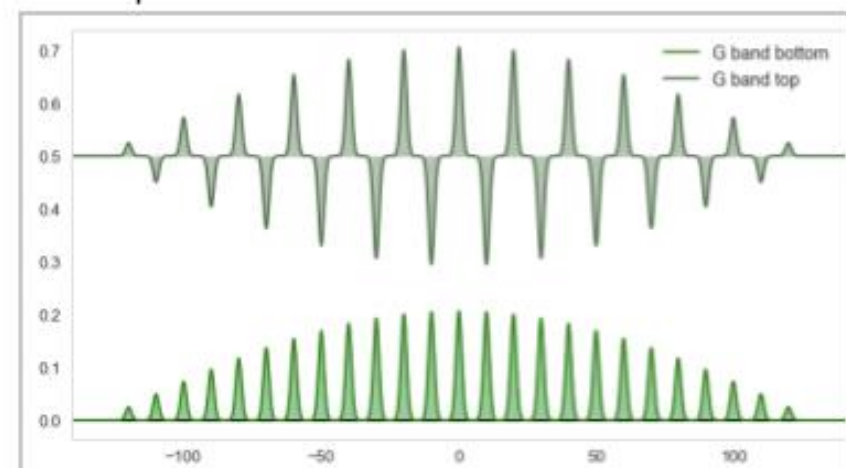
$$\text{Dispersion} = -12.90 \text{ KHz}$$



$|\psi|^2$



ϕ



EPR theory vs. experiment

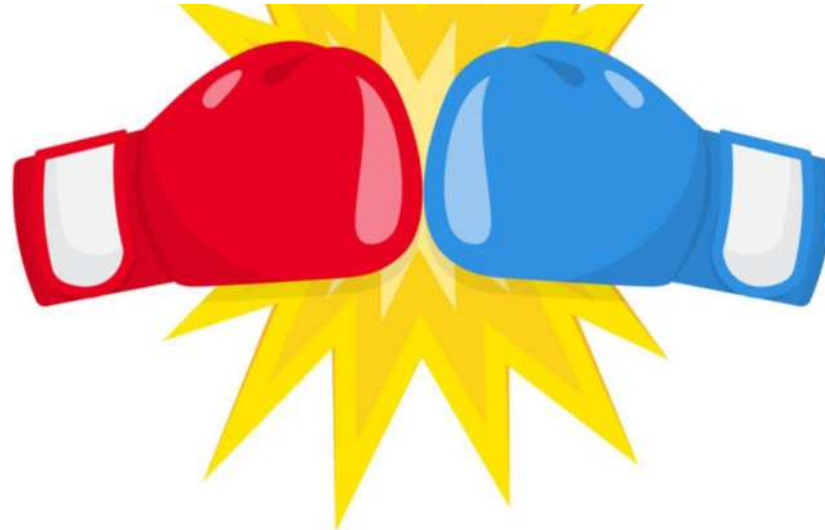
arXiv.org > quant-ph > arXiv:2010.00620

Quantum Physics

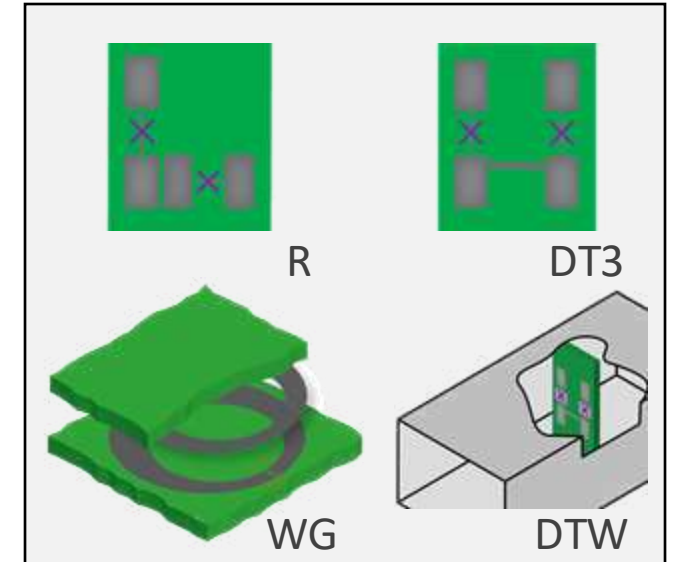
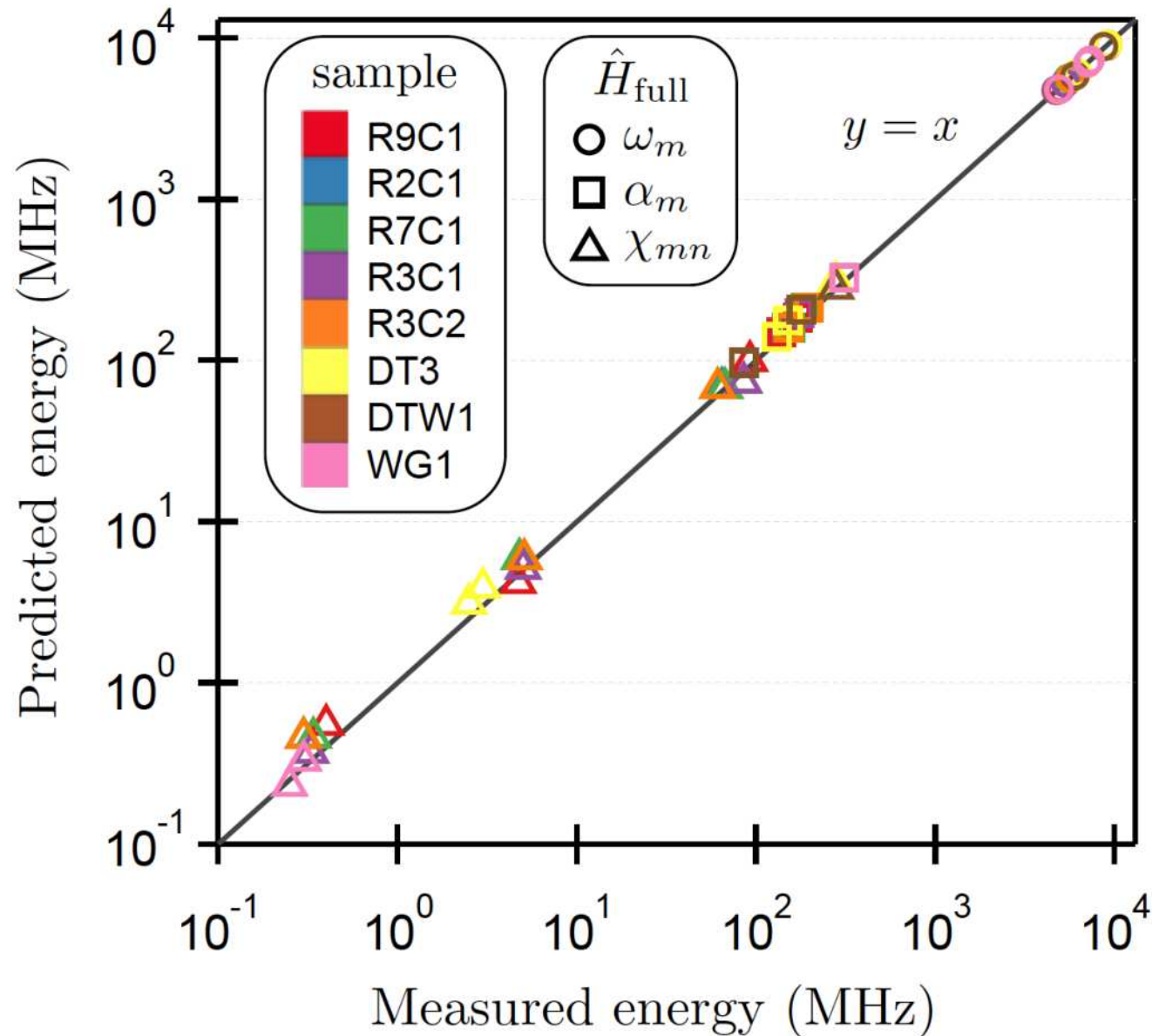
[Submitted on 1 Oct 2020]

Energy-participation quantization of Josephson circuits

Zlatko K. Minev, Zaki Leghtas, Shantanu O. Mundhada, Lysander Christakis, Ioan M. Pop, Michel H. Devoret



Theory vs. experiment: agreement over 5 orders of magnitude



R: Minev *et al.* (2018)

WG: Minev *et al.* (2013, 2016)

DT3, DTW: Minev *et al.* (2019)

Planar devices & comparison to other methods

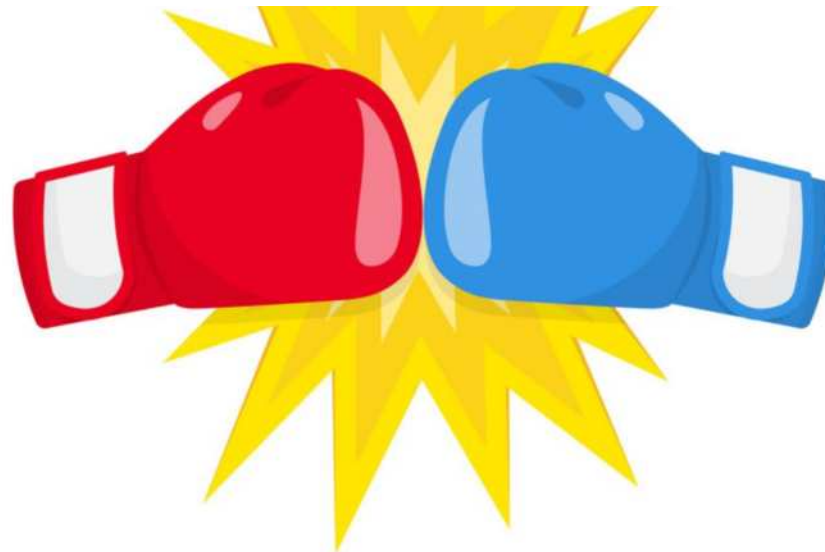
arXiv > quant-ph > arXiv:2103.10344 Search... Help | Adv

Quantum Physics

[Submitted on 18 Mar 2021]

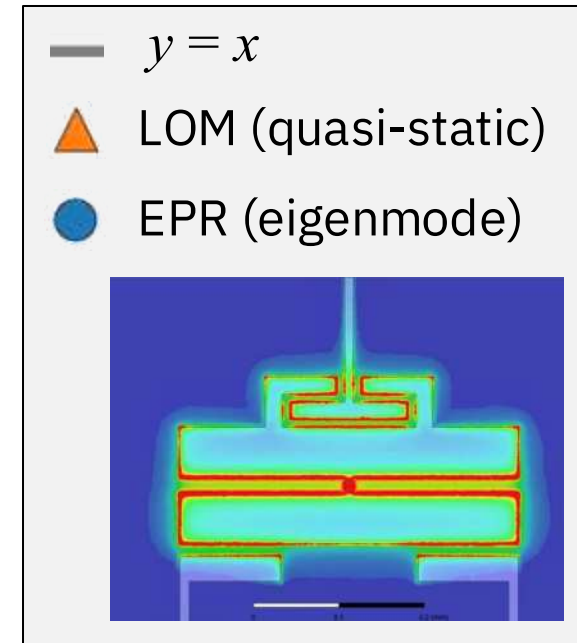
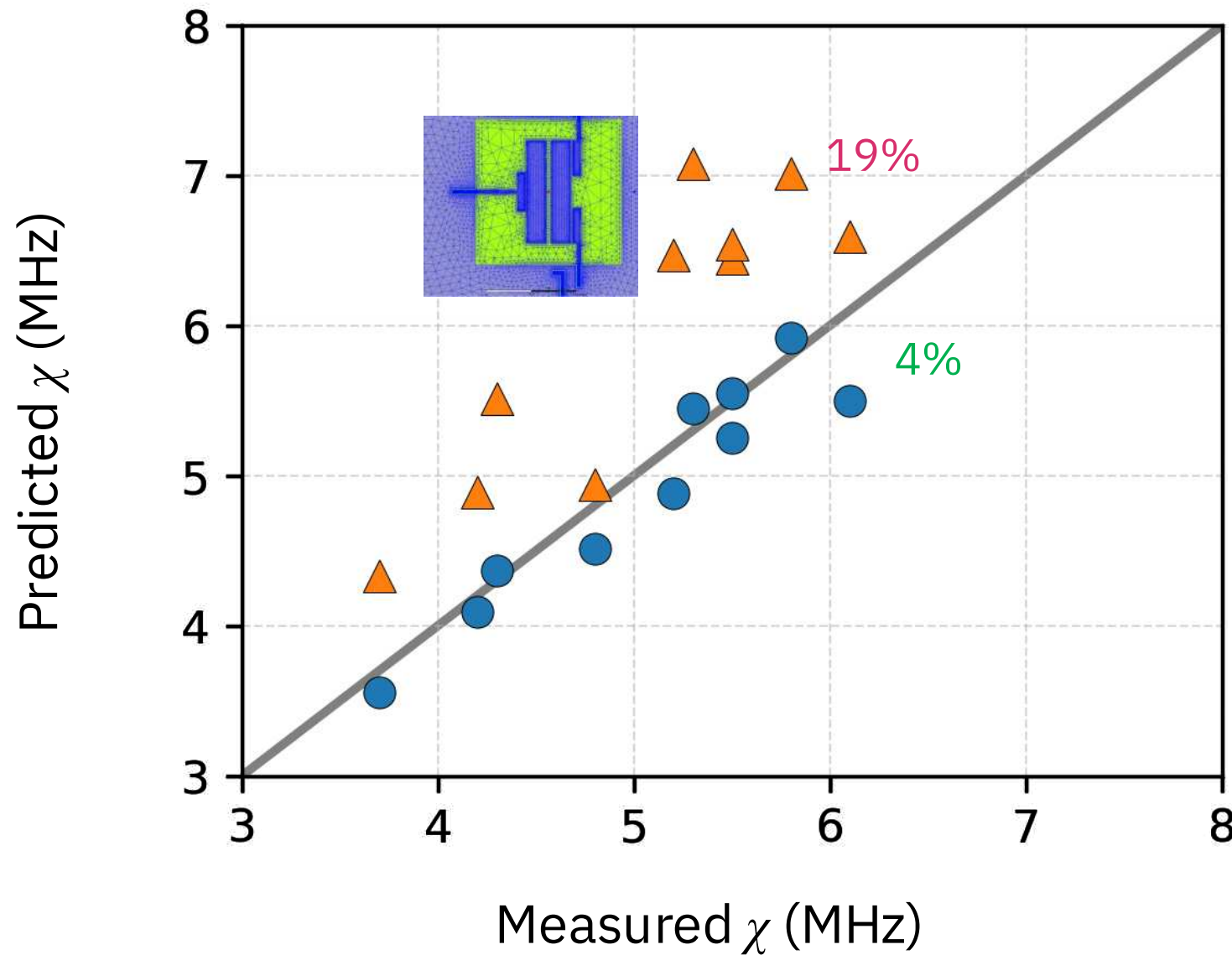
Circuit quantum electrodynamics (cQED) with modular quasi-lumped models

Zlatko K. Minev, Thomas G. McConkey, Maika Takita, Antonio D. Corcoles, Jay M. Gambetta

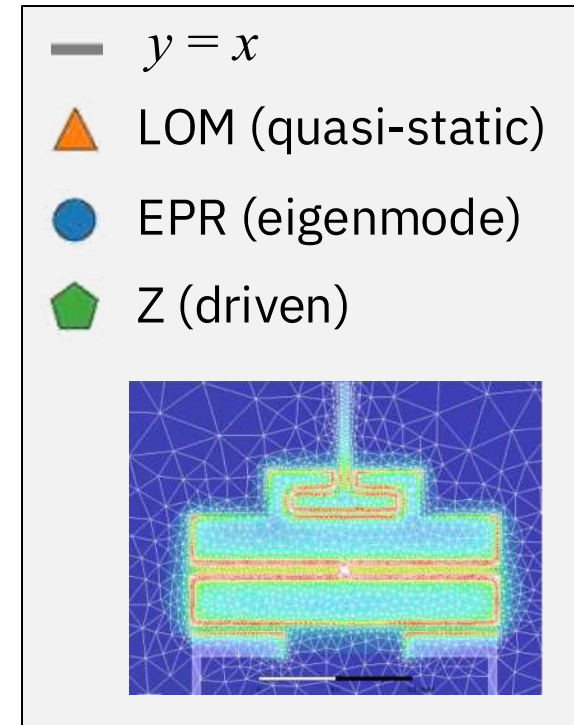
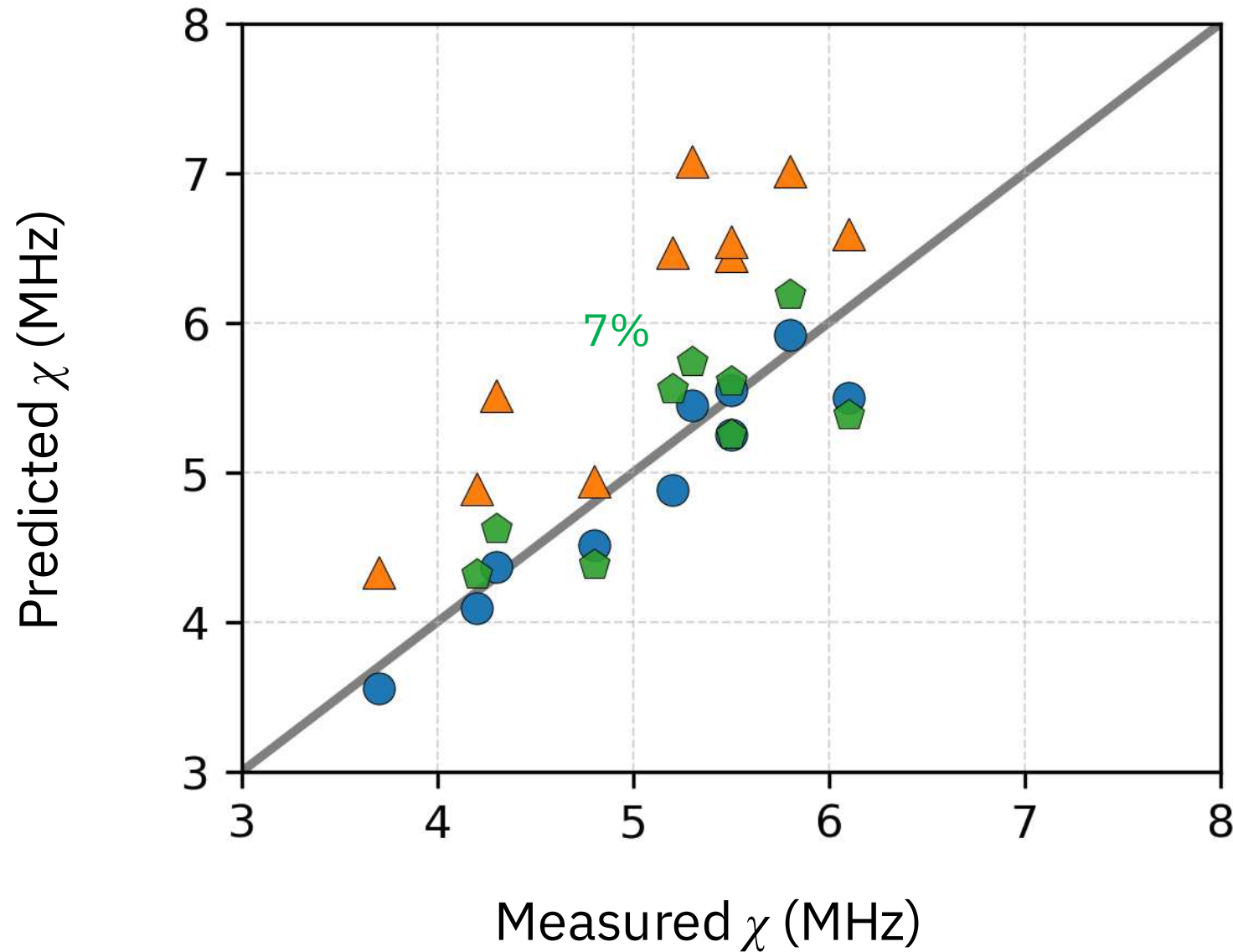


Automated with
Qiskit | quantum device
design

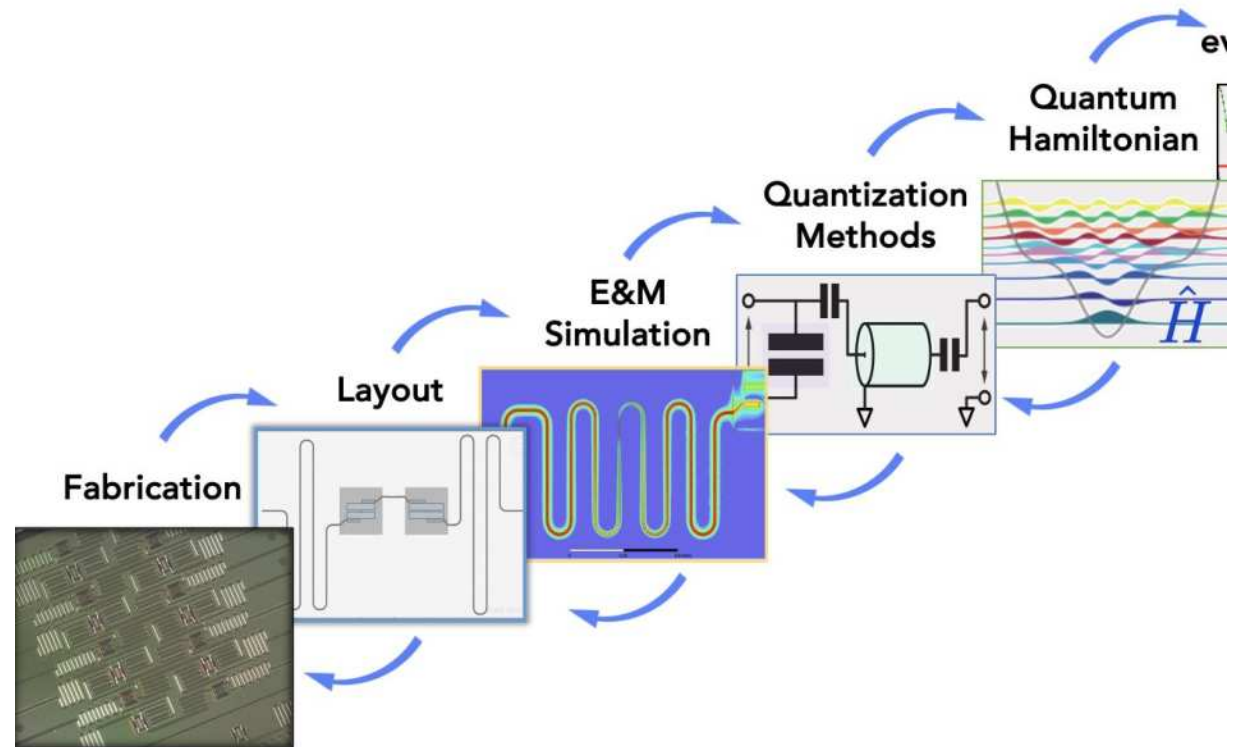
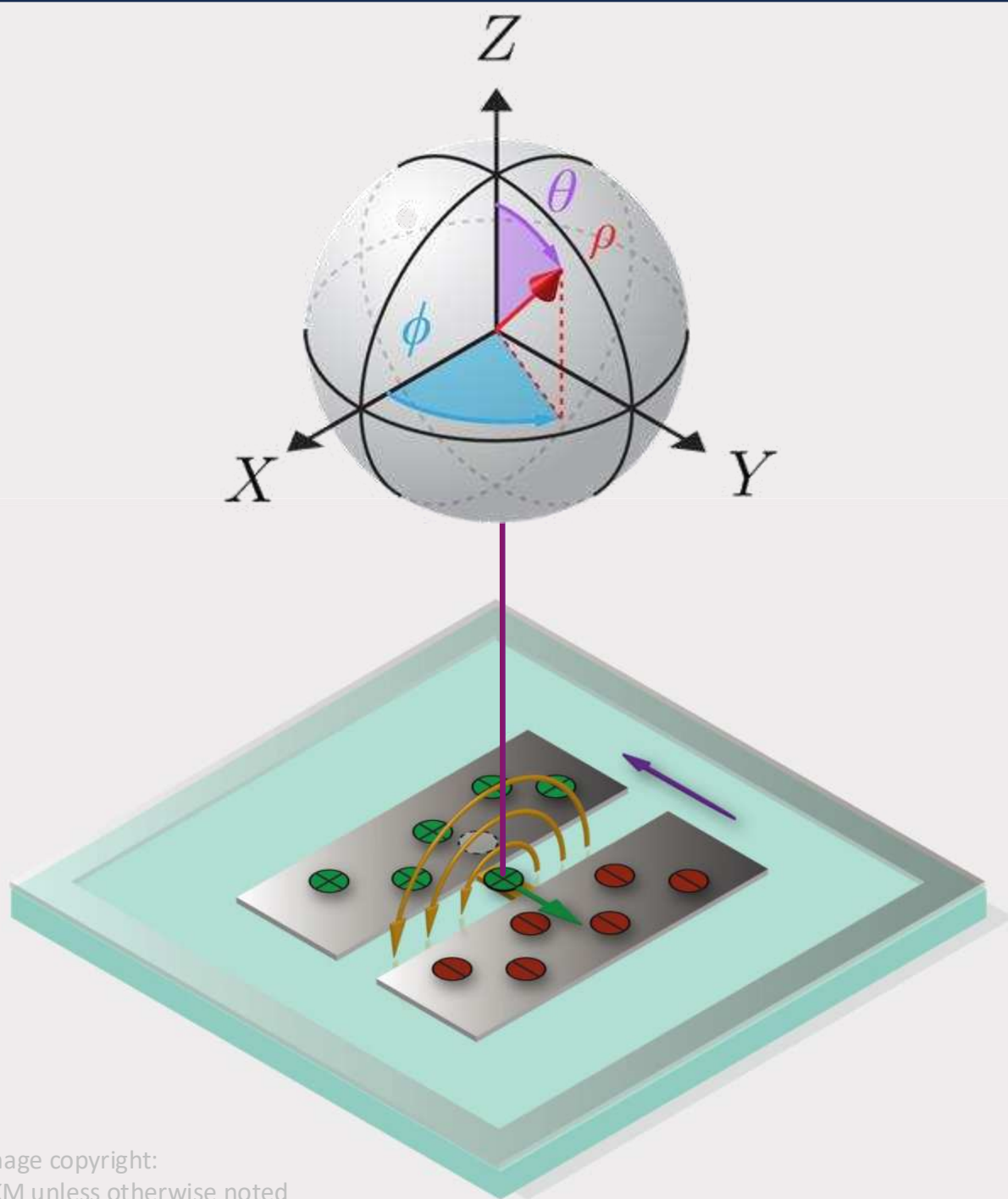
Measured vs. predicted: qubit-readout cross-Kerr



Measured vs. predicted: qubit-readout cross-Kerr



Putting it together Automation



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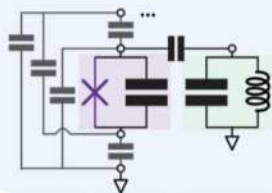


Quantization methods library

quasi-static

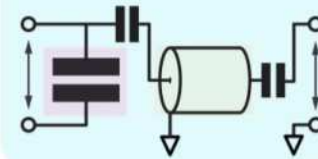
full-wave

lumped



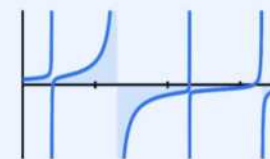
Yurke & Denker (1984), Devoret (1997), Burkard et al. (2004), Koch et al. ...

quasi-lumped



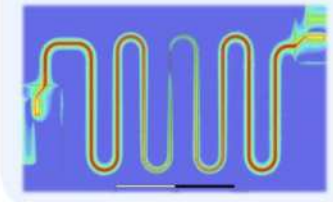
Malekakhlagh et al. (2017, 2019), Gely et al. (2019), Parra-Rodriguez et al. (2019), Minev et al. (2021), ...

impedance



Nigg et al. (2012), Bourassa et al. (2012), Solgun et al. (2014, 2015, 2017) ...

energy



Minev (2018)
Minev et al. (2020)

more speed



more information, accuracy



Zlatko Minev
IBM Quantum

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Quantization methods library: Tutorials

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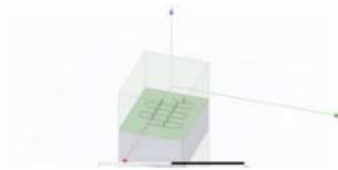
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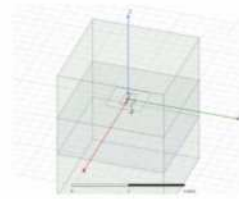
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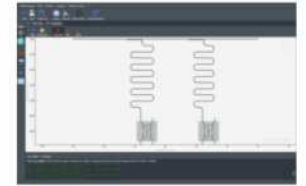
Eigenmode and EPR



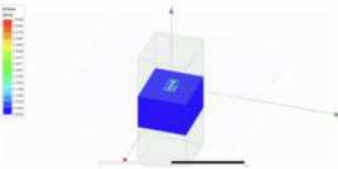
Impedance Analysis



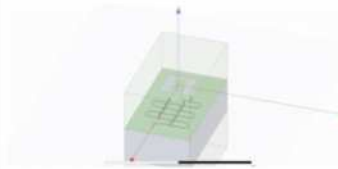
Capacitance Matrix and LOM Analysis



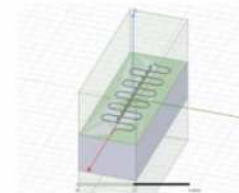
Sweep a option in QComponent, get Ansys HFSS driven-modal scatter matrix, impedance matrix, and admittance matrix.



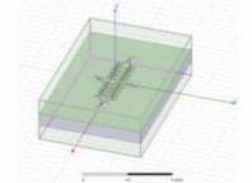
Analyzing and Tuning a Transmon Qubit



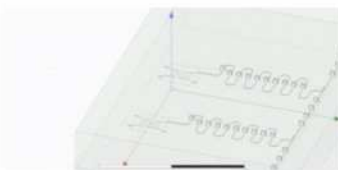
Analyzing a transmon and a resonator



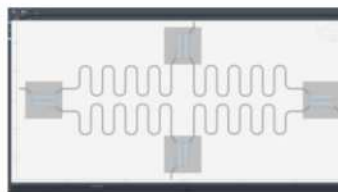
Analyzing a resonator



How to Render a Metal Design into Ansys



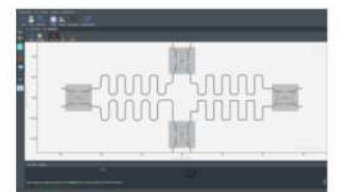
Driven Modal Simulation and S-Parameters



Sweep an option in QComponent, get capacitance matrix from Ansys Q3D and pyEPR



Sweep an option in QComponent, get Ansys HFSS eigenmode solution matrix.



How to Render a Metal Design into GDS

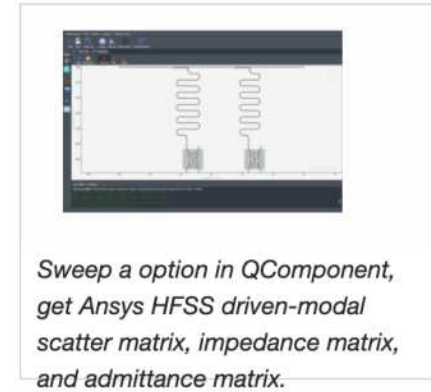
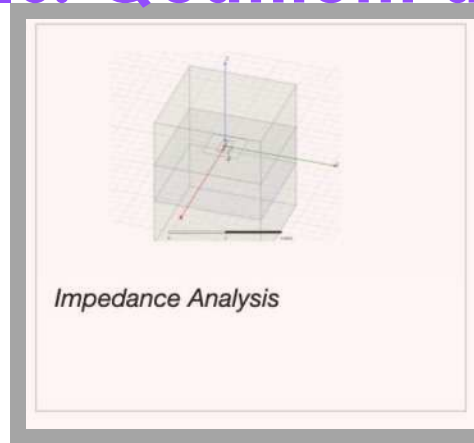
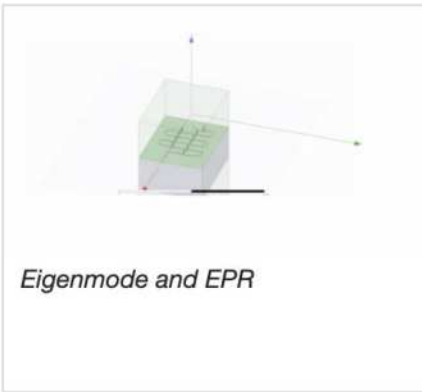
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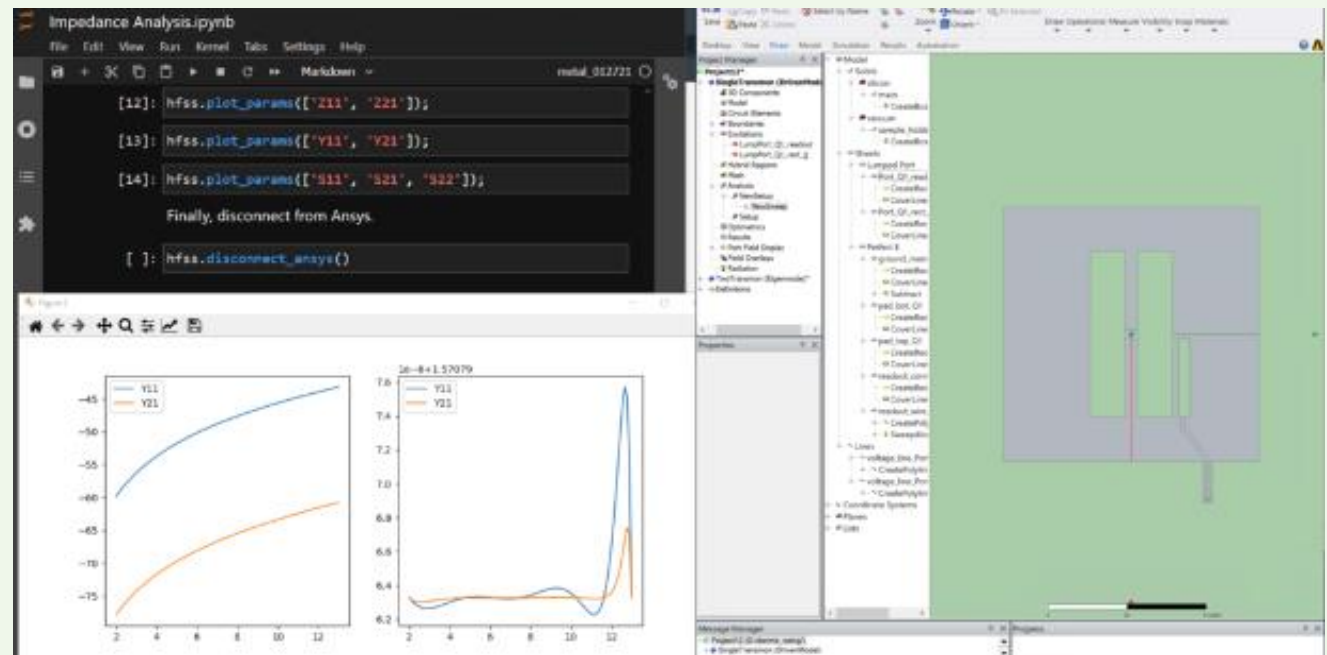
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S, Z, Y Impedance Scattering

arXiv:1204.0587 ...



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Sweep a option in QComponent,
get Ansys HFSS driven-modal
scatter matrix, impedance matrix,
and admittance matrix.

Eigenmode

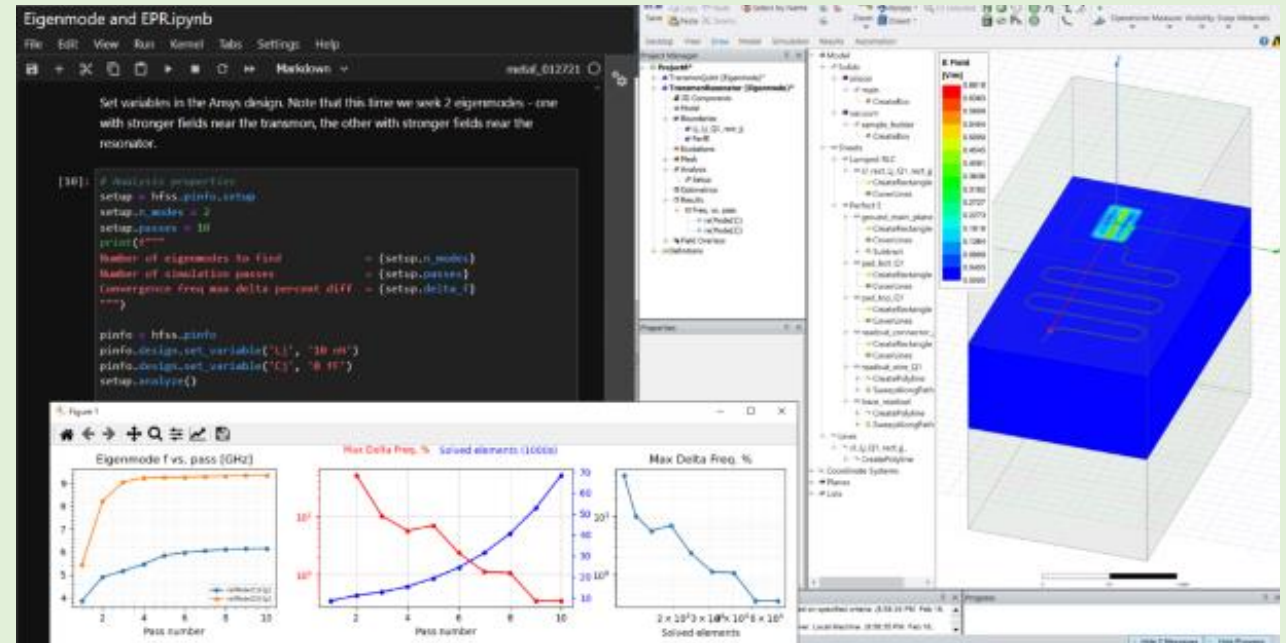
Energy-participation quantization of Josephson circuits

[Zlatko K. Minev](#) , [Zaki Leghtas](#), [Shantanu O. Mundhada](#), [Lysander Christakis](#), [Ioan M. Pop](#)
& [Michel H. Devoret](#)

[npj Quantum Information](#) **7**, Article number: 131 (2021) | [Cite this article](#)

[Minev arXiv: 1902.10355](#)

[Minev arXiv: 2010.00620](#)



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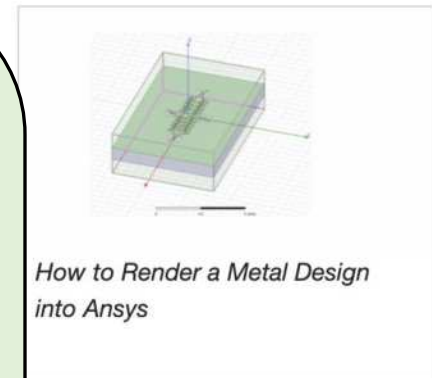
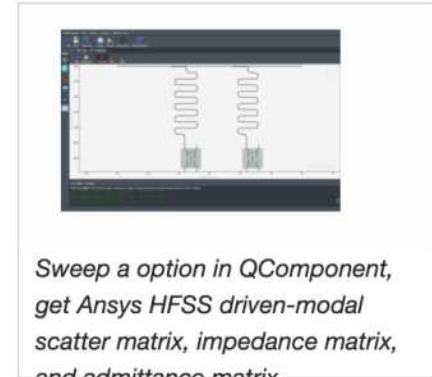
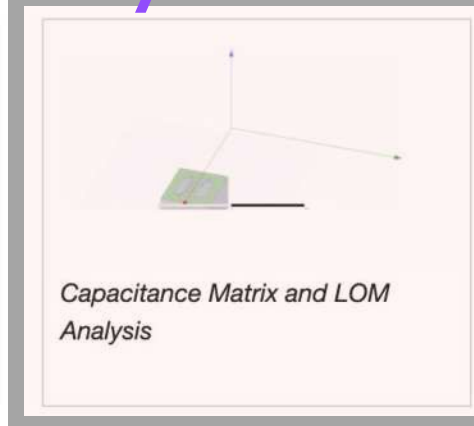
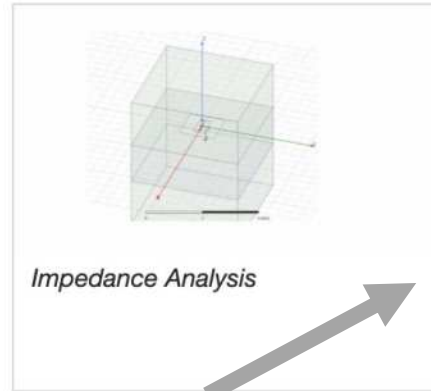
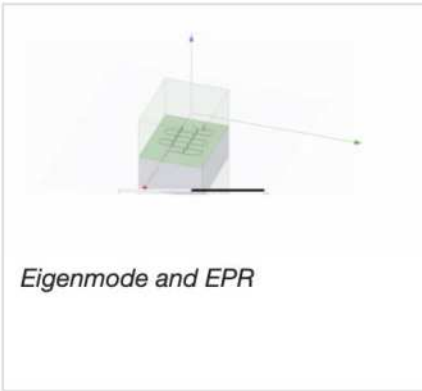
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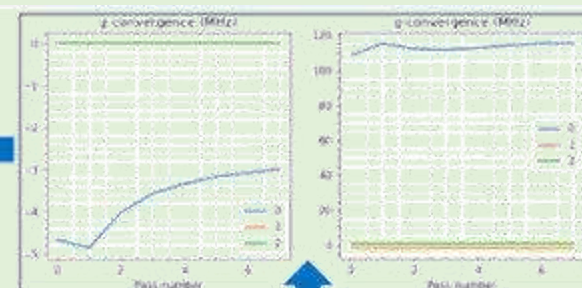
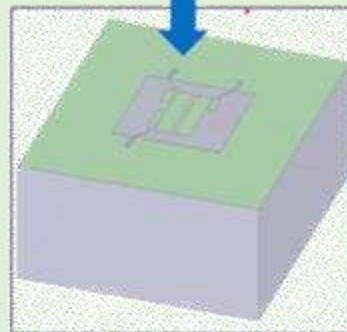
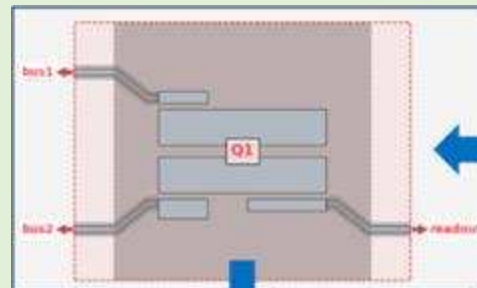


Tutorials: Quantum analysis library



Capacitive

arXiv:2103.10344 ...



```

4 Using the analysis results, get capacitance matrix as a dataframe
qld.get_capacitance_matrix()
    
```

	bus1_connector_pad_Q1	bus2_connector_pad_Q1	ground
bus1_connector_pad_Q1	47.71247	-0.38203	
bus2_connector_pad_Q1	-0.38203	51.80766	
ground_main_plane	-33.11632	-35.62303	
pad_bot_Q1	-1.34246	-12.58801	
pad_top_Q1	-12.12145	-1.59640	
readout_connector_pad_Q1	-0.17280	-0.96276	

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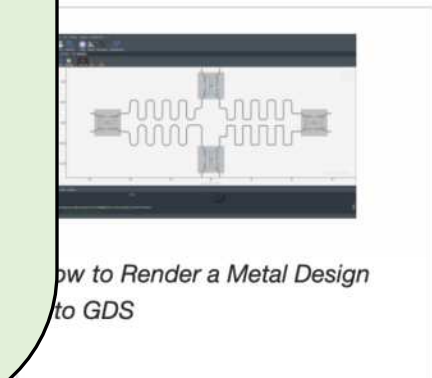
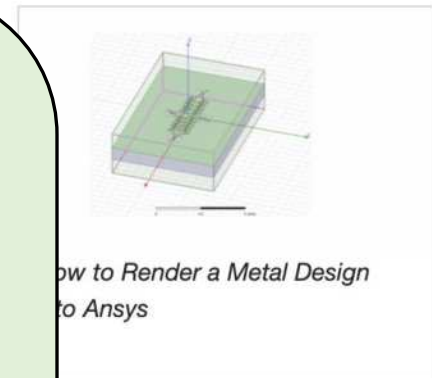
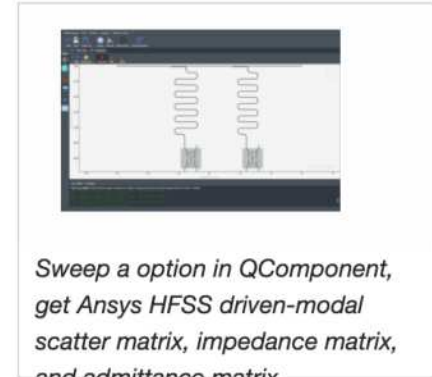
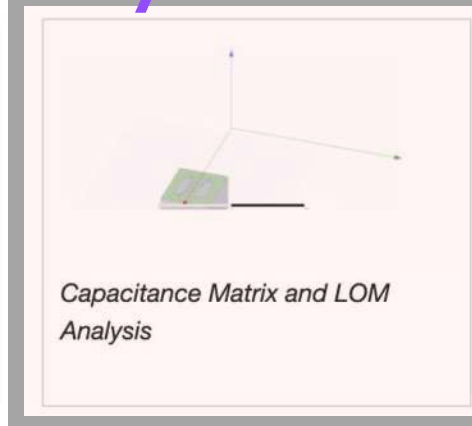
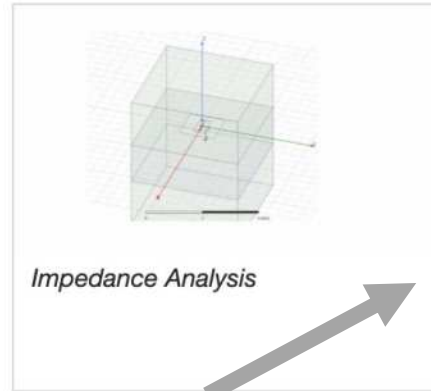
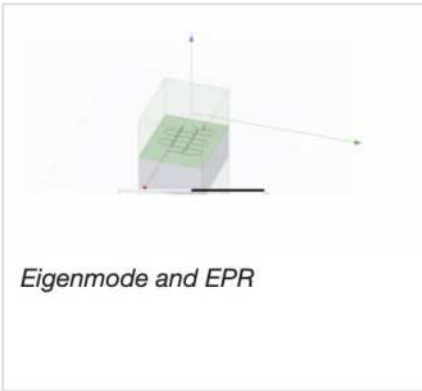
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Cornell University

arXiv.org > quant-ph > arXiv:2103.10344

Quantum Physics

Circuit quantum electrodynamics (cQED) with modular quasi-lumped models

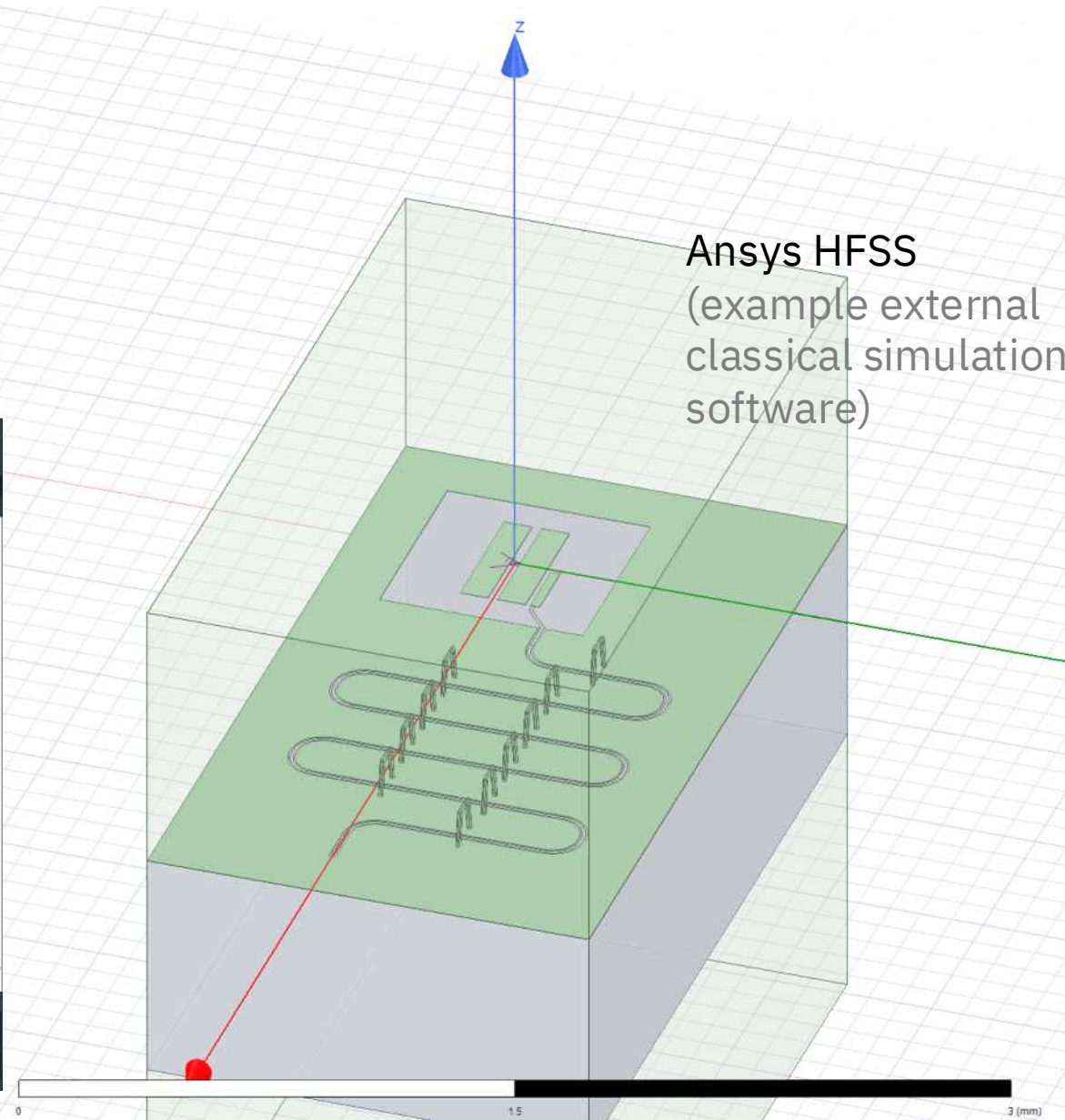
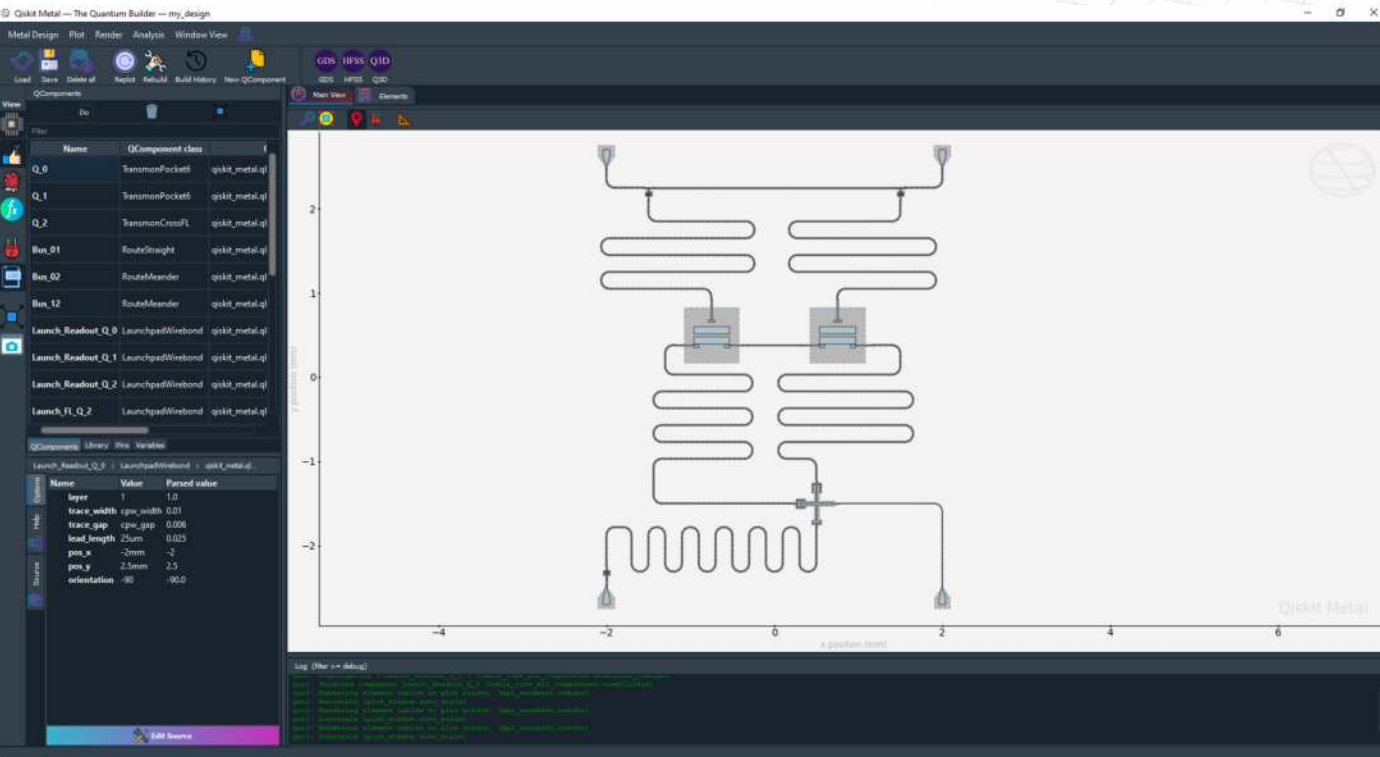
Zlatko K. Minev,* Thomas G. McConkey, Maika Takita, Antonio Corcoles, and Jay M. Gambetta
IBM Quantum, IBM T.J. Watson Research Center, Yorktown Heights, US

WIP: General capacitive analysis code
map to known building blocks: e.g.,
transmon, fluxonium, zero-pi, etc.

Automated with
Qiskit | quantum device
design

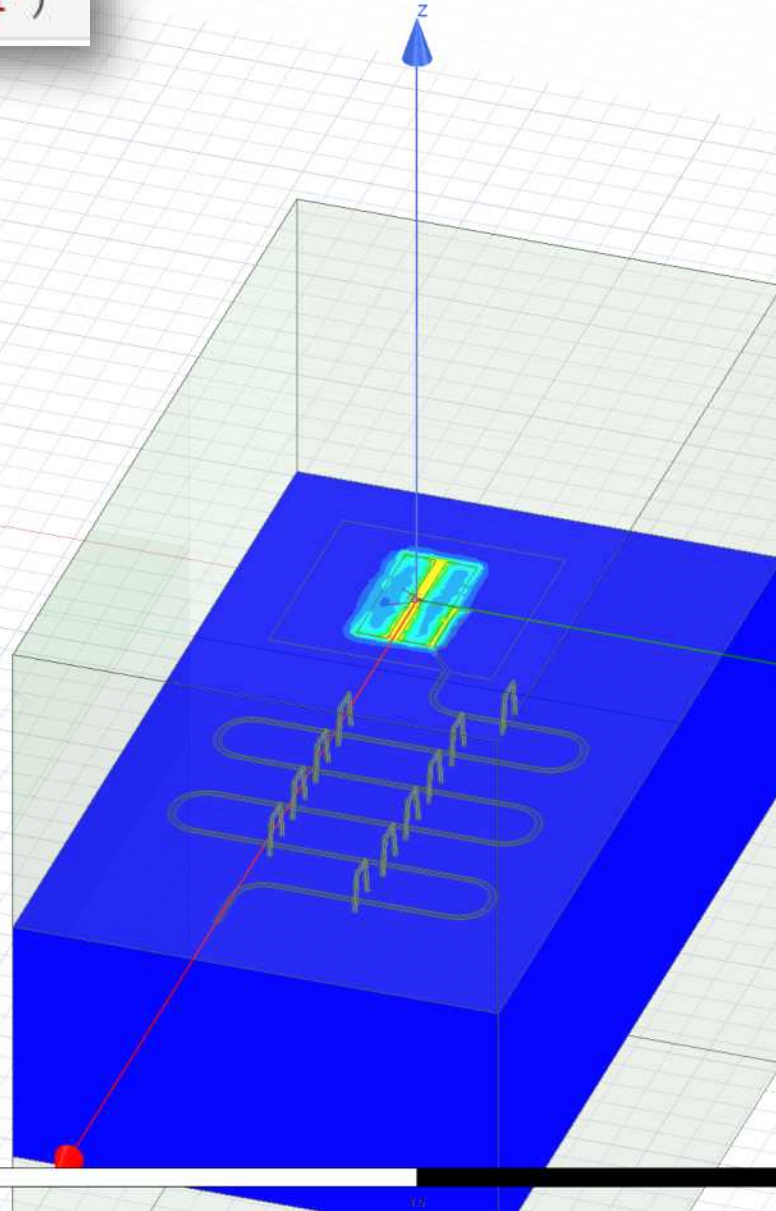
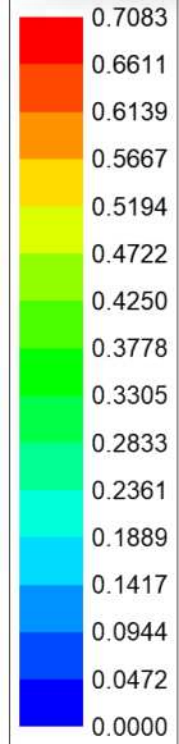


Ansyz HFSS
(example external
classical simulation
software)



```
metal.analysis.lumped_model.analyze('Q1')
```

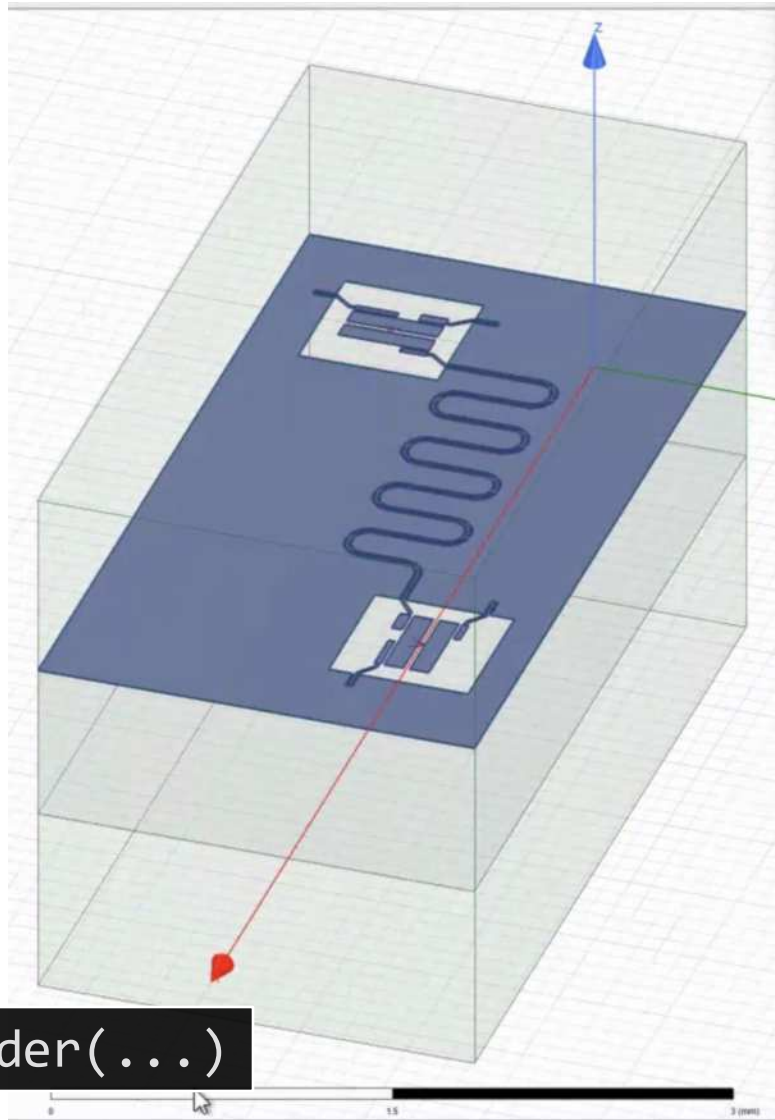
[V/m]



Ansyz HFSS

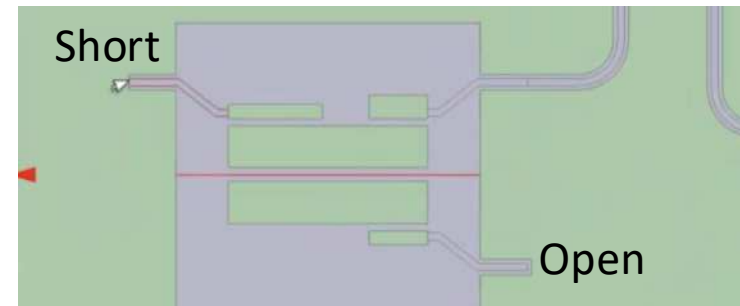
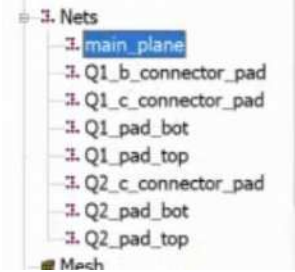
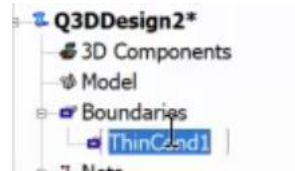
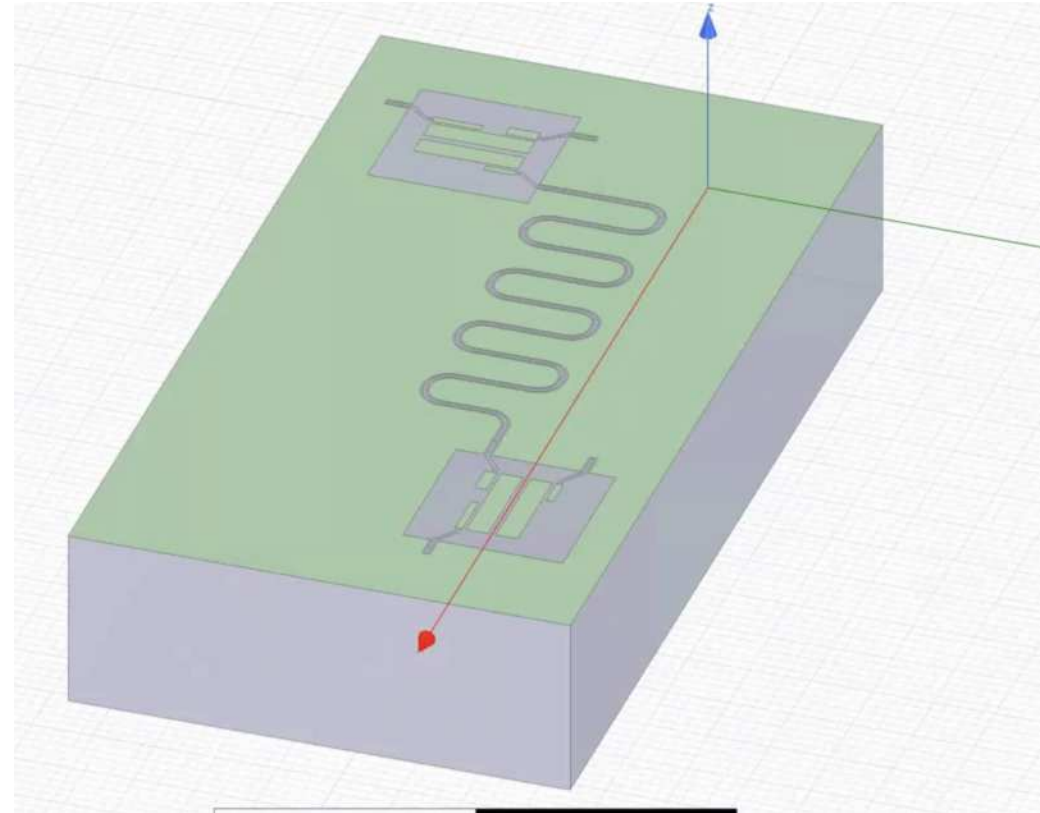
Qiskit Metal render & electromagnetic analysis

For example: Ansys HFSS

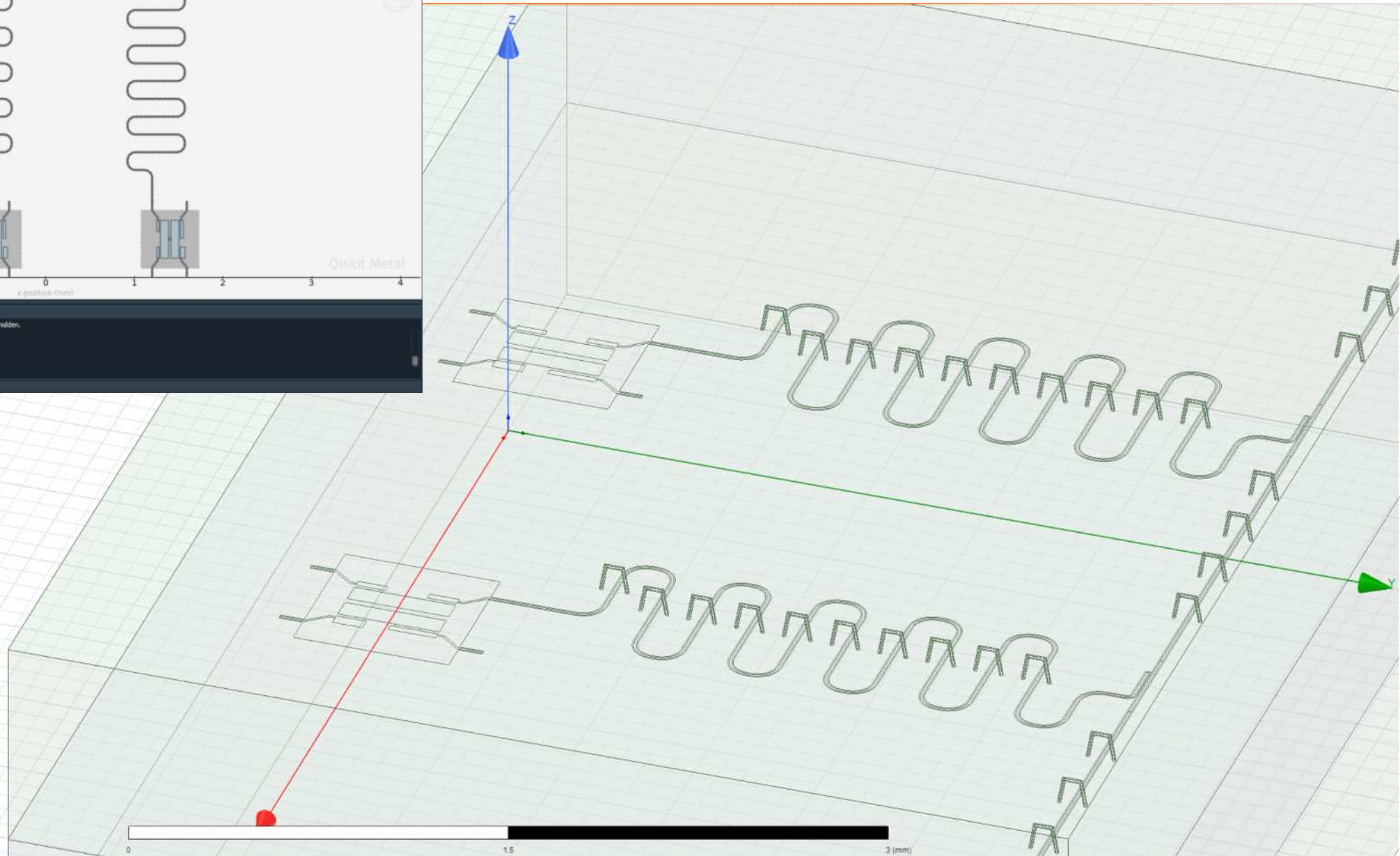
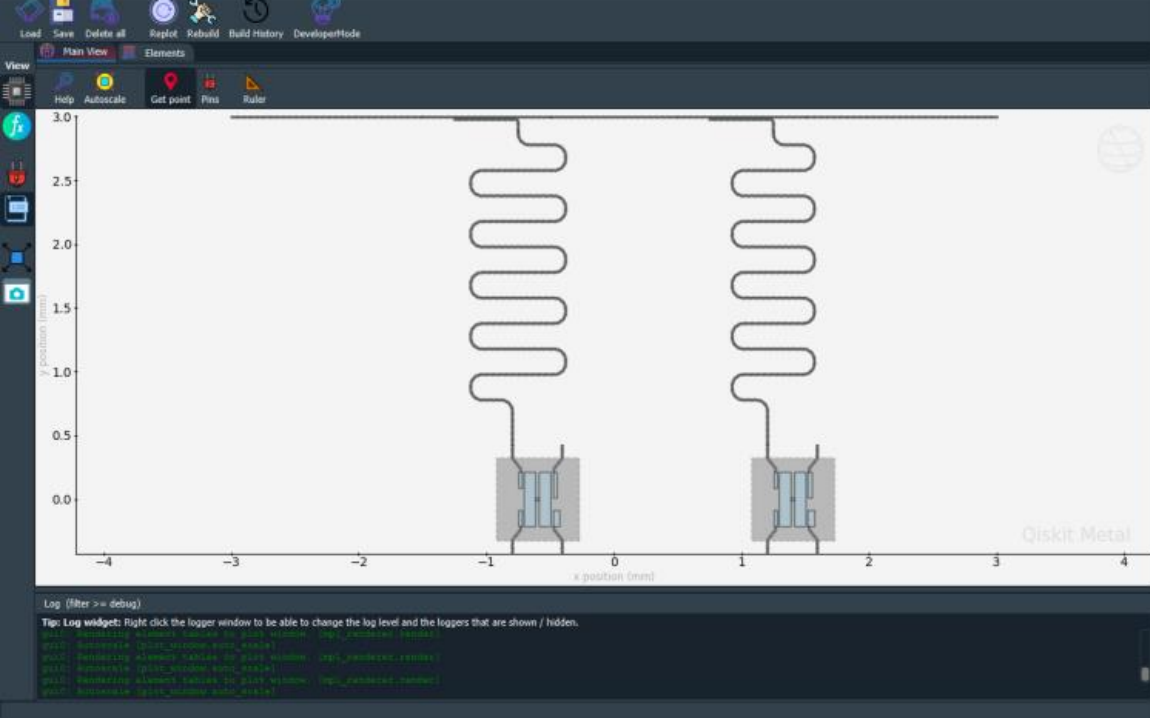


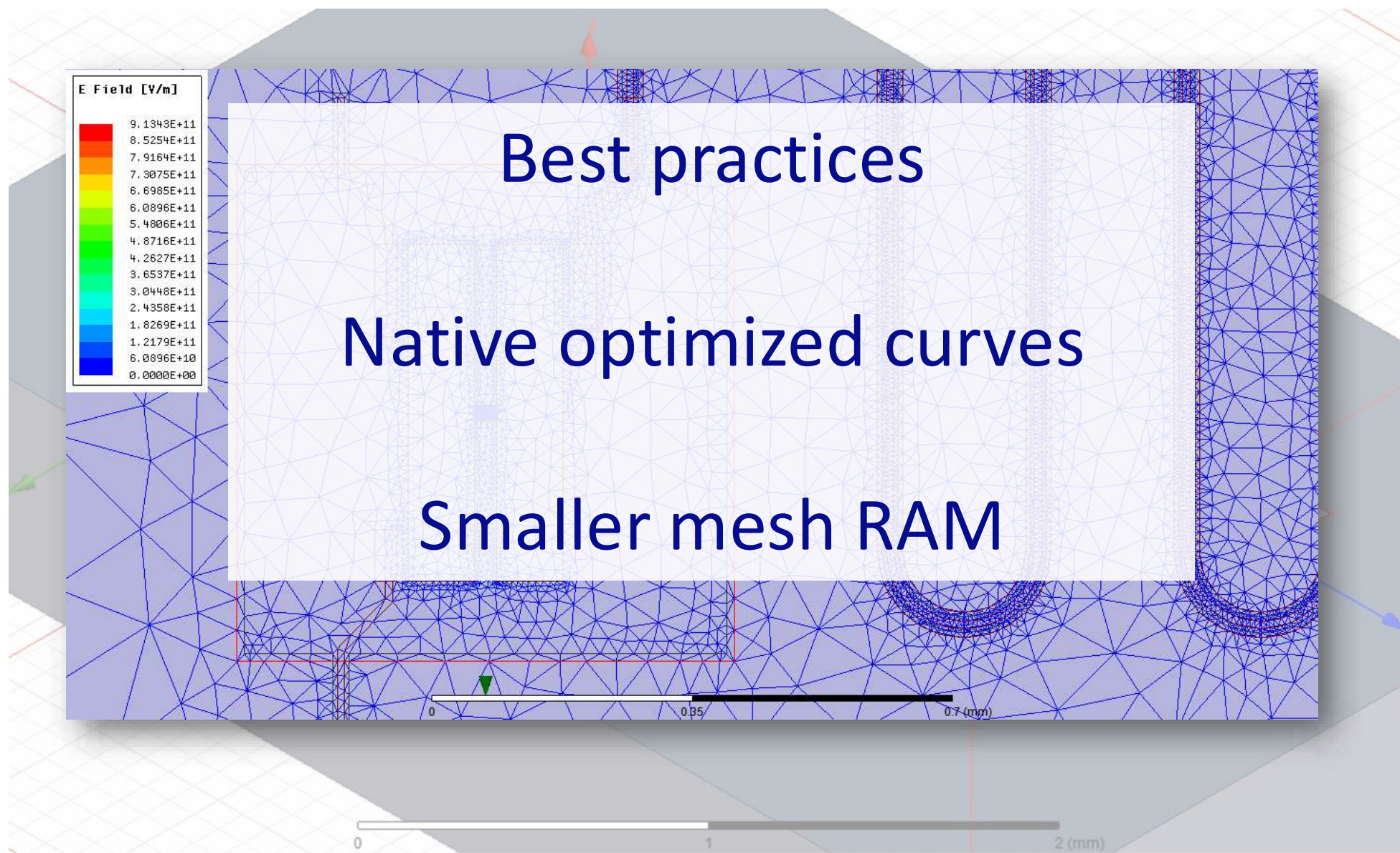
```
design.render(...)
```

Q3D

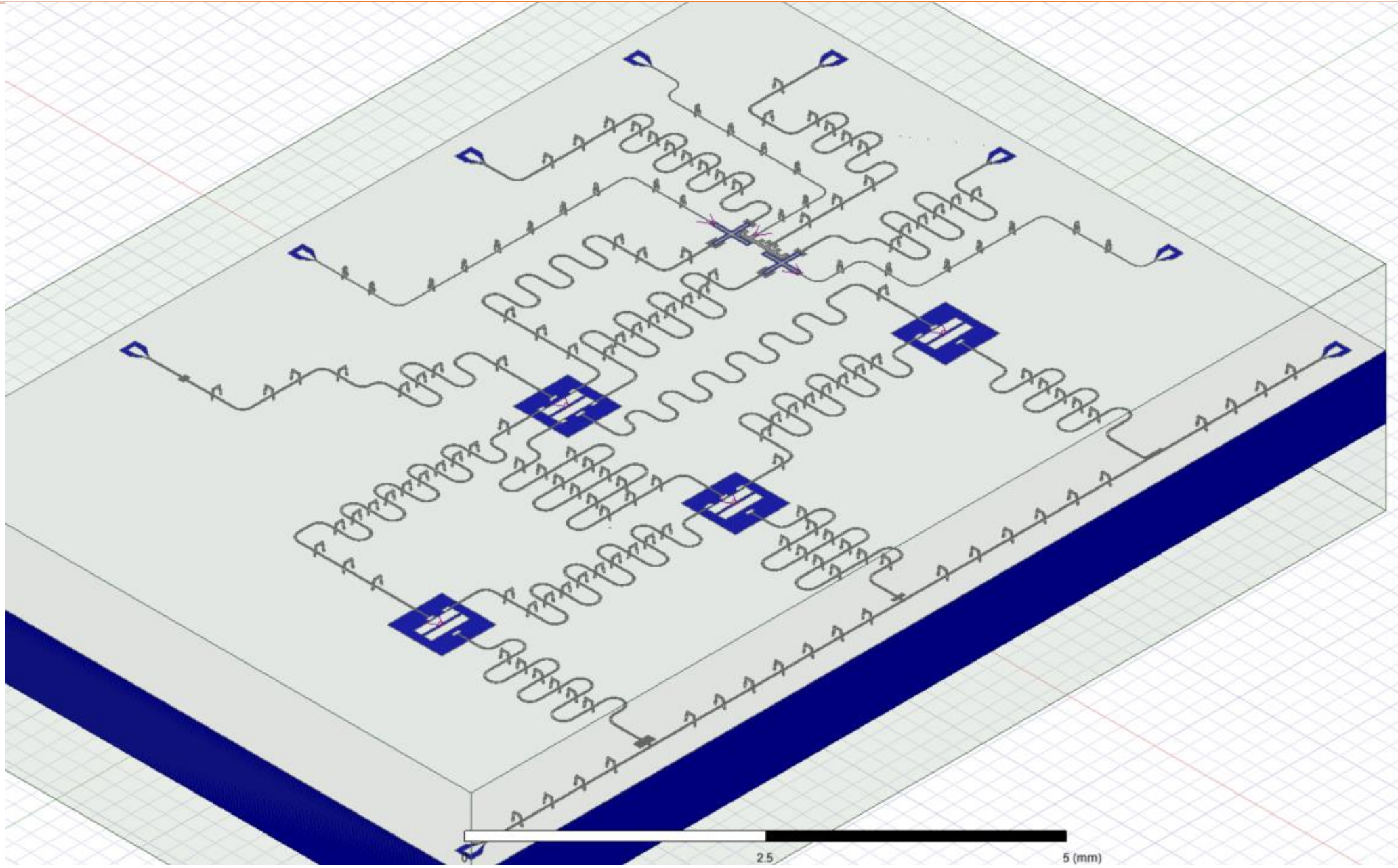


More complex designs

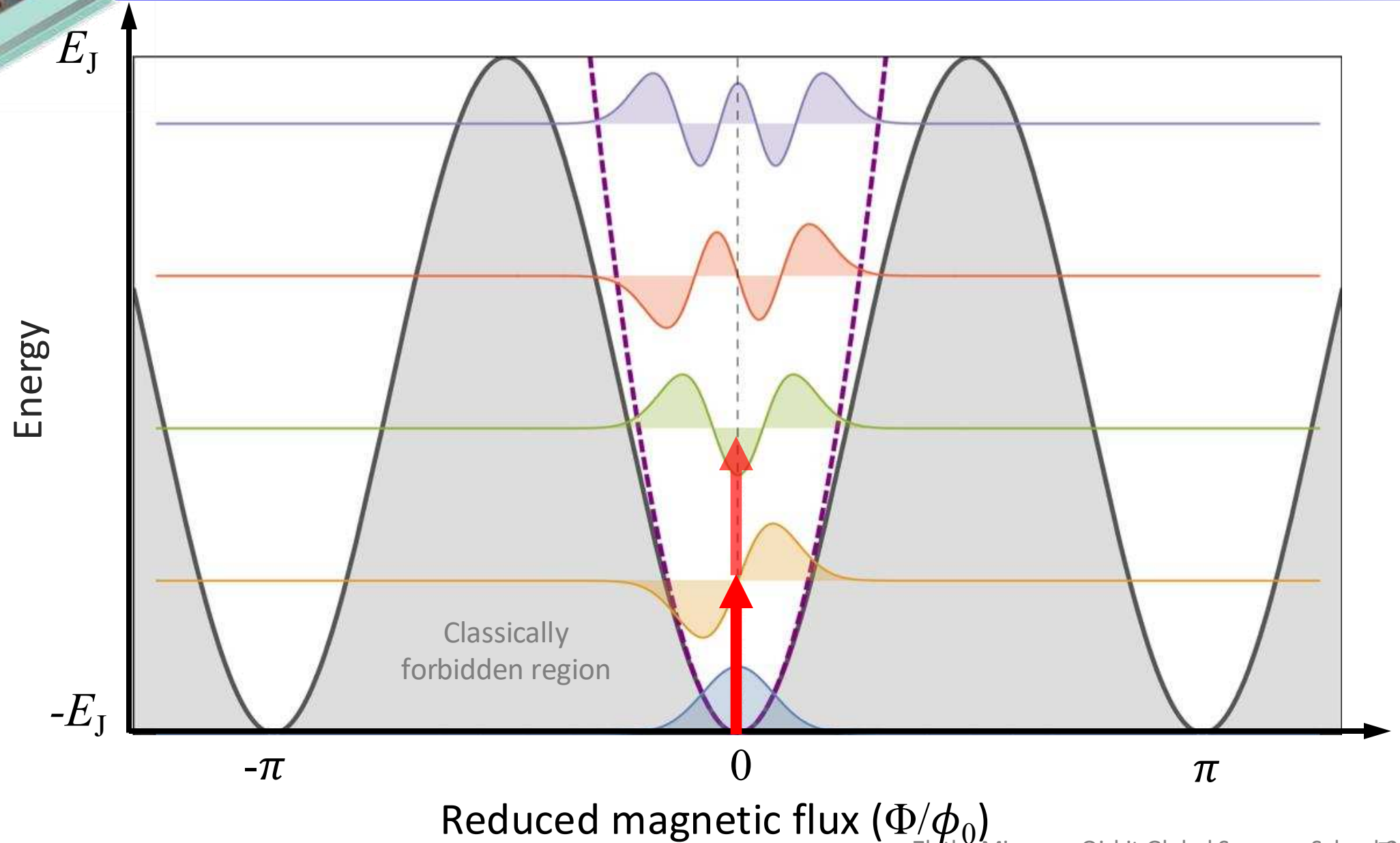




More complex quantum chips – see next tutorial

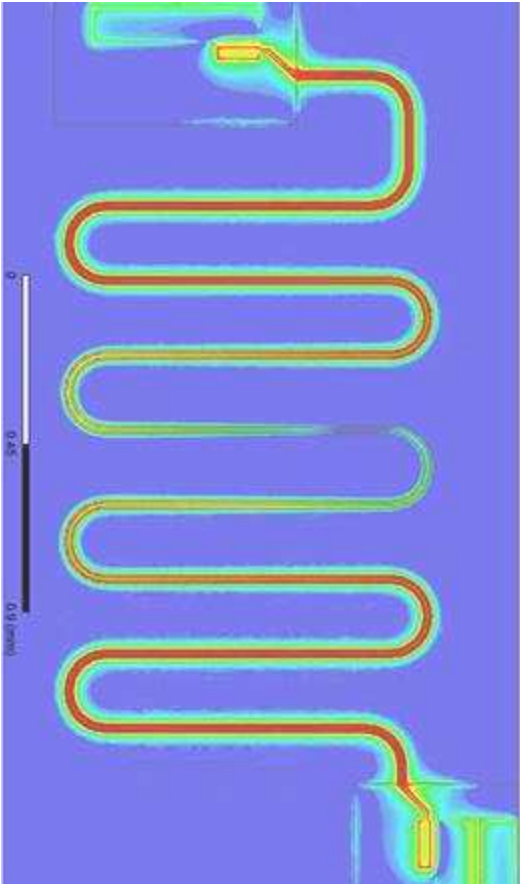


Exploring a transmon qubit

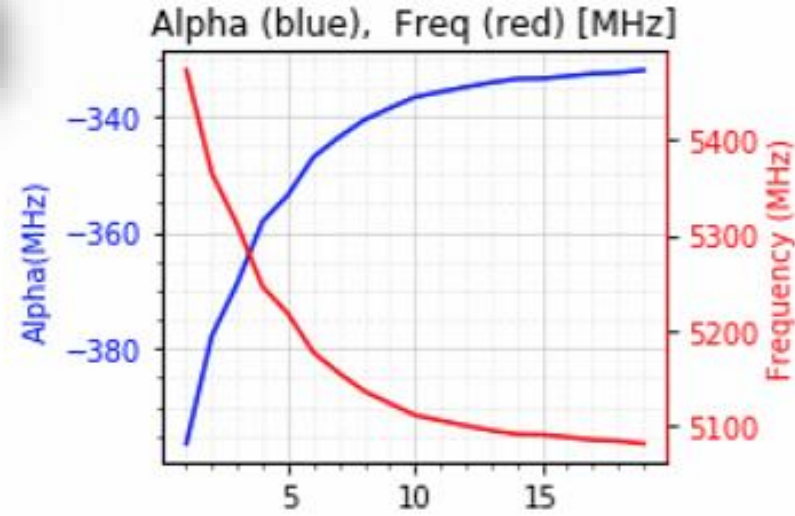


Automated analysis and reports

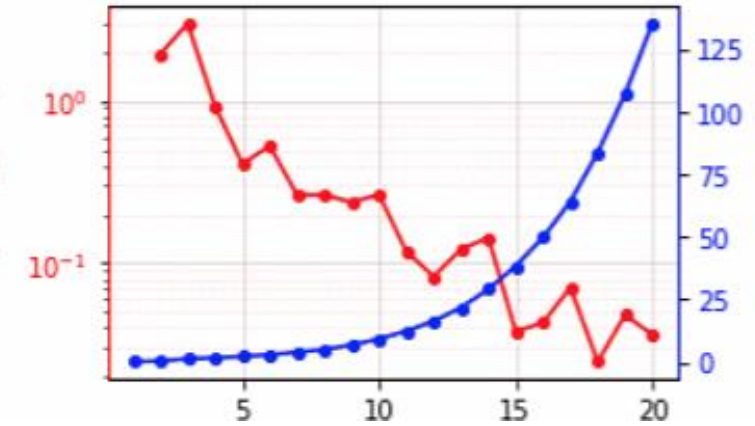
```
metal.analysis.lumped_model.analyze('Q1')
```



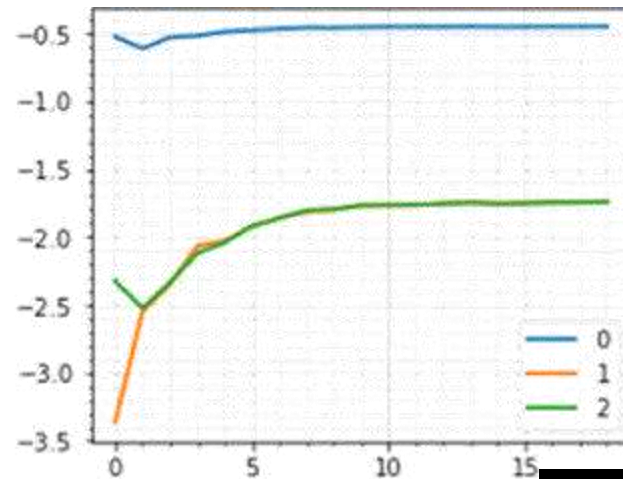
Qubit frequency & anharmonicity



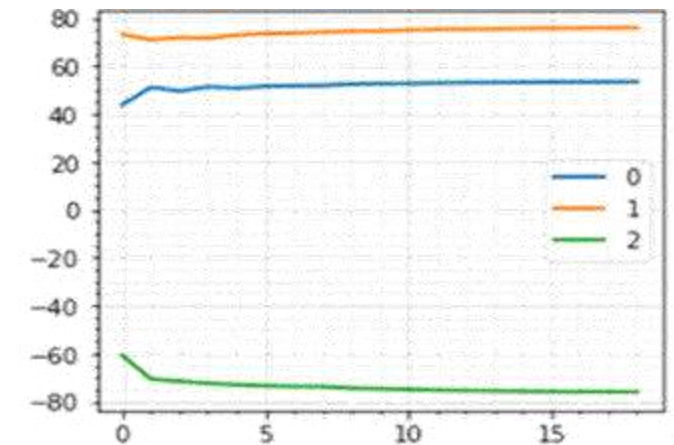
FE simulation convergence



cross-Kerr χ coupling (MHz)

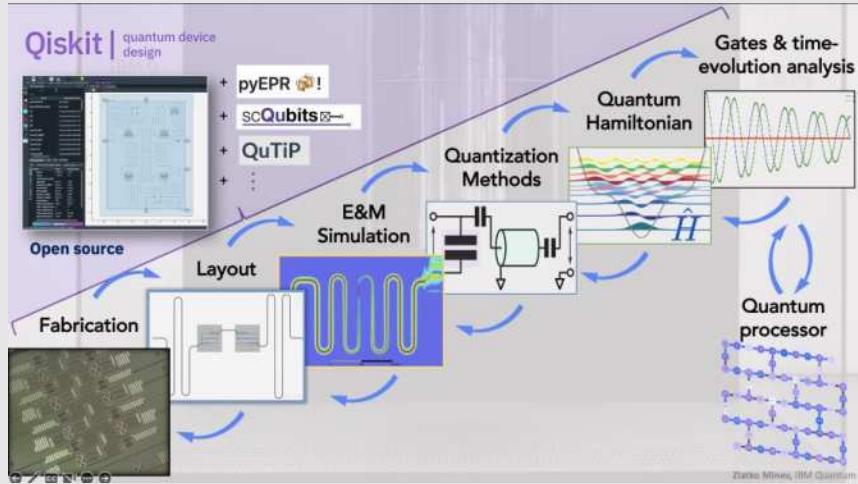


Linear g coupling (MHz)



Analysis pass number

Building it together: Open Source Community



Open source

Call for community participation!
Contact us!



Sadman Ahmed
Shanto



Eli Levenson-Falk



Jens Koch



T.C. Thann



B. Joecker



C. Warran

You?

...

Lab work with Dr. Murat Sarihan, Sadman Shanto, and more!

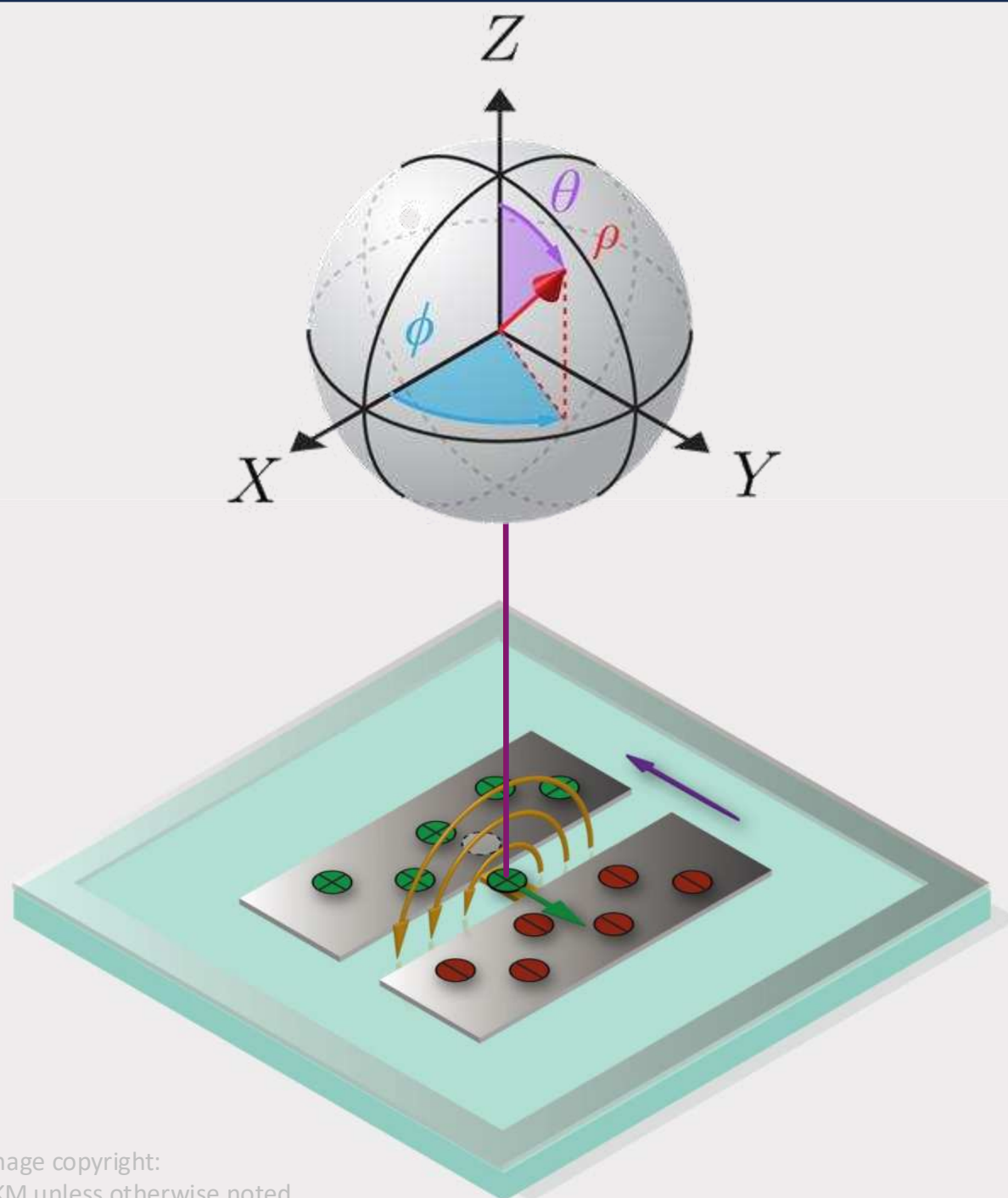


@zlatko_minev



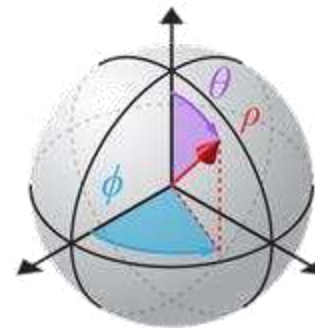
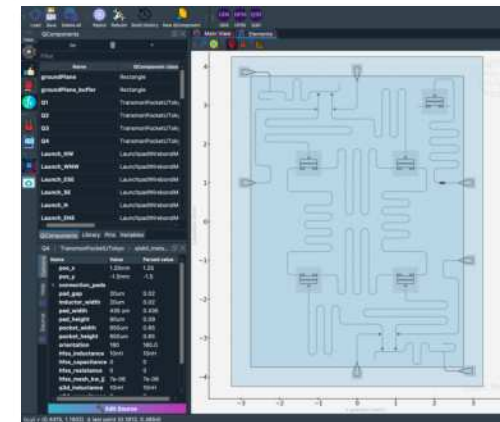
zlatko-minev.com/blog

Zlatko Minev



Overview of Quantum Hardware Design

Energy, Circuits, and Metal



Slides!

Zlatko K. Minev

Current Affiliation:

Google Quantum AI; Assoc. Fellow @ CIFAR

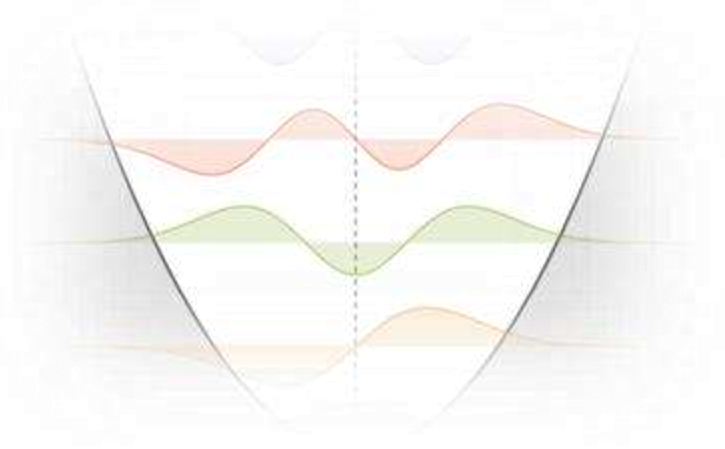
Former Affiliation: IBM Quantum; Yale University



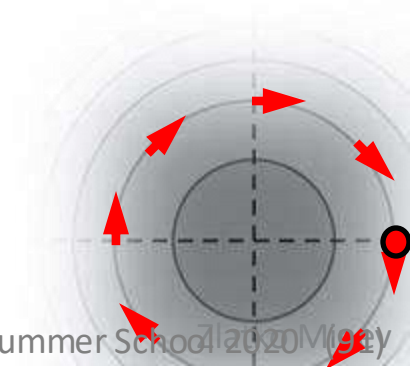
@zlatko_minev



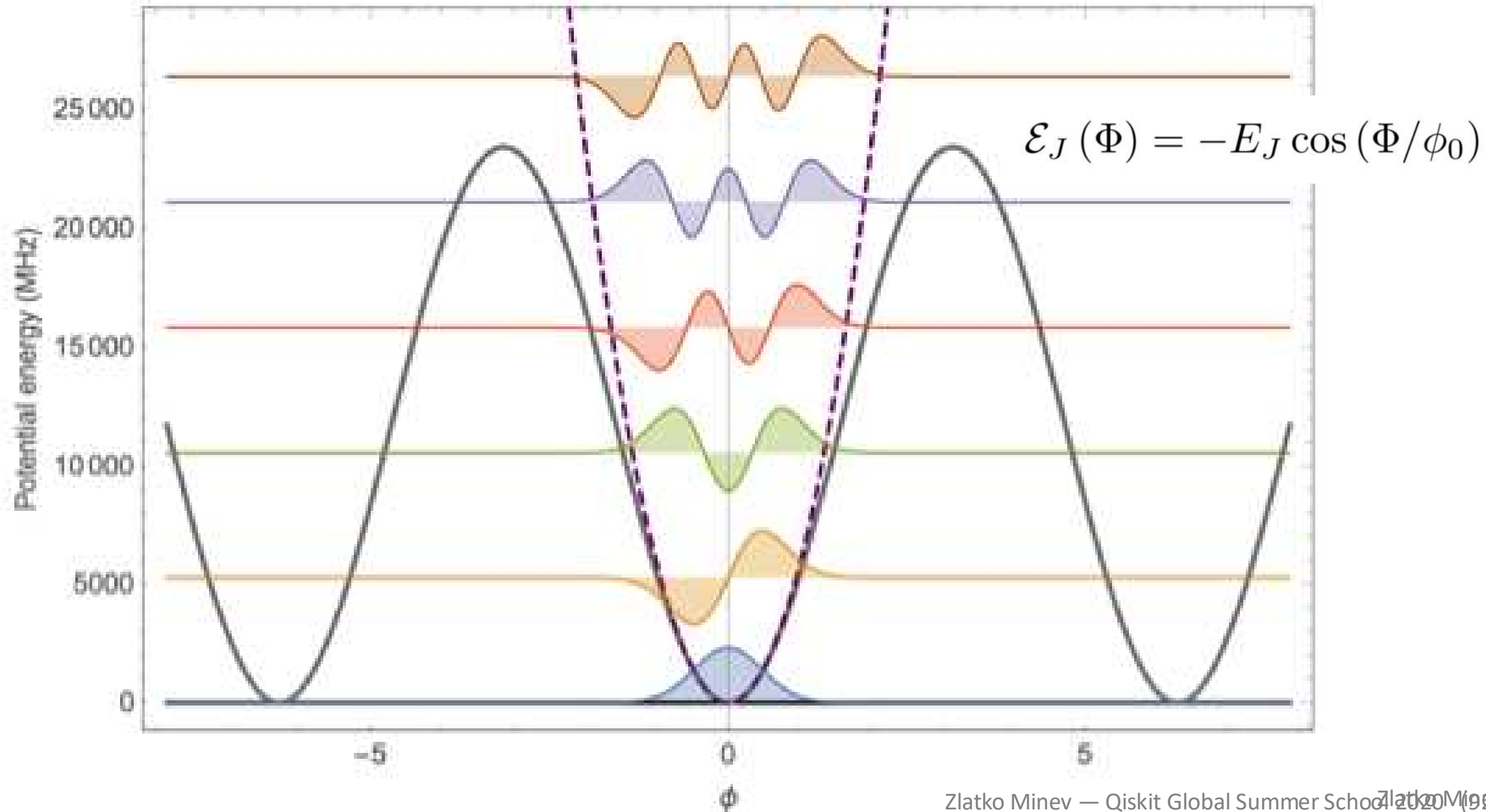
zlatko-minev.com/blog



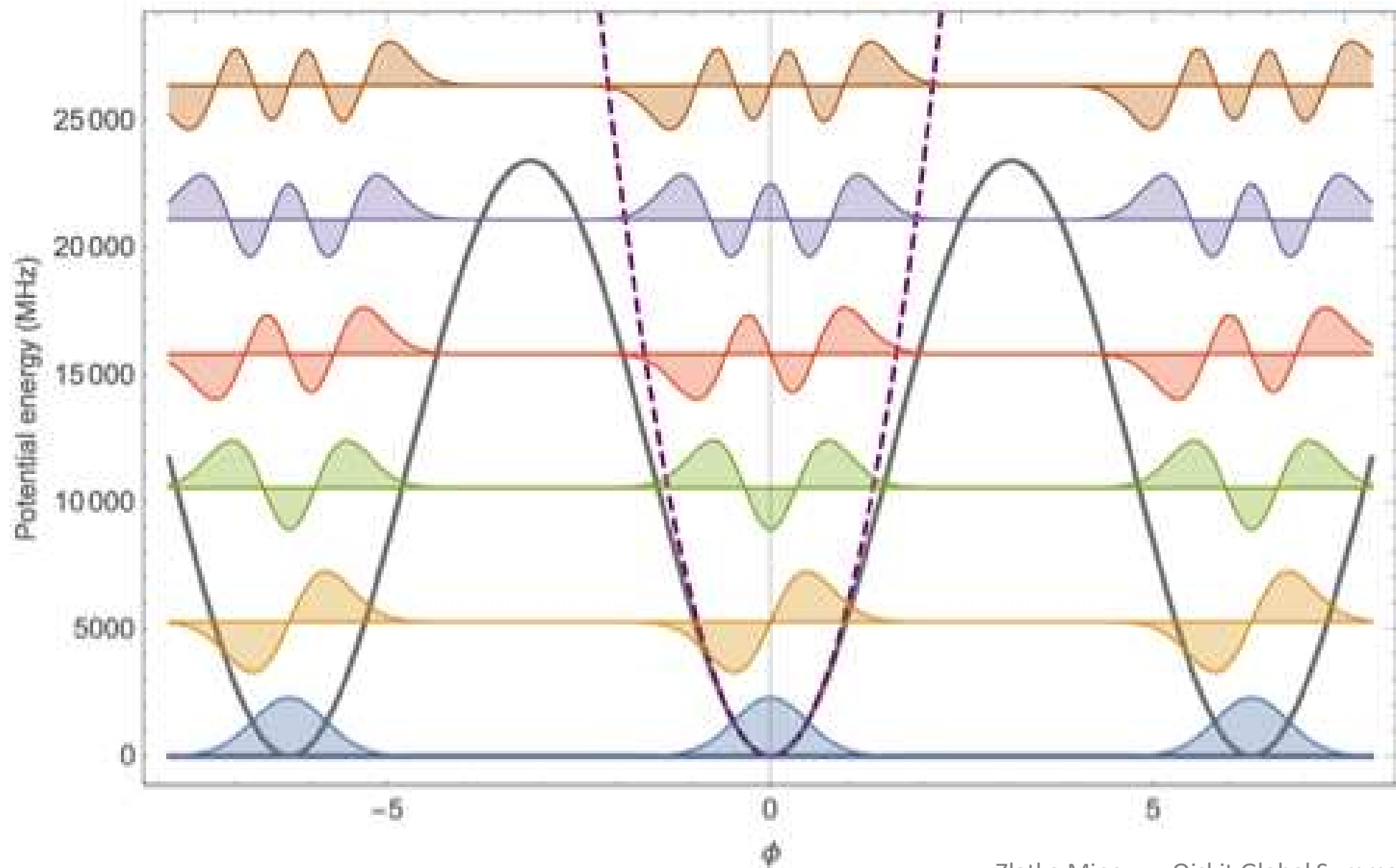
Is the junction phase/flux
compact or not?



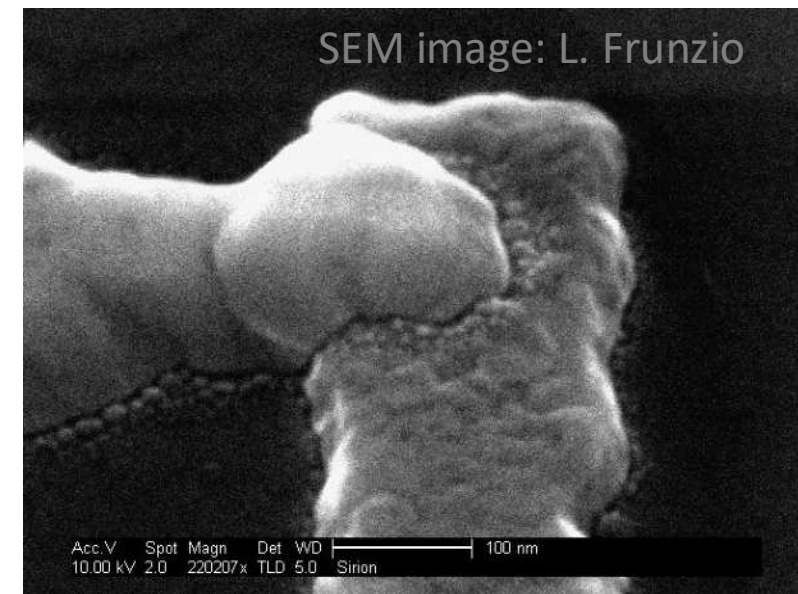
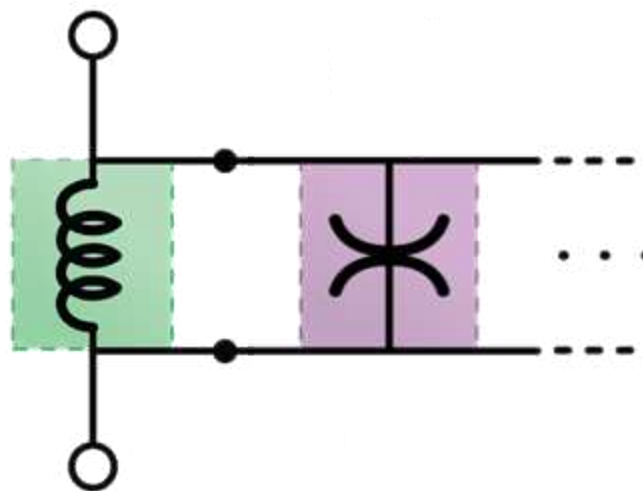
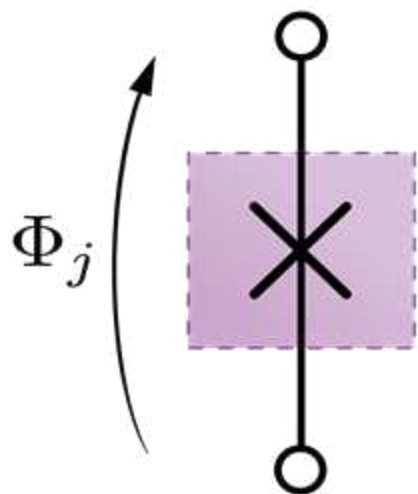
Harmonic Approximation



Tight-binding model



Simple non-linear dipole: Josephson junction



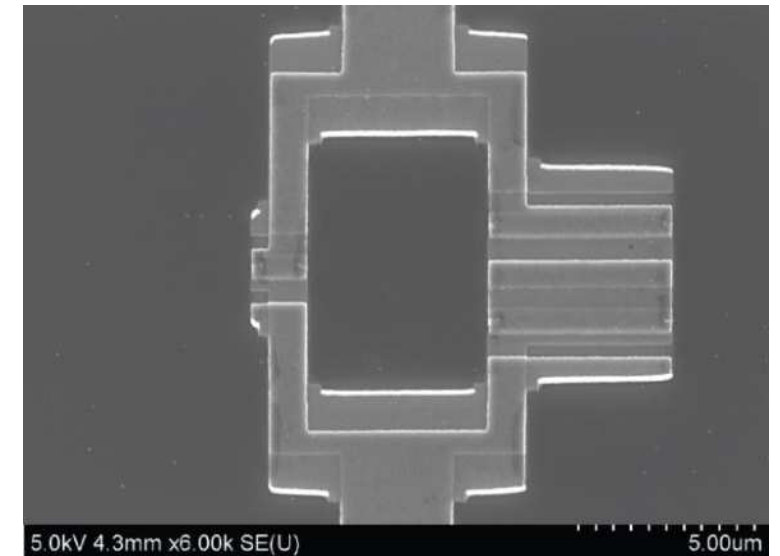
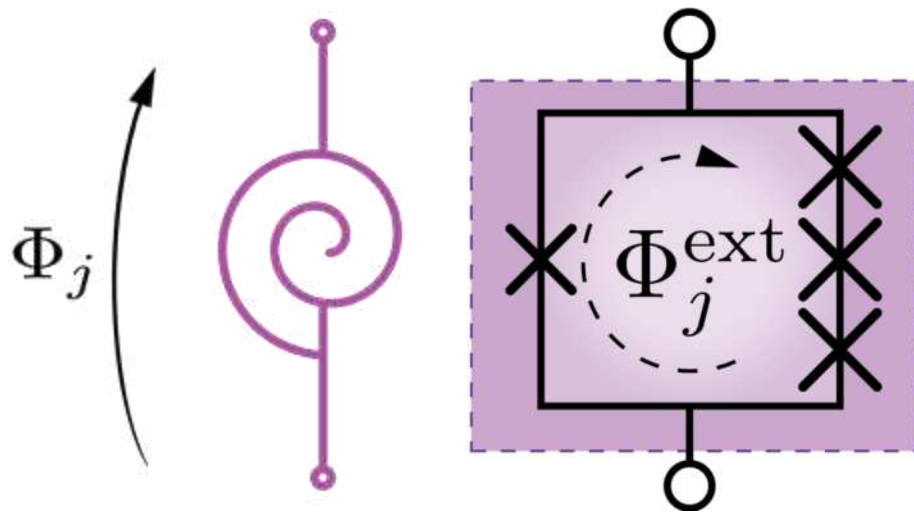
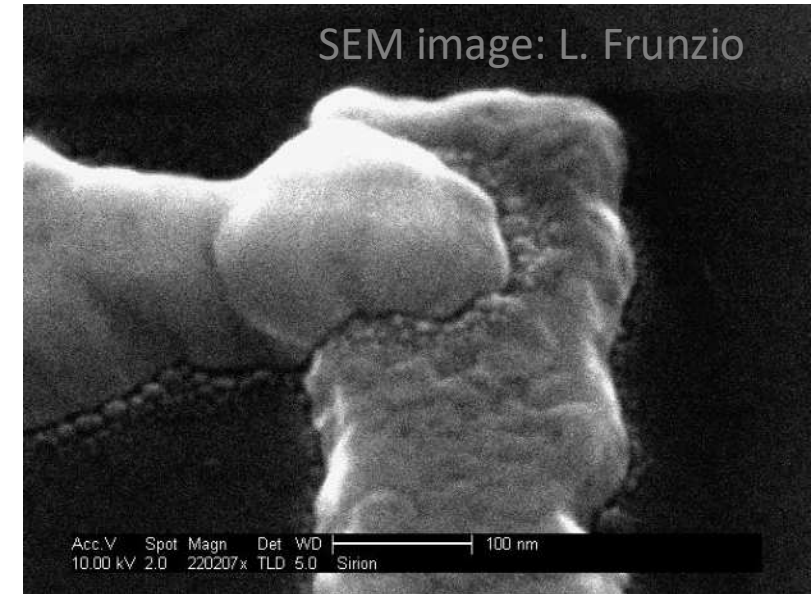
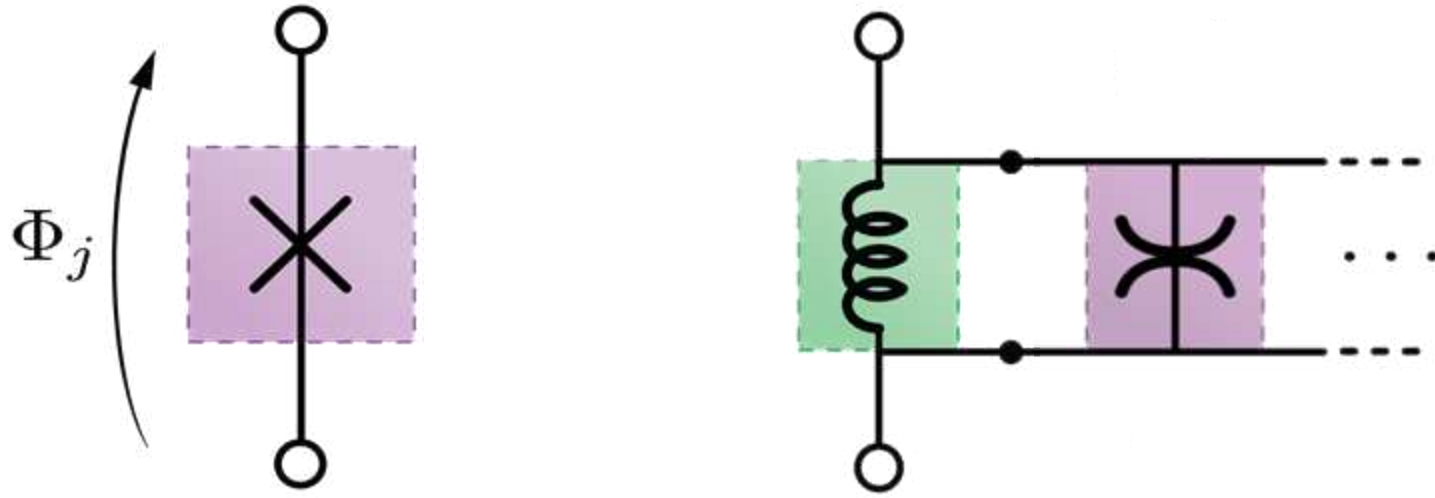
$$\mathcal{E}_j(\Phi_j) = E_j (1 - \cos(\Phi_j/\phi_0))$$

$$= \mathcal{E}_j^{\text{lin}}(\Phi_j) + \mathcal{E}_j^{\text{nl}}(\Phi_j)$$

$$= \frac{E_j}{2} \left(\frac{\Phi_j}{\phi_0} \right)^2 - \frac{E_j}{4!} \left(\frac{\Phi_j}{\phi_0} \right)^4 + \mathcal{O}(\Phi_j^6)$$

$$\phi_0 \equiv \hbar/2e$$

Composite-circuit non-linear dipole

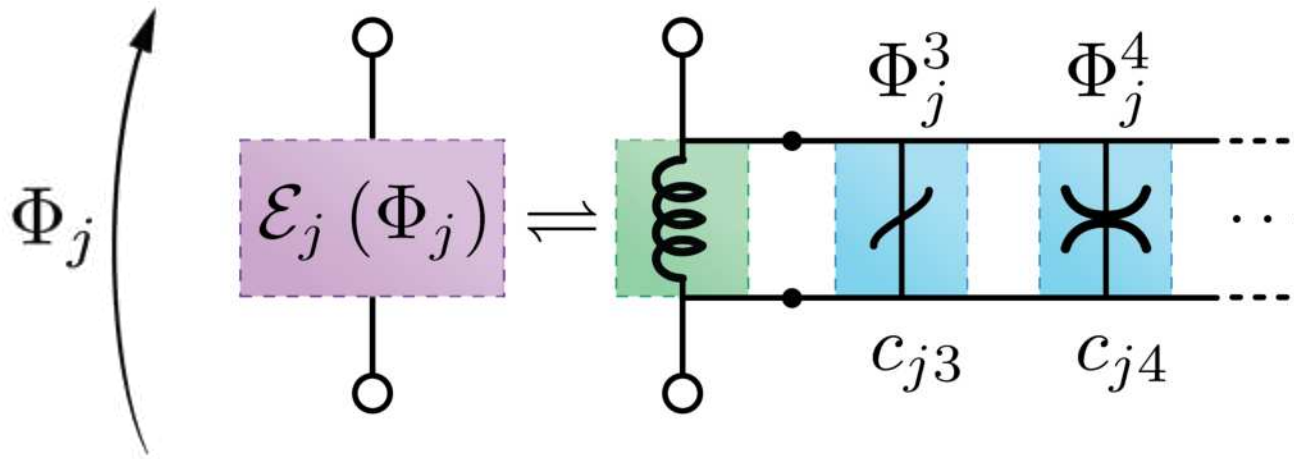


SEM image: Frattini & Sivak; see [APL \(2017\)](#)

General non-linear dipole

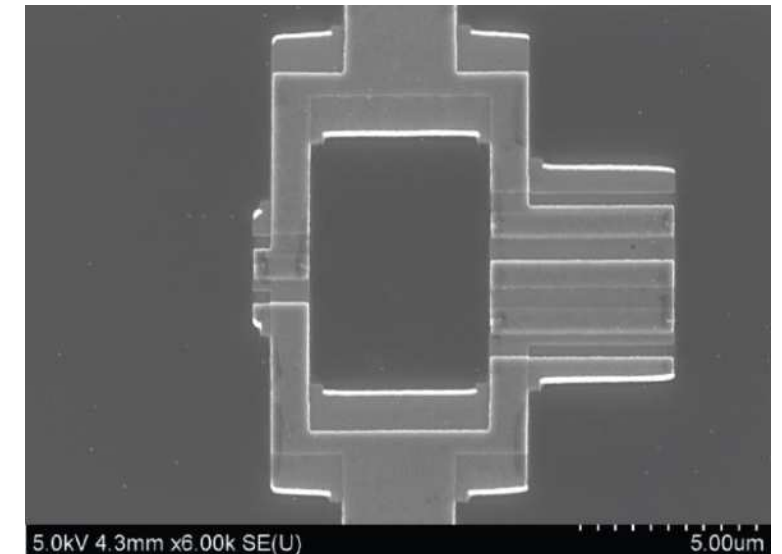
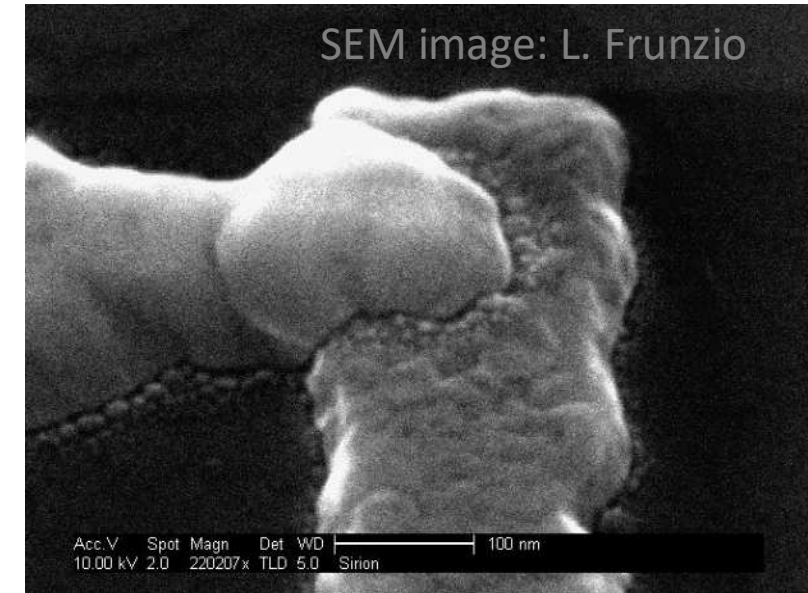
Energy function

$$\mathcal{E}_j(\Phi_j; \Phi_j^{\text{ext}}) = \mathcal{E}_j^{\text{lin}}(\Phi_j) + \mathcal{E}_j^{\text{nl}}(\Phi_j)$$



$$E_j \sum_{p=3}^{\infty} c_{jp} \left(\frac{\Phi_j}{\phi_0} \right)^p \quad (\text{expansion not necessary})$$

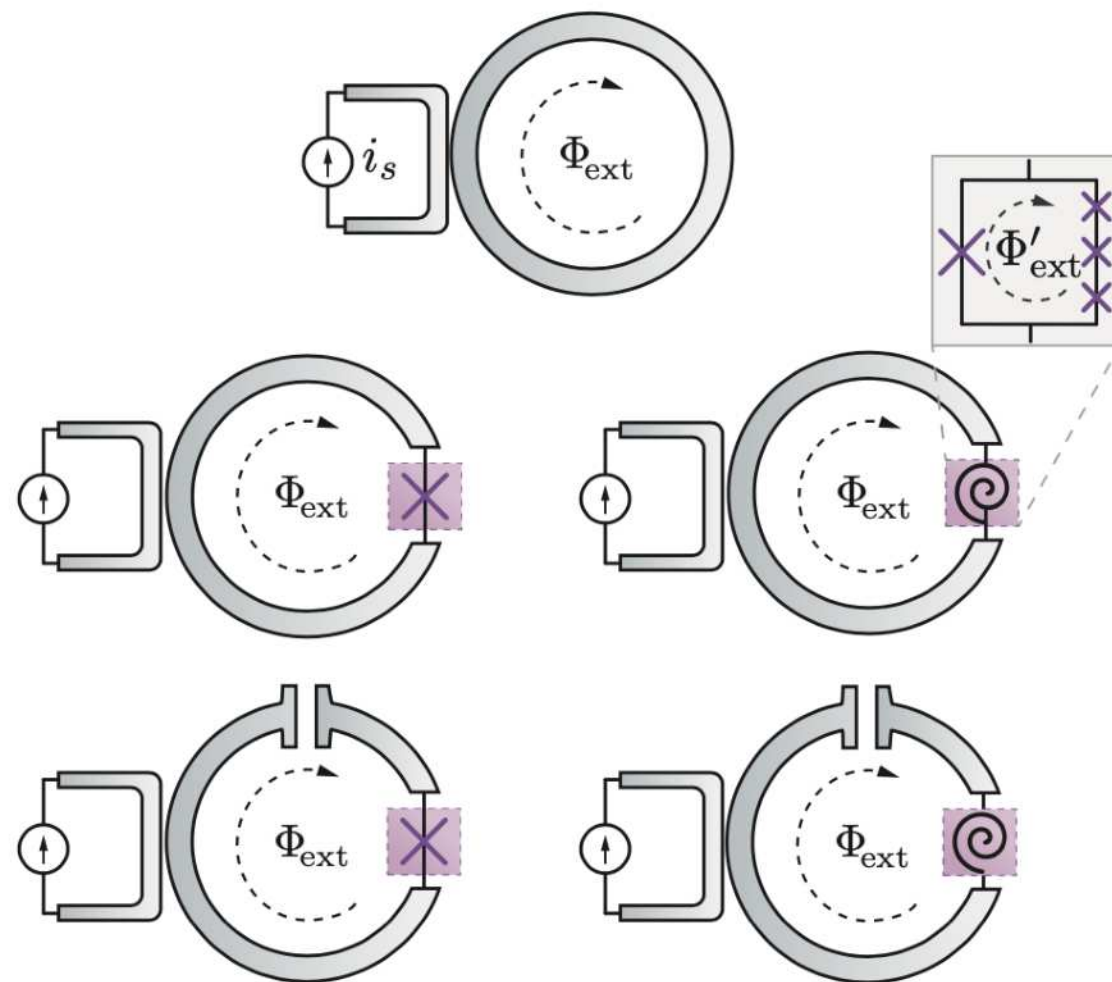
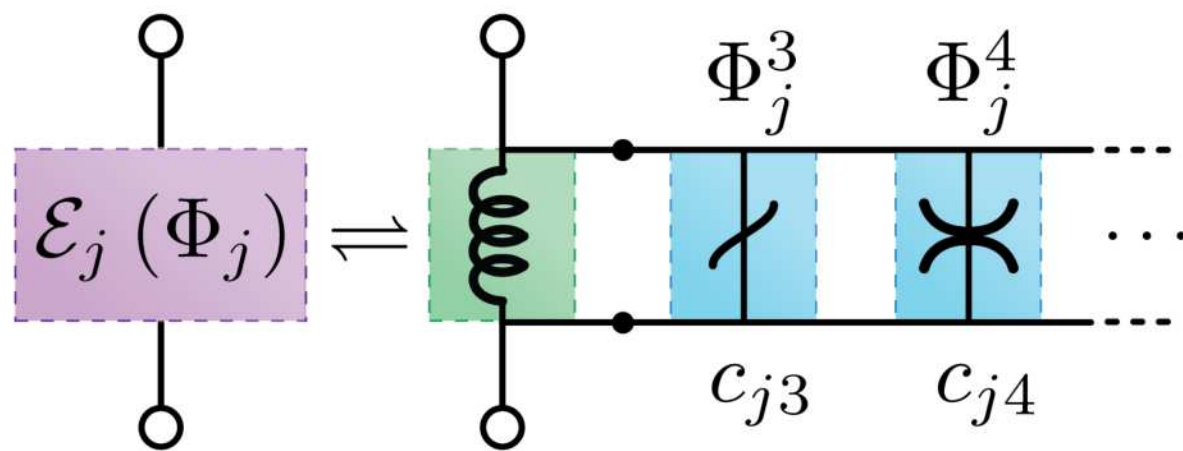
series with respect to operational equilibrium point

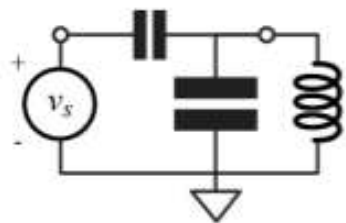


SEM image: Frattini & Sivak; see [APL \(2017\)](#)

Bias conditions

$$\mathcal{E}_j(\Phi_j; \Phi_j^{\text{ext}}) = \mathcal{E}_j^{\text{lin}}(\Phi_j) + \mathcal{E}_j^{\text{nl}}(\Phi_j)$$





Charge dispersion with EPR

$$\omega_\theta = 6.50 \text{ GHz}$$

$$\alpha^{(4)} = 322.00 \text{ MHz}$$

$$\phi_{\text{ZPF}} = 0.45$$

$$E_J = 16.40 \text{ GHz}$$

$$E_C = 0.32 \text{ GHz} \quad (E_J/E_C = 50.90)$$

$$\text{Dispersion} = -12.90 \text{ KHz}$$

